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Abstract Algebra

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I covered this material in a two-semester graduate course in abstract algebra in 2004-05, rethinking the material from scratch, ignoring traditional prejudices.

I wrote proofs which are natural outcomes of the viewpoint. A viewpoint is good if taking it up means that there is less to remember. *Robustness*, as opposed to *fragility*, is a desirable feature of an argument. It is burdensome to be clever. Since it is non-trivial to arrive at a viewpoint that allows proofs to seem easy, such a viewpoint is revisionist. However, this is a *good* revisionism, as opposed to much worse, destructive revisionisms which are nevertheless popular, most notably the misguided impulse to logical perfection [sic]. Logical streamlining is not the same as optimizing for performance.

The worked examples are meant to be model solutions for many of the standard traditional exercises. I no longer believe that everyone is obliged to redo everything themselves. Hopefully it is possible to learn from others' efforts.

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Introduction

Abstract Algebra is not a conceptually well-defined body of material, but a conventional name that refers roughly to one of the several lists of things that mathematicians need to know to be competent, effective, and sensible. This material fits a two-semester beginning graduate course in abstract algebra. It is a *how-to* manual, not a monument to traditional icons. Rather than an encyclopedic reference, it tells a story, with plot-lines and character development propelling it forward.

The main novelty is that most of the standard exercises in abstract algebra are given here as *worked* examples. Some additional exercises are given, which are variations on the worked examples. The reader might contemplate the examples before reading the solutions, but this is not mandatory. The examples are given to *assist*, not necessarily *challenge*. The point is *not* whether or not the reader can do the problems on their own, since all of these are at least fifty years old, but, rather, whether the *viewpoint* is assimilated. In particular, it often happens that a logically correct solution is conceptually regressive, and should not be considered satisfactory.

I promote an efficient, abstract viewpoint whenever it is purposeful to abstract things, especially when letting go of appealing but irrelevant details is advantageous. Some things often not mentioned in an algebra course are included. Some naive set theory, developing ideas about ordinals, is occasionally useful, and the abstraction of this setting makes the set theory seem less farfetched or baffling than it might in a more elementary context. Equivalents of the Axiom of Choice are described. Quadratic reciprocity is useful in understanding quadratic and cyclotomic extensions of the rational numbers, and I give the proof by Gauss' sums. An economical proof of Dirichlet's theorem on primes in arithmetic progressions is included, with discussion of relevant complex analysis, since existence of primes satisfying linear congruence conditions comes up in practice. Other small enrichment topics are treated briefly at opportune moments in examples and exercises. Again, algebra is not a unified or linearly ordered body of knowledge, but only a rough naming convention for an ill-defined and highly variegated landscape of ideas. Further, as with all parts of the basic graduate mathematics curriculum, many important things are inevitably left out. For algebraic geometry or algebraic number theory, much more commutative algebra is useful than is presented here. Only vague hints of representation theory are detectable here.

Far more systematic emphasis is given to finite fields, cyclotomic polynomials (divisors of $x^n - 1$), and cyclotomic fields than is usual, and less emphasis is given to *abstract* Galois theory. Ironically, there are many more explicit Galois theory *examples* here than in sources that emphasize abstract Galois theory. After proving Lagrange's theorem and the Sylow theorem, the pure theory of finite groups is not especially emphasized. After all, the Sylow theorem is not interesting because it allows classification of groups of small order, but because its *proof* illustrates *group actions on sets*, a ubiquitous mechanism in mathematics. A strong and recurring theme is the characterization of objects by *(universal) mapping properties*, rather than by goofy constructions. Nevertheless, formal category theory does not appear. A greater emphasis is put on linear and multilinear algebra, while doing little with general commutative algebra apart from Gauss' lemma and Eisenstein's criterion, which are immediately useful.

Students need good role models for writing mathematics. This is a reason for the complete write-ups of solutions to many examples, since most traditional situations do *not* provide students with *any* models for solutions to the standard problems. This is bad. Even worse, lacking full solutions written by a practiced hand, inferior and regressive solutions may propagate. I do not always insist that students give solutions in the style I wish, but it is very desirable to provide beginners with good examples.

The reader is assumed to have *some* prior acquaintance with introductory abstract algebra and linear algebra, not to mention other standard courses that are considered preparatory for graduate school. This is not so much for specific information as for maturity.

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1. The integers

1.1 Unique factorization
1.2 Irrationalities
1.3 $\mathbb{Z}/m$, the integers mod m
1.4 Fermat's little theorem, Euler's theorem
1.5 Sun-Ze's theorem
1.6 Worked examples

1.1 *Unique factorization*

Let $\mathbb{Z}$ denote the integers. Say d **divides** m, equivalently, that m is a **multiple** of d, if there exists an integer q such that $m = qd$. Write $d|m$ if d divides m.

It is easy to prove, from the definition, that if $d|x$ and $d|y$ then $d|(ax + by)$ for any integers x, y, a, b: let $x = rd$ and $y = sd$, and

$$ax + by = a(rd) + b(sd) = d \cdot (ar + bs)$$

1.1.1 Theorem: Given an integer N and a non-zero integer m there are unique integers q and r, with $0 \leq r < |m|$ such that

$$N = q \cdot m + r$$

The integer r is the **reduction modulo** m of N.

Proof: Let S be the set of all non-negative integers expressible in the form $N - sm$ for some integer s. The set S is non-empty, so by well-ordering has a least element $r = N - qm$. Claim that $r < |m|$. If not, then still $r - |m| \geq 0$, and also

$$r - |m| = (N - qm) - |m| = N - (q \pm 1)m$$

(with the sign depending on the sign of m) is still in the set S, contradiction. For uniqueness, suppose that both $N = qm + r$ and $N = q'm + r'$. Subtract to find

$$r - r' = m \cdot (q' - q)$$

Thus, $r - r'$ is a multiple of m. But since $-|m| < r - r' < |m|$ we have $r = r'$. And then $q = q'$. ///

1.1.2 Remark: The conclusion of the theorem is that in $\mathbb{Z}$ one can divide and obtain a remainder *smaller* than the divisor. That is, $\mathbb{Z}$ is **Euclidean**.

As an example of nearly trivial things that can be proven about divisibility, we have:

A divisor d of n is **proper** if it is neither $\pm n$ nor ± 1. A positive integer p is **prime** if it has no proper divisors and if $p > 1$.

1.1.3 Proposition: A positive integer n is prime if and only if it is not divisible by any of the integers d with $1 < d \leq \sqrt{n}$.

Proof: Suppose that n has a proper factorization $n = d \cdot e$, where $d \leq e$. Then

$$d = \frac{n}{e} \leq \frac{n}{d}$$

gives $d^2 \leq n$, so $d \leq \sqrt{n}$. ///

1.1.4 Remark: The previous proposition suggests that to test an integer n for primality we attempt to divide n by all integers $d = 2, 3, \ldots$ in the range $d \leq \sqrt{n}$. If no such d divides n, then n is prime. This procedure is **trial division**.

Two integers are **relatively prime** or **coprime** or **mutually prime** if for every integer d if $d|m$ and $d|n$ then $d = \pm 1$.

An integer d is a **common divisor** of integers $n_1, \ldots, n_m$ if d divides each n_i. An integer N is a **common multiple** of integers $n_1, \ldots, n_m$ if N is a multiple of each. The following peculiar characterization of the greatest common divisor of two integers is fundamental.

1.1.5 Theorem: Let m, n be integers, not both zero. Among all *common* divisors of m, n there is a unique $d > 0$ such that for *every* other common divisor e of m, n we have $e|d$. This d is the *greatest common divisor* of m, n, denoted $\gcd(m, n)$. And

$$\gcd(mn) = \text{least positive integer of the form } xm + yn \text{ with } x, y \in \mathbb{Z}$$

Proof: Let $D = x_o m + y_o n$ be the least positive integer expressible in the form $xm + yn$. First, we show that any divisor d of both m and n divides D. Let $m = m'd$ and $n = n'd$ with $m', n' \in \mathbb{Z}$. Then

$$D = x_o m + y_o n = x_o(m'd) + y_o(n'd) = (x_o m' + y_o n') \cdot d$$

which presents D as a multiple of d.

On the other hand, let $m = qD + r$ with $0 \leq r < D$. Then

$$0 \leq r = m - qD = m - q(x_o m + y_o n) = (1 - qx_o) \cdot m + (-y_o) \cdot n$$

That is, r is expressible as $x'm + y'n$. Since $r < D$, and since D is the smallest positive integer so expressible, $r = 0$. Therefore, $D|m$, and similarly $D|n$. ///

Similarly:

1.1.6 Corollary: Let m, n be integers, not both zero. Among all *common* multiples of m, n there is a unique positive one N such that for *every* other common multiple M we have $N|M$. This multiple N is the *least common multiple* of m, n, denoted $\text{lcm}(m, n)$. In particular,

$$\text{lcm}(m, n) = \frac{mn}{\gcd(m, n)}$$

Proof: Let

$$L = \frac{mn}{\gcd(m,n)}$$

First we show that L is a multiple of m and n. Indeed, let

$$m = m' \cdot \gcd(m,n) \qquad n = n' \cdot \gcd(m,n)$$

Then

$$L = m \cdot n' = m' \cdot n$$

expresses L as an integer multiple of m and of n. On the other hand, let M be a multiple of both m and n. Let $\gcd(m,n) = am + bn$. Then

$$1 = a \cdot m' + b \cdot n'$$

Let $N = rm$ and $N = sn$ be expressions of N as integer multiples of m and n. Then

$$N = 1 \cdot N = (a \cdot m' + b \cdot n') \cdot N = a \cdot m' \cdot sn + b \cdot n' \cdot rm = (as + br) \cdot L$$

as claimed. ///

The innocent assertion and perhaps odd-seeming argument of the following are essential for what follows. Note that the key point is the peculiar characterization of the *gcd*, which itself comes from the Euclidean property of $\mathbb{Z}$.

1.1.7 Theorem: A prime p divides a product ab if and only if $p|a$ or $p|b$.

Proof: If $p|a$ we are done, so suppose p does not divide a. Since p is prime, and since $\gcd(p,a) \neq p$, it must be that $\gcd(p,a) = 1$. Let r, s be integers such that $1 = rp + sa$, and let $ab = kp$. Then

$$b = b \cdot 1 = b(rp + sa) = p \cdot (rb + sk)$$

so b is a multiple of p. ///

Granting the theorem, the proof of unique factorization is nearly an afterthought:

1.1.8 Corollary: *(Unique Factorization)* Every integer n can be written in an *essentially unique* way (up to reordering the factors) as $\pm$ a product of primes:

$$n = \pm\, p_1^{e_1}\, p_2^{e_2} \dots p_m^{e_m}$$

with positive integer exponents and primes $p_1 < \dots < p_m$.

Proof: For *existence,* suppose $n > 1$ is the least integer *not* having a factorization. Then n cannot be prime itself, or just '$n = n$' is a factorization. Therefore n has a proper factorization $n = xy$ with $x, y > 1$. Since the factorization is *proper*, both x and y are strictly smaller than n. Thus, x and y both can be factored. Putting together the two factorizations gives the factorization of n, contradicting the assumption that there exist integers lacking prime factorizations.

Now *uniqueness.* Suppose

$$q_1^{e_1} \dots q_m^{e_m} = N = p_1^{f_1} \dots p_n^{f_n}$$

where $q_1 < \dots < q_m$ are primes, and $p_1 < \dots < p_n$ are primes, and the exponents e_i and f_i are positive integers. Since q_1 divides the left-hand side of the equality, it divides the right-hand side. Therefore, q_1 must divide one of the factors on the right-hand side. So q_1 must divide some p_i. Since p_i is prime, it must be that $q_1 = p_i$.

If $i > 1$ then $p_1 < p_i$. And p_1 divides the left-hand side, so divides one of the q_j, so is some q_j, but then

$$p_1 = q_j \geq q_1 = p_i > p_1$$

which is impossible. Therefore, $q_1 = p_1$.

Without loss of generality, $e_1 \leq f_1$. Thus, by dividing through by $q_1^{e_1} = p_1^{e_1}$, we see that the corresponding exponents e_1 and f_1 must also be equal. Then do induction. ///

1.1.9 Example: The simplest meaningful (and standard) example of the failure of unique factorization into primes is in the collection of numbers

$$\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$$

The relation

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{5})$$

gives two different-looking factorizations of 6. We must verify that 2, 3, $1 + \sqrt{-5}$, and $1 - \sqrt{-5}$ are *primes* in R, in the sense that they cannot be further factored.

To prove this, we use *complex conjugation*, denoted by a bar over the quantity to be conjugated: for real numbers a and b,

$$\overline{a + b\sqrt{-5}} = a - b\sqrt{-5}$$

For α, β in R,

$$\overline{\alpha \cdot \beta} = \overline{\alpha} \cdot \overline{\beta}$$

by direct computation. Introduce the **norm**

$$N(\alpha) = \alpha \cdot \overline{\alpha}$$

The multiplicative property

$$N(\alpha \cdot \beta) = N(\alpha) \cdot N(\beta)$$

follows from the corresponding property of conjugation:

$$N(\alpha) \cdot N(\beta) = \alpha\overline{\alpha}\beta\overline{\beta} = (\alpha\beta) \cdot (\overline{\alpha}\,\overline{\beta})$$

$$= (\alpha\beta) \cdot (\overline{\alpha\beta}) = N(\alpha\beta)$$

Note that $0 \leq N(\alpha) \in \mathbb{Z}$ for α in R.

Now suppose $2 = \alpha\beta$ with α, β in R. Then

$$4 = N(2) = N(\alpha\beta) = N(\alpha) \cdot N(\beta)$$

By unique factorization in $\mathbb{Z}$, $N(\alpha)$ and $N(\beta)$ must be $1, 4$, or $2, 2$, or $4, 1$. The middle case is impossible, since no norm can be 2. In the other two cases, one of α or β is ± 1, and the factorization is not *proper*. That is, 2 cannot be factored further in $\mathbb{Z}[\sqrt{-5}]$. Similarly, 3 cannot be factored further.

If $1 + \sqrt{-5} = \alpha\beta$ with α, β in R, then again

$$6 = N\left(1 + \sqrt{-5}\right) = N(\alpha\beta) = N(\alpha) \cdot N(\beta)$$

Again, the integers $N(\alpha)$ and $N(\beta)$ must either be $1, 6$, $2, 3$, $3, 2$, or $6, 1$. Since the norm cannot be 2 or 3, the middle two cases are impossible. In the remaining two cases, one of α or β is ± 1, and the factorization is not *proper*. That is, $1 + \sqrt{-5}$ cannot be factored further in R. Neither can $1 - \sqrt{-5}$. Thus,

$$6 = 2 \cdot 3 = \left(1 + \sqrt{-5}\right)\left(1 - \sqrt{5}\right)$$

is a factorization of 6 in two different ways *into primes* in $\mathbb{Z}[\sqrt{-5}]$.

1.1.10 Example: The **Gaussian integers**

$$\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$$

where $i^2 = -1$ do have a Euclidean property, and thus have unique factorization. Use the *integer-valued* norm

$$N(a + bi) = a^2 + b^2 = (a + bi) \cdot \overline{(a + bi)}$$

It is important that the notion of size be integer-valued and respect multiplication. We claim that, given $\alpha, \delta \in \mathbb{Z}[i]$ there is $q \in \mathbb{Z}[i]$ such that

$$N(\alpha - q \cdot \delta) < N(\delta)$$

Since N is *multiplicative* (see above), we can divide through by δ inside

$$\mathbb{Q}(i) = \{a + bi : a, b, \in \mathbb{Q}\}$$

(where $\mathbb{Q}$ is the rationals) to see that we are asking for $q \in \mathbb{Z}[i]$ such that

$$N(\frac{\alpha}{\delta} - q) < N(1) = 1$$

That is, given $\beta = \alpha/\delta$ in $\mathbb{Q}(i)$, we must be able to find $q \in \mathbb{Z}[i]$ such that

$$N(\beta - q) < 1$$

With $\beta = a + bi$ with $a, b \in \mathbb{Q}$, let

$$a = r + f_1 \qquad b = s + f_2$$

with $r, s \in \mathbb{Z}$ and f_1, f_2 rational numbers with

$$|f_i| \leq \frac{1}{2}$$

That this is possible is a special case of the fact that any *real* number is at distance at most $1/2$ from some integer. Then take

$$q = r + si$$

Then

$$\beta - q = (a + bi) - (r + si) = f_1 + if_2$$

and

$$N(\beta - q) = N(f_1 + if_2) = f_1^2 + f_2^2 \leq \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2} < 1$$

Thus, indeed $\mathbb{Z}[i]$ has the Euclidean property, and, by the same proof as above, has unique factorization.

1.2 *Irrationalities*

The usual proof that there is no square root of 2 in the rationals $\mathbb{Q}$ uses a little bit of unique factorization, in the notion that it is possible to put a fraction into lowest terms, that is, having relatively prime numerator and denominator.

That is, given a fraction a/b (with $b \neq 0$), letting $a' = a/\gcd(a, b)$ and $b' = b/\gcd(a, b)$, one can and should show that $\gcd(a', b') = 1$. That is, $a'b/b'$ is **in lowest terms**. And

$$\frac{a'}{b'} = \frac{a}{b}$$

1.2.1 Example: Let p be a prime number. We claim that there is no $\sqrt{p}$ in the rationals $\mathbb{Q}$. Suppose, to the contrary, that $a/b = \sqrt{p}$. Without loss of generality, we can assume that $\gcd(a, b) = 1$. Then, squaring and multiplying out,

$$a^2 = pb^2$$

Thus, $p|a^2$. Since $p|cd$ implies $p|c$ or $p|d$, necessarily $p|a$. Let $a = pa'$. Then

$$(pa')^2 = pb^2$$

or

$$pa'^2 = b^2$$

Thus, $p|b$, contradicting the fact that $\gcd(a, b) = 1$. ///

The following example illustrates a possibility that will be subsumed later by *Eisenstein's criterion*, which is itself an application of *Newton polygons* attached to polynomials.

1.2.2 Example: Let p be a prime number. We claim that there is no rational solution to

$$x^5 + px + p = 0$$

Indeed, suppose that a/b were a rational solution, in lowest terms. Then substitute and multiply through by b^5 to obtain

$$a^5 + pab^4 + pb^5 = 0$$

From this, $p|a^5$, so, since p is prime, $p|a$. Let $a = pa'$. Then

$$(pa')^5 + p(pa')b^4 + pb^5 = 0$$

or

$$p^4a'^5 + p^2a'b^4 + b^5 = 0$$

From this, $p|b^5$, so $p|b$ since p is prime. This contradicts the lowest-terms hypothesis.

1.3 $\mathbb{Z}/m$, *the integers mod* m

Recall that a *relation* R on a set S is a subset of the cartesian product $S \times S$. Write

$$x \; R \; y$$

if the ordered pair (x, y) lies in the subset R of $S \times S$. An **equivalence relation** R on a set S is a relation satisfying
• **Reflexivity**: $x \; R \; x$ for all $x \in S$
• **Symmetry**: If $x \; R \; y$ then $y \; R \; x$
• **Transitivity**: If $x \; R \; y$ and $y \; R \; z$ then $x \; R \; z$

A common notation for an equivalence relation is

$$x \sim y$$

that is, with a tilde rather than R.

Let $\sim$ be an equivalence relation on a set S. For $x \in S$, the $\sim$ - **equivalence class** $\bar{x}$ containing x is the subset

$$\bar{x} = \{x' \in S : x' \sim x\}$$

The **set of equivalence classes** of $\sim$ on S is denoted by

$$S/\sim$$

(as a quotient). Every element $z \in S$ is contained in an equivalence class, namely the equivalence class $\bar{z}$ of all $s \in S$ so that $s \sim z$. Given an equivalence class A inside S, an x in the set S such that $\bar{x} = A$ is a **representative** for the equivalence class. That is, any element of the subset A is a representative.

A set $\mathcal{S}$ of non-empty subsets of a set S whose union is the whole S, and which are mutually disjoint, is a **partition** of S. One can readily check that the equivalence classes of an equivalence relation on a set S form a partition of S, and, conversely, any partition of S defines an equivalence relation by positing that $x \sim y$ if and only if they lie in the same set of the partition. ///

If two integers x, y differ by a multiple of a non-zero integer m, that is, if $m|(x-y)$, then x is **congruent to** y **modulo** m, written

$$x \equiv y \bmod m$$

Such a relation a **congruence** modulo m, and m is the **modulus**. When Gauss first used this notion 200 years ago, it was sufficiently novel that it deserved a special notation, but, now that the novelty has worn off, we will simply write

$$x = y \bmod m$$

and (unless we want special emphasis) simply say that x is **equal to** y modulo m.

1.3.1 Proposition: (For fixed modulus m) equality modulo m is an equivalence relation. ///

Compatibly with the general usage for equivalence relations, the **congruence class** (or **residue class** or **equivalence class**) of an integer x modulo m, denoted $\bar{x}$ (with only implicit reference to m) is the set of all integers equal to x mod m:

$$\bar{x} = \{y \in \mathbb{Z} : y = x \bmod m\}$$

The **integers mod** m, denoted $\mathbb{Z}/m$, is the collection of *congruence classes* of integers modulo m. For some $X \in \mathbb{Z}/m$, a choice of ordinary integer x so that $\bar{x} = X$ is a **representative** for the congruence class X.

1.3.2 Remark: A popular but unfortunate notation for $\mathbb{Z}/m$ is $\mathbb{Z}_m$. We will not use this notation. It is unfortunate because for primes p the notation $\mathbb{Z}_p$ is the *only* notation for the *p-adic integers*.

1.3.3 Remark: On many occasions, the bar is dropped, so that x-mod-m may be written simply as 'x'.

1.3.4 Remark: The traditionally popular collection of representatives for the equivalence classes modulo m, namely

$$\{\bar{0}, \bar{1}, \bar{2}, \dots \overline{m-2}, \overline{m-1}\}$$

is not the only possibility.

The benefit Gauss derived from the explicit notion of congruence was that congruences behave much like equalities, thus allowing us to benefit from our prior experience with equalities. Further, but not surprisingly with sufficient hindsight, congruences behave nicely with respect to the basic operations of addition, subtraction, and multiplication:

1.3.5 Proposition: Fix the modulus m. If $x = x' \bmod m$ and $y = y' \bmod m$, then

$$x + y = x' + y' \bmod m$$

$$xy = x'y' \bmod m$$

Proof: Since $m|(x'-x)$ there is an integer k such that $mk = x' - x$. Similarly, $y' = y + \ell m$ for some integer ℓ. Then

$$x' + y' = (x + mk) + (y + m\ell) = x + y + m \cdot (k + \ell)$$

Thus, $x' + y' = x + y \bmod m$. And

$$x' \cdot y' = (x + mk) \cdot (y + m\ell) = x \cdot y + xm\ell + mky + mk \cdot m\ell = x \cdot y + m \cdot (k + \ell + mk\ell)$$

Thus, $x'y' = xy \bmod m$. ///

As a corollary, congruences *inherit* many basic properties from ordinary arithmetic, simply because $x = y$ implies $x = y \bmod m$:
• *Distributivity:* $x(y + z) = xy + xz \bmod m$
• *Associativity of addition:* $(x + y) + z = x + (y + z) \bmod m$
• *Associativity of multiplication:* $(xy)z = x(yz) \bmod m$
• *Property of* 1*:* $1 \cdot x = x \cdot 1 = x \bmod m$
• *Property of* 0*:* $0 + x = x + 0 = x \bmod m$

In this context, a **multiplicative inverse mod** m to an integer a is an integer b (if it exists) such that

$$a \cdot b = 1 \bmod m$$

1.3.6 Proposition: An integer a has a multiplicative inverse modulo m if and only if $\gcd(a, m) = 1$.

Proof: If $\gcd(a, m) = 1$ then there are r, s such that $ra + sm = 1$, and

$$ra = 1 - sm = 1 \bmod m$$

The other implication is easy. ///

In particular, note that if a is invertible mod m then any a' in the residue class of a mod m is likewise invertible mod m, and any other element b' of the residue class of an inverse b is also an inverse. Thus, it makes sense to refer to elements of $\mathbb{Z}/m$ as being invertible or not. Notation:

$$(\mathbb{Z}/m)^\times = \{\bar{x} \in \mathbb{Z}/m : \gcd(x, m) = 1\}$$

This set $(\mathbb{Z}/m)^\times$ is the **multiplicative group** or **group of units** of $\mathbb{Z}/m$.

1.3.7 Remark: It is easy to verify that the set $(\mathbb{Z}/m)^\times$ is **closed under multiplication** in the sense that $a, b \in (\mathbb{Z}/m)^\times$ implies $ab \in (\mathbb{Z}/m)^\times$, and is **closed under inverses** in the sense that $a \in (\mathbb{Z}/m)^\times$ implies $a^{-1} \in (\mathbb{Z}/m)^\times$.

1.3.8 Remark: The superscript is not an 'x' but is a 'times', making a reference to multiplication and multiplicative inverses mod m. Some sources write $\mathbb{Z}/m^*$, but the latter notation is inferior, as it is too readily confused with other standard notation (for *duals*).

1.4 *Fermat's Little Theorem*

1.4.1 Theorem: Let p be a prime number. Then for any integer x

$$x^p = x \bmod p$$

Proof: First, by the Binomial Theorem

$$(x + y)^p = \sum_{0 \le i \le p} \binom{p}{i} x^i \, y^{p-i}$$

In particular, the binomial coefficients are *integers*. Now we can show that the prime p divides the binomial coefficients

$$\binom{p}{i} = \frac{p!}{i!\ (p-i)!}$$

with $1 \leq i \leq p-1$. We have

$$\binom{p}{i} \cdot i! \cdot (p-i)! = p!$$

(Since we know that the binomial coefficient is an integer, the following argument makes sense.) The prime p divides the right-hand side, so divides the left-hand side, but does not divide $i!$ nor $(p-i)!$ (for $0 < i < p$) since these two numbers are products of integers smaller than p and (hence) not divisible by p. Again using the fact that $p|ab$ implies $p|a$ or $p|b$, p does not divide $i! \cdot (p-i)!$, so p must divide the binomial coefficient.

Now we prove Fermat's Little Theorem for *positive* integers x by induction on x. Certainly $1^p = 1 \bmod p$. Now suppose that we know that

$$x^p = x \bmod p$$

Then

$$(x+1)^p = \sum_{0 \leq i \leq p} \binom{p}{i} x^i\, 1^{p-i} = x^p + \sum_{0<i<p} \binom{p}{i} x^i + 1$$

All the coefficients in the sum in the middle of the last expression are divisible by p, so

$$(x+1)^p = x^p + 0 + 1 = x + 1 \bmod p$$

This proves the theorem for positive x. ///

1.4.2 Example: Let p be a prime with $p = 3 \bmod 4$. Suppose that a is a **square modulo** p, in the sense that there exists an *integer* b such that

$$b^2 = a \bmod p$$

Such b is a **square root modulo** p of a. Then we claim that $a^{(p+1)/4}$ is a square root of a mod p. Indeed,

$$\left(a^{(p+1)/4}\right)^2 = \left((b^2)^{(p+1)/4}\right)^2 = b^{p+1} = b^p \cdot b = b \cdot b \bmod p$$

by Fermat. Then this is a mod p. ///

1.4.3 Example: Somewhat more generally, let q be a prime, and let p be another prime with $p = 1 \bmod q$ but $p \neq 1 \bmod q^2$.

$$r = q^{-1} \bmod \frac{p-1}{q}$$

Then if a is a qth power modulo p, a qth root of a mod p is given by the formula

$$q^{th} \text{ root of } a \bmod p = a^r \bmod p$$

If a is *not* a qth power mod p then this formula does *not* product a qth root.

1.4.4 Remark: For prime q and prime $p \neq 1 \bmod q$ there is an even simpler formula for q^{th} roots, namely let

$$r = q^{-1} \bmod p - 1$$

and then

$$q^{th} \text{ root of } a \bmod p = a^r \bmod p$$

Further, as can be seen from the even-easier proof of this formula, *everything* mod such p is a qth power.

For a positive integer n, the **Euler phi-function** $\varphi(n)$ is the number of integers b so that $1 \le b \le n$ and $\gcd(b, n) = 1$. Note that

$$\varphi(n) = \text{cardinality of } (\mathbb{Z}/n)^\times$$

1.4.5 Theorem: *(Euler)* For x relatively prime to a positive integer n,

$$x^{\varphi(n)} = 1 \bmod n$$

1.4.6 Remark: The special case that n is prime is Fermat's Little Theorem.

Proof: Let $G = (\mathbb{Z}/m)^\times$, for brevity. First note that the product

$$P = \prod_{g \in G} g = \text{product of all elements of } G$$

is again in G. Thus, P has a multiplicative inverse mod n, although we do not try to identify it. Let x be an element of G. Then we claim that the map $f : G \longrightarrow G$ defined by

$$f(g) = xg$$

is a bijection of G to itself. First, check that f really maps G to itself: for x and g both invertible mod n,

$$(xg)(g^{-1}x^{-1}) = 1 \bmod n$$

Next, injectivity: if $f(g) = f(h)$, then $xg = xh \bmod n$. Multiply this equality by $x^{-1} \bmod n$ to obtain $g = h \bmod n$. Last, surjectivity: given $g \in G$, note that $f(x^{-1}g) = g$.

Then

$$P = \prod_{g \in G} g = \prod_{g \in G} f(g)$$

since the map f merely permutes the elements of G. Then

$$P = \prod_{g \in G} f(g) = \prod_{g \in G} xg = x^{\varphi(n)} \prod_{g \in G} g = x^{\varphi(n)} \cdot P$$

Since P is invertible mod n, multiply through by $P^{-1} \bmod n$ to obtain

$$1 = x^{\varphi(n)} \bmod n$$

This proves Euler's Theorem. ///

1.4.7 Remark: This proof of Euler's theorem, while subsuming Fermat's Little Theorem as a special case, strangely uses fewer specifics. There is no mention of binomial coefficients, for example.

1.4.8 Remark: The argument above is a prototype example for the basic Lagrange's Theorem in basic group theory.

1.5 Sun-Ze's theorem

The result of this section is sometimes known as the **Chinese Remainder Theorem**. Indeed, the earliest results (including and following Sun Ze's) were obtained in China, but such sloppy attribution is not good. Sun Ze's result was obtained before 850, and the statement below was obtained by Chin Chiu Shao about 1250. Such results, with virtually the same proofs, apply much more generally.

1.5.1 Theorem: *(Sun-Ze)* Let m and n be relatively prime positive integers. Let r and s be integers such that

$$rm + sn = 1$$

Then the function

$$f : \mathbb{Z}/m \times \mathbb{Z}/n \longrightarrow \mathbb{Z}/mn$$

defined by

$$f(x, y) = y \cdot rm + x \cdot sn$$

is a bijection. The inverse map

$$f^{-1} : \mathbb{Z}/mn \longrightarrow \mathbb{Z}/m \times \mathbb{Z}/n$$

is

$$f^{-1}(z) = (x\text{-mod-}m, y\text{-mod-}n)$$

Proof: First, the peculiar characterization of $\gcd(m, n)$ as the smallest positive integer expressible in the form $rm + sn$ assures (since here $\gcd(m, n) = 1$) that integers r and s exist such that $rm + sn = 1$. Second, the function f is well-defined, that is, if $x' = x + am$ and $y' = y + bn$ for integers a and b, then still

$$f(x', y') = f(x, y)$$

Indeed,

$$f(x', y') = y'rm + x'sn = (y + an)rm + (x + am)sn$$
$$= yrm + xsn + mn(ar + bs) = f(x, y) \bmod mn$$

proving the well-definedness.

To prove surjectivity of f, for any integer z, let $x = z$ and $y = z$. Then

$$f(x, y) = zrm + zsn = z(rm + sn) = z \cdot 1 \bmod mn$$

(To prove injectivity, we *could* use the fact that $\mathbb{Z}/m \times \mathbb{Z}/n$ and $\mathbb{Z}/mn$ are finite sets of the same size, so a surjective function is necessarily injective, but a more direct argument is more instructive.) Suppose

$$f(x', y') = f(x, y)$$

Then modulo m the yrm and $y'rm$ are 0, so

$$xsn = x'sn \bmod m$$

From $rm + sn = 1 \bmod mn$ we obtain $sn = 1 \bmod m$, so

$$x = x' \bmod m$$

Symmetrically,

$$y = y' \bmod n$$

giving injectivity.

Finally, by the same reasoning,

$$f(x,y) = yrm + xsn = y \cdot 0 + x \cdot 1 \bmod m = x \bmod m$$

and similarly

$$f(x,y) = yrm + xsn = y \cdot 1 + x \cdot 0 \bmod n = y \bmod n$$

This completes the argument. ///

1.5.2 Remark: The above result is the simplest prototype for a very general result.

1.6 *Worked examples*

1.6.1 Example: Let D be an integer that is not the square of an integer. Prove that there is no $\sqrt{D}$ in $\mathbb{Q}$.

Suppose that a, b were integers ($b \neq 0$) such that $(a/b)^2 = D$. The fact/principle we intend to invoke here is that fractions can be put in *lowest terms*, in the sense that the numerator and denominator have greatest common divisor 1. This follows from *existence* of the *gcd*, and from the fact that, if $\gcd(a,b) > 1$, then let $c = a/\gcd(a,b)$ and $d = b/\gcd(a,b)$ and we have $c/d = a/b$. Thus, still $c^2/d^2 = D$. One way to proceed is to prove that c^2/d^2 is still in lowest terms, and thus cannot be an integer unless $d = \pm 1$. Indeed, if $\gcd(c^2, d^2) > 1$, this *gcd* would have a prime factor p. Then $p|c^2$ implies $p|c$, and $p|d^2$ implies $p|d$, by the critical proven property of primes. Thus, $\gcd(c,d) > 1$, contradiction.

1.6.2 Example: Let p be prime, $n > 1$ an integer. Show (directly) that the equation $x^n - px + p = 0$ has no rational root (where $n > 1$).

Suppose there were a rational root a/b, without loss of generality in lowest terms. Then, substituting and multiplying through by b^n, one has

$$a^n - pb^{n-1}a + pb^n = 0$$

Then $p|a^n$, so $p|a$ by the property of primes. But then p^2 divides the first two terms, so must divide pb^n, so $p|b^n$. But then $p|b$, by the property of primes, contradicting the lowest-common-terms hypothesis.

1.6.3 Example: Let p be prime, b an integer not divisible by p. Show (directly) that the equation $x^p - x + b = 0$ has no rational root.

Suppose there were a rational root c/d, without loss of generality in lowest terms. Then, substituting and multiplying through by d^p, one has

$$c^p - d^{p-1}c + bd^p = 0$$

If $d \neq \pm 1$, then some prime q divides d. From the equation, $q|c^p$, and then $q|c$, contradiction to the lowest-terms hypothesis. So $d = 1$, and the equation is

$$c^p - c + b = 0$$

By Fermat's Little Theorem, $p|c^p - c$, so $p|b$, contradiction.

1.6.4 Example: Let r be a positive integer, and p a prime such that $\gcd(r, p-1) = 1$. Show that every b in $\mathbb{Z}/p$ has a unique r^{th} root c, given by the formula

$$c = b^s \bmod p$$

where $rs = 1 \bmod (p-1)$. [*Corollary of Fermat's Little Theorem.*]

1.6.5 Example: Show that $R = \mathbb{Z}[\sqrt{-2}]$ and $\mathbb{Z}[\frac{1+\sqrt{-7}}{2}]$ are Euclidean.

First, we consider $R = \mathbb{Z}[\sqrt{-D}]$ for $D = 1, 2, \ldots$. Let $\omega = \sqrt{-D}$. To prove Euclidean-ness, note that the Euclidean condition that, given $\alpha \in \mathbb{Z}[\omega]$ and non-zero $\delta \in \mathbb{Z}[\omega]$, there exists $q \in \mathbb{Z}[\omega]$ such that

$$|\alpha - q \cdot \delta| < |\delta|$$

is equivalent to

$$|\alpha/\delta - q| < |1| = 1$$

Thus, it suffices to show that, given a complex number α, there is $q \in \mathbb{Z}[\omega]$ such that

$$|\alpha - q| < 1$$

Every complex number α can be written as $x + y\omega$ with real x and y. The simplest approach to analysis of this condition is the following. Let m, n be integers such that $|x - m| \leq 1/2$ and $|y - n| \leq 1/2$. Let $q = m + n\omega$. Then $\alpha - q$ is of the form $r + s\omega$ with $|r| \leq 1/2$ and $|s| \leq 1/2$. And, then,

$$|\alpha - q|^2 = r^2 + Ds^2 \leq \frac{1}{4} + \frac{D}{4} = \frac{1+D}{4}$$

For this to be strictly less than 1, it suffices that $1 + D < 4$, or $D < 3$. This leaves us with $\mathbb{Z}[\sqrt{-1}]$ and $\mathbb{Z}[\sqrt{-2}]$.

In the second case, consider $Z[\omega]$ where $\omega = (1 + \sqrt{-D})/2$ and $D = 3 \bmod 4$. (The latter condition assures that $\mathbb{Z}[x]$ works the way we hope, namely that everything in it is expressible as $a + b\omega$ with $a, b \in \mathbb{Z}$.) For D=3 (the Eisenstein integers) the previous approach still works, but fails for $D = 7$ and for $D = 11$. Slightly more cleverly, realize that first, given complex α, integer n can be chosen such that

$$-\sqrt{D}/4 \leq \text{imaginary part}(\alpha - n\omega) \leq +\sqrt{D}/4$$

since the imaginary part of ω is $\sqrt{D}/2$. *Then* choose integer m such that

$$-1/2 \leq \text{ real part}(\alpha - n\omega - m) \leq 1/2$$

Then take $q = m + n\omega$. We have chosen q such that $\alpha - q$ is in the *rectangular* box of complex numbers $r + s\sqrt{-7}$ with

$$|r| \leq 1/2 \quad \text{and} \quad |s| \leq 1/4$$

Yes, 1/4, not 1/2. Thus, the size of $\alpha - q$ is at most

$$1/4 + D/16$$

The condition that this be strictly less than 1 is that $4 + D < 16$, or $D < 12$ (and $D = 1 \bmod 4$). This gives $D = 3, 7, 11$.

1.6.6 Example: Let $f : X \longrightarrow Y$ be a function from a set X to a set Y. Show that f has a left inverse if and only if it is injective. Show that f has a right inverse if and only if it is surjective. (Note where, if anywhere, the Axiom of Choice is needed.)

1.6.7 Example: Let $h : A \longrightarrow B$, $g : B \longrightarrow C$, $f : C \longrightarrow D$. Prove the associativity

$$(f \circ g) \circ h = f \circ (g \circ h)$$

Two functions are equal if and only if their values (for the same inputs) are the same. Thus, it suffices to evaluate the two sides at $a \in A$, using the definition of composite:

$$((f \circ g) \circ h)(a) = (f \circ g)(h(a)) = f(g((h(a))) = f((g \circ h)(a)) = (f \circ (g \circ h))(a)$$

1.6.8 Example: Show that a set is infinite if and only if there is an injection of it to a proper subset of itself. Do not set this up so as to trivialize the question.

The other definition of *finite* we'll take is that a set S is finite if there is a surjection to it from one of the sets

$$\{\},\ \{1\},\ \{1,2\},\ \{1,2,3\},\ \dots$$

And a set is *infinite* if it has no such surjection.

We find a denumerable subset of an infinite set S, as follows. For infinite S, since S is not empty (or there'd be a surjection to it from $\{\}$), there is an element s_1. Define

$$f_1 : \{1\} \longrightarrow S$$

by $f(1) = s_1$. This cannot be surjective, so there is $s_2 \neq s_1$. Define

$$f_2 : \{1,2\} \longrightarrow S$$

by $f(1) = s_1$, $f(2) = s_2$. By induction, for each natural number n we obtain an injection $f_n : \{1,\dots\} \longrightarrow S$, and distinct elements $s_1, 2_2, \dots$. Let S' be the complement to $\{s_1, s_2, \dots\}$ in S. Then define $F : S \longrightarrow S$ by

$$F(s_i) = s_{i+1} \quad F(s') = s' \text{ (for } s' \in S')$$

This is an injection to the proper subset $S - \{s_1\}$.

On the other hand, we claim that no set $\{1,\dots,n\}$ admits an injection to a proper subset of itself. If there were such, by Well-Ordering there would be a least n such that this could happen. Let f be an injection of $S = \{1,\dots,n\}$ to a proper subset of itself.

By hypothesis, f restricted to $S' = \{1,2,\dots,n-1\}$ does *not* map S' to a proper subset of itself. The restriction of an injective function is still injective. Thus, either $f(i) = n$ for some $1 \leq i < n$, or $f(S')$ is the *whole* set S'. In the former case, let j be the least element not in the image $f(S)$. (Since $f(i) = n$, $j \neq n$, but this doesn't matter.) Replace f by $\pi \circ f$ where π is the permutation of $\{1,\dots,n\}$ that interchanges j and n and leaves everything else fixed. Since permutations are bijections, this $\pi \circ f$ is still an injection of S to a proper subset. Thus, we have reduced to the second case, that $f(S') = S'$. By injectivity, $f(n)$ can't be in S', but then $f(n) = n$, and the image $f(S)$ is not a proper subset of S after all, contradiction. ///

In a similar vein, one can *prove* the Pigeon-Hole Principle, namely, that for $m < n$ a function

$$f : \{1,\dots,n\} \longrightarrow \{1,\dots,m\}$$

cannot be injective. Suppose this is false. Let n be the smallest such that there is $m < n$ with an injective map as above. The restriction of an injective map is still injective, so f on $\{1,\dots,n-1\}$ is still injective. By the minimality of n, it must be that $n - 1 = m$, and that f restricted to $\{1,\dots,m\}$ is a bijection of that set to itself. But then there is no possibility for $f(n)$ in $\{1,\dots,m\}$ without violating the injectivity. Contradiction. Thus, there is no such injection to a smaller set.

Exercises

1.1 Let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1 x + a_0$ be a polynomial with integer coefficients a_i. Show that if $f(x) = 0$ has a root in $\mathbb{Q}$, then this root is an integer dividing a_0.

1.2 Show that $x^2 - y^2 = 102$ has no solutions in integers

1.3 Show that $x^3 - y^3 = 3$ has no solutions in integers.

1.4 Show that $x^3 + y^3 - z^3 = 4$ has no solutions in integers.

1.5 Show that $x^2 + 3y^2 + 6z^3 - 9w^5 = 2$ has no solutions in integers.

1.6 The defining property of *ordered pair* (a, b) is that $(a, b) = (a', b')$ if and only if $a = a'$ and $b = b'$. Show that the set-theoretic construction $(a, b) = \{\{a\}, \{a, b\}\}$ succeeds in making an object that behaves as an ordered pair is intended. (*Hint:* Beware: if $x = y$, then $\{x, y\} = \{x\}$.)

1.7 Let p be a prime, and q a positive integer power of p. Show that p divides the binomial coefficients $\binom{q}{i} = q!/i!(q-i)!$ for $0 < i < q$.

1.8 Show that the greatest common divisor of non-zero integers x, y, z is the smallest positive integer expressible as $ax + by + cz$ for integers a, b, c.

1.9 Let m, n be relatively prime integers. Without using factorizations, prove that $m|N$ and $n|N$ implies $mn|N$.

1.10 (*A warm-up to Hensel's lemma*) Let $p > 2$ be a prime. Suppose that b is an integer not divisible by p such that there is a solution y to the equation $y^2 = b \bmod p$. Show (by induction on n) that for $n \geq 1$ there is a unique $x \bmod p^n$ such that $x = b \bmod p$ and

$$x^p = b \bmod p^n$$

1.11 (*Another warm-up to Hensel's lemma*) Let $p > 2$ be a prime. Let y be an integer such that $y \equiv 1 \bmod p$. Show (by induction on n) that for $n \geq 1$ there is a unique $x \bmod p^n$ so that

$$x^p = y \bmod p^n$$

1.12 Let φ be Euler's phi-function, equal to the number of integers ℓ such that $1 \leq \ell < n$ with ℓ relatively prime to n. Show that for a positive integer n

$$n \;=\; \sum_{d|n,\ d>0} \varphi(d)$$

2. Groups I

2.1 Groups
2.2 Subgroups, Lagrange's theorem
2.3 Homomorphisms, kernels, normal subgroups
2.4 Cyclic groups
2.5 Quotient groups
2.6 Groups acting on sets
2.7 The Sylow theorem
2.8 Trying to classify finite groups, part I
2.9 Worked examples

2.1 *Groups*

The simplest, but not most immediately intuitive, object in abstract algebra is a *group*. Once introduced, one can see this structure nearly everywhere in mathematics. [1]

By definition, a **group** G is a set with an **operation** $g * h$ (formally, a function $G \times G \longrightarrow G$), with a special element e called **the identity**, and with properties:
• *The property of the identity*: for all $g \in G$, $e * g = g * e = g$.
• *Existence of* **inverses**: for all $g \in G$ there is $h \in G$ (the **inverse** of g) such that $h * g = g * h = e$.
• *Associativity*: for all $x, y, z \in G$, $x * (y * z) = (x * y) * z$.

If the operation $g * h$ is **commutative**, that is, if

$$g * h = h * g$$

then the group is said to be **abelian**. [2] In that case, often, but not necessarily, the operation is written

[1] Further, the notion of group proves to be more than a mere *descriptive* apparatus. It provides unification and synthesis for arguments and concepts which otherwise would need individual development. Even more, abstract structure theorems for groups provide *predictive* indications, in the sense that we know something in advance about groups we've not yet seen.

[2] After N.H. Abel, who in his investigation of the solvability by radicals of algebraic equations came to recognize

as *addition*. And when the operation is written as addition, the identity is often written as 0 instead of e.

In many cases the group operation is written as multiplication or simply as juxtaposition

$$g * h = g \cdot h = gh$$

This does not *preclude* the operation being abelian, but only denies the *presumption* that the operation is abelian. If the group operation is written as multiplication, then often the identity is denoted as 1 rather than e. Unless written additively, the **inverse** [3] of an element g in the group is denoted

$$\text{inverse of } g = g^{-1}$$

If the group operation is written as *addition*, then the inverse is denoted

$$\text{inverse of } g = -g$$

Many standard mathematical items with natural operations are groups: The set $\mathbb{Z}$ of integers $\mathbb{Z}$ with addition $+$ is an abelian group. The set $n\mathbb{Z}$ of multiples of an integer n, with addition, is an abelian group. The set $\mathbb{Z}/m$ of integers mod m, with addition mod m as the operation is an abelian group. The set $\mathbb{Z}/m^\times$ of integers mod m *relatively prime to* m, with multiplication mod m as the operation is an abelian group.

The set $\mathbb{Z}$ of integers with operation being *multiplication* is *not* a group, because there are no inverses. [4] The closest we can come is the set $\{1, -1\}$ with multiplication.

Other things which we'll define formally only a bit later are groups: vector spaces with vector addition are abelian groups. The set $GL(2, \mathbb{R})$ of invertible 2-by-2 real matrices, with group law matrix multiplication, is a non-abelian group. Here the identity is the matrix

$$1_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The existence of inverses is part of the definition. The *associativity* of matrix multiplication is not entirely obvious from the definition, but can either be checked by hand or inferred from the fact that composition of functions is associative.

A more abstract example of a group is the set S_n of **permutations** of a set with n elements (n an integer), where *permutation* means *bijection to itself*. Here the operation is *composition* (as functions) of permutations. If there are more than two things in the set, S_n is non-abelian.

Some nearly trivial uniqueness issues should be checked: [5]
• *(Uniqueness of identity)* If $f \in G$ and $f * g = g * f = g$ for all g in G, then $f = e$.
• *(Uniqueness of inverses)* For given $g \in G$, if $a * g = e$ and $g * b = e$, then $a = b$.

Proof: For the first assertion,

$$\begin{array}{rcll} f & = & f * e & \text{(property of } e\text{)} \\ & = & e & \text{(assumed property of } e\text{)} \end{array}$$

---

the significance of commutativity many decades before the notion of *group* was formalized.

[3] once we prove its uniqueness!

[4] The fact that there are multiplicative inverses in the larger set $\mathbb{Q}^\times$ of non-zero rational numbers is beside the point, since these inverses are not inside the given set $\mathbb{Z}$.

[5] These are the sort of properties which, if they were *not* provable from the definition of group, would probably need to be added to the definition. We are fortunate that the innocent-looking definition does in fact yield these results.

which was claimed. For the second, similarly,

$$a = a * e = a * (g * b) = (a * g) * b = e * b = b$$

where we use, successively, the property of the identity, the defining property of b, associativity, the defining property of a, and then the property of the identity again. ///

2.1.1 Remark: These uniqueness properties justify speaking of *the* inverse and *the* identity.

2.2 *Subgroups*

Subgroups are subsets of groups which are groups *in their own right*, in the following sense. A subset H of a group G is said to be a **subgroup** if, with the same operation and identity element as that used in G, it is a group.

That is, if H contains the identity element $e \in G$, if H contains inverses of all elements in it, and if H contains products of any two elements in it, then H is a subgroup.

Common terminology is that H is **closed under inverses** if for $h \in H$ the inverse h^{-1} is in H, and **closed under the group operation** if $h_1, h_2 \in H$ implies $h_1 * h_2$ is in H. [6]

Note that the associativity of the operation is assured since the operation was *assumed* associative for G itself to be a group.)

The subset $\{e\}$ of a group G is always a subgroup, termed **trivial**. A subgroup of G other than the trivial subgroup and the group G itself is **proper**.

2.2.1 Proposition: The intersection $\bigcap_{H \in S} H$ of any collection of subgroups of a group G is again a subgroup of G.

Proof: Since the identity e of G lies in each H, it lies in their intersection. If h lies in H for every $H \in S$, then h^{-1} lies in H for every $H \in S$, so h^{-1} is in the intersection. Similarly, if h_1, h_2 are both in H for every $H \in S$, so is their product, and then the product is in the intersection. ///

Given a set X of elements in a group G, the **subgroup generated by** [7] X is defined to be

$$\text{subgroup generated by } X \ = \langle X \rangle = \bigcap_{H \supset X} H$$

where H runs over *subgroups* of G containing X. The previous proposition ensures that this really is a subgroup. If $X = \{x_1, \ldots, x_n\}$ we may, by abuse of notation, write also

$$\langle X \rangle = \langle x_1, \ldots, x_n \rangle$$

and refer to the subgroup generated by $x_1, \ldots, x_n$ rather than by the subset X.

A **finite group** is a group which (as a set) is finite. The **order** of a finite group is the number of elements in it. Sometimes the order of a group G is written as $|G|$ or $o(G)$. The first real theorem in group theory is

[6] In reality, the very notion of *operation* includes the assertion that the output is again in the set. Nevertheless, the property is important enough that extra emphasis is worthwhile.

[7] Later we will see a constructive version of this notion. Interestingly, or, perhaps, disappointingly, the more constructive version is surprisingly complicated. Thus, the present quite non-constructive definition is useful, possibly essential.

2.2.2 Theorem: *(Lagrange)* [8] Let G be a *finite* group. Let H be a subgroup of G. Then the order of H *divides* the order of G.

Proof: For $g \in G$, the **left coset** of H by g or **left translate** of H by g is

$$gH = \{gh : h \in H\}$$

(Similarly, the **right coset** of H by g or **right translate** of H by g is $Hg = \{hg : h \in H\}$.)

First, we will prove that the collection of all left cosets of H is a *partition* of G. Certainly $x = x \cdot e \in xH$, so every element of G lies in a left coset of H. Now suppose that $xH \cap yH \neq \phi$ for $x, y \in G$. Then for some $h_1, h_2 \in H$ we have $xh_1 = yh_2$. Multiply both sides of this equality on the right by h_2^{-1} to obtain

$$(xh_1)h_2^{-1} = (yh_2)h_2^{-1} = y$$

Let $z = h_1 h_2^{-1}$ for brevity. Since H is a *subgroup*, $z \in H$. Then

$$yH = \{yh : h \in H\} = \{(xz)h : h \in H\} = \{x(zh) : h \in H\}$$

Thus, $yH \subset xH$. Since the relationship between x and y is symmetrical, also $xH \subset yH$, and $xH = yH$. Thus, the left cosets of H in G partition G.

Next, show that the cardinalities of the left cosets of H are identical, by demonstrating a *bijection* from H to xH for any $x \in G$. Define

$$f(g) = xg$$

This maps H to yH, and if $f(g) = f(g')$, then

$$xg = xg'$$

from which left multiplication by x^{-1} gives $g = g'$. For *surjectivity*, note that the function f was arranged so that

$$f(h) = xh$$

Thus, all left cosets of H have the same number of elements as H.

So G is the disjoint union of the left cosets of H. From this, $|H|$ divides $|G|$. ///

The **index** $[G : H]$ of a subgroup H in a group G is the number of disjoint (left or right) cosets of H in G. Thus, Lagrange's theorem says

$$|G| = [G : H] \cdot |H|$$

For a single element g of a group G, one can verify that

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$$

where $g^0 = e$, and

$$g^n = \begin{cases} \underbrace{g * g * \ldots * g}_{n} & (0 < n \in Z) \\ \underbrace{g^{-1} * g^{-1} * \ldots * g^{-1}}_{|n|} & (0 > n \in Z) \end{cases}$$

[8] Since the notion of abstract group did not exist until about 1890, Lagrange, who worked in the late 18th and early 19th centuries, could not have proven the result as it is stated. However, his work in number theory repeatedly used results of this sort, as did Gauss's of about the same time. That is, Lagrange and Gauss recognized the principle without having a formal framework for it.

One might do the slightly tedious induction proof of the fact that, for all choices of sign of integers m, n,

$$g^{m+n} = g^m * g^n$$

$$(g^m)n = g^{mn}$$

That is, the so-called *Laws of Exponents* are provable properties. And, thus, $\langle g\rangle$ really is a subgroup. For various reasons, a (sub)group which can be generated by a single element is called a **cyclic subgroup**. Note that a cyclic group is necessarily abelian.

The smallest positive integer n (if it exists) such that

$$g^n = e$$

is the **order** or **exponent** of g, often denoted by $|g|$ or $o(g)$. If there is no such n, say that the order of g is *infinite*. [9]

2.2.3 Proposition: Let g be a finite-order element of a group G, with order n. Then the order of g (as group *element*) is equal to the order of $\langle g\rangle$ (as subgroup). In particular,

$$\langle g\rangle = \{g^0, g^1, g^2, \ldots, g^{n-1}\}$$

and, for arbitrary integers i, j,

$$g^i = g^j \quad \text{if and only if} \quad i = j \bmod n$$

Proof: The last assertion implies the first two. On one hand, if $i = j \bmod n$, then write $i = j + \ell n$ and compute

$$g^i = g^{j+\ell n} = g^j \cdot (g^n)^\ell = g^j \cdot e^\ell = g^j \cdot e = g^j$$

On the other hand, suppose that $g^i = g^j$. Without loss of generality, $i \le j$, and $g^i = g^j$ implies $e = g^{j-i}$. Let

$$j - i = q \cdot n + r$$

where $0 \le r < n$. Then

$$e = g^{j-i} = g^{qn+r} = (g^n)^q \cdot g^r = e^q \cdot g^r = e \cdot g^r = g^r$$

Therefore, since n is the least such that $g^n = e$, necessarily $r = 0$. That is, $n|j - i$. ///

2.2.4 Corollary: *(of Lagrange's theorem)* The order $|g|$ of an element g of a finite group G divides the order of G. [10]

Proof: We just proved that $|g| = |\langle g\rangle|$, which, by Lagrange's theorem, divides $|G|$. ///

Now we can recover Euler's theorem as an example of the latter corollary of Lagrange's theorem:

2.2.5 Corollary: *(Euler's theorem, again)* Let n be a positive integer. For $x \in \mathbb{Z}$ relatively prime to n,

$$x^{\varphi(n)} = 1 \bmod n$$

[9] Yes, this use of the term *order* is in conflict with the use for subgroups, but we immediately prove their compatibility.

[10] One can also imitate the direct proof of Euler's theorem, and produce a proof of this corollary at least for finite abelian groups.

Proof: The set $\mathbb{Z}/n^\times$ of integers mod n relatively prime to n is a group with $\varphi(n)$ elements. By Lagrange, the order k of $g \in \mathbb{Z}/n^\times$ divides $\varphi(n)$. Therefore, $\varphi(n)/k$ is an integer, and

$$g^{\varphi(n)} = (g^k)^{\varphi(n)/k} = e^{\varphi(n)/k} = e$$

as desired. ///

The idea of Euler's theorem can be abstracted. For a group G, the smallest positive integer ℓ so that for every $g \in G$

$$g^\ell = e$$

is the **exponent** of the group G. It is not clear from the definition that there really is such a positive integer ℓ. Indeed, for *infinite* groups G there may not be. But for *finite* groups the mere finiteness allows us to characterize the exponent:

2.2.6 Corollary: *(of Lagrange's theorem)* Let G be a finite group. Then the exponent of G divides the order $|G|$ of G.

Proof: From the definition, the exponent is the least common multiple of the orders of the elements of G. From Lagrange's theorem, each such order is a divisor of $|G|$. The least common multiple of any collection of divisors of a fixed number is certainly a divisor of that number. ///

2.3 *Homomorphisms, kernels, normal subgroups*

Group homomorphisms are the maps of interest among groups.

A *function* (or *map*)

$$f : G \longrightarrow H$$

from one group G to another H is a **(group) homomorphism** if the *group operation is preserved* in the sense that

$$f(g_1 g_2) = f(g_1)\ f(g_2)$$

for all $g_1, g_2 \in G$. Let e_G be the identity in G and e_H the identity in H. The **kernel** of a homomorphism f is

$$\text{kernel of } f \ = \ker f = \{g \in G : f(g) = e_H\}$$

The **image** of f is just like the image of any function:

$$\text{image of } f \ = \text{im} f = \{h \in H : \text{ there is } g \in G \text{ so that } f(g) = h\}$$

2.3.1 Theorem: Let $f : G \longrightarrow H$ be a group homomorphism. Let e_G be the identity in G and let e_H be the identity in H. Then
• Necessarily f carries the identity of G to the identity of H: $f(e_G) = e_H$.
• For $g \in G$, $f(g^{-1}) = f(g)^{-1}$.
• The *kernel* of f is a subgroup of G.
• The *image* of f is a subgroup of H.
• Given a subgroup K of H, the *pre-image*

$$f^{-1}(K) = \{g \in G : f(g) \in K\}$$

of K under f is a subgroup of G.
• A group homomorphism $f : G \longrightarrow H$ is *injective* if and only if the kernel is *trivial* (that is, is the trivial subgroup $\{e_G\}$).

Proof: The image $f(e_G)$ has the property

$$f(e_G) = f(e_G \cdot e_G) = f(e_G) \cdot f(e_G)$$

Left multiplying by $f(e_G)^{-1}$ (whatever this may be),

$$f(e_G)^{-1} \cdot f(e_G) = f(e_G)^{-1} \cdot (f(e_G) \cdot f(e_G))$$

Simplifying,

$$e_H = (f(e_G)^{-1} \cdot f(e_G)) \cdot f(e_G) = e_H \cdot f(e_G) = f(e_G)$$

so the identity in G is mapped to the identity in H.

To check that the image of an inverse is the inverse of an image, compute

$$f(g^{-1}) \cdot f(g) = f(g^{-1} \cdot g) = f(e_G) = e_H$$

using the fact just proven that the identity in G is mapped to the identity in H.

Now prove that the kernel is a subgroup of G. The identity lies in the kernel since, as we just saw, it is mapped to the identity. If g is in the kernel, then g^{-1} is also, since, as just showed, $f(g^{-1}) = f(g)^{-1}$. Finally, suppose both x, y are in the kernel of f. Then

$$f(xy) = f(x) \cdot f(y) = e_H \cdot e_H = e_H$$

Let X be a subgroup of G. Let

$$f(X) = \{f(x) : x \in X\}$$

To show that $f(X)$ is a subgroup of H, we must check for presence of the identity, closure under taking inverses, and closure under products. Again, $f(e_G) = e_H$ was just proven. Also, we showed that $f(g)^{-1}) = f(g^{-1})$, so the image of a subgroup is closed under inverses. And $f(xy) = f(x)f(y)$ by the defining property of a group homomorphism, so the image is closed under multiplication.

Let K be a subgroup of H. Let x, y be in the pre-image $f^{-1}(K)$. Then

$$f(xy) = f(x) \cdot f(y) \in K \cdot K = K$$

$$f(x^{-1}) = f(x)^{-1} \in K$$

And already $f(e_G) = e_H$, so the pre-image of a subgroup is a group.

Finally, we prove that a homomorphism $f : G \longrightarrow H$ is injective if and only if its kernel is trivial. First, if f is injective, then at most one element can be mapped to $e_H \in H$. Since we know that at least e_G is mapped to e_H by such a homomorphism, it must be that *only* e_G is mapped to e_H. Thus, the kernel is trivial. On the other hand, suppose that the kernel is trivial. We will suppose that $f(x) = f(y)$, and show that $x = y$. Left multiply $f(x) = f(y)$ by $f(x)^{-1}$ to obtain

$$e_H = f(x)^{-1} \cdot f(x) = f(x)^{-1} \cdot f(y)$$

By the homomorphism property,

$$e_H = f(x)^{-1} \cdot f(y) = f(x^{-1}y)$$

Thus, $x^{-1}y$ is in the kernel of f, so (by assumption) $x^{-1}y = e_G$. Left multiplying this equality by x and simplifying, we get $y = x$. ///

If a group homomorphism $f : G \longrightarrow H$ is *surjective*, then H is said to be a **homomorphic image** of G. If a group homomorphism $f : G \longrightarrow H$ has an inverse homomorphism, then f is said to be an **isomorphism**, and G and H are said to be **isomorphic**, written

$$G \approx H$$

For groups, if a group homomorphism is a *bijection*, then it has an inverse which is a group homomorphism, so is an isomorphism.

2.3.2 Remark: Two groups that are *isomorphic* are considered to be 'the same', in the sense that any *intrinsic* group-theoretic assertion about one is also true of the other.

A subgroup N of a group G is **normal** [11] or **invariant** [12] if, for every $g \in G$,

$$gNg^{-1} = N$$

where the notation is

$$gNg^{-1} = \{gng^{-1} : n \in N\}$$

This is readily seen to be equivalent to the condition that

$$gN = Ng$$

for all $g \in G$. Evidently in an abelian group G *every* subgroup is normal. It is not hard to check that *intersections of normal subgroups are normal*.

2.3.3 Proposition: The kernel of a homomorphism $f : G \longrightarrow H$ is a normal subgroup.

Proof: For $n \in \ker f$, using things from just above,

$$f(gng^{-1}) = f(g)\, f(n)\, f(g^{-1}) = f(g)\, e_H\, f(g)^{-1} = f(g)\, f(g)^{-1} = e_H$$

as desired. ///

A group with no *proper* normal subgroups is **simple**. Sometimes this usage is restricted to apply only to groups *not* of orders which are prime numbers, since (by Lagrange) such groups have no proper subgroups whatsoever, much less normal ones.

2.4 *Cyclic groups*

Finite groups generated by a single element are easy to understand. The collections of all *subgroups* and of all *generators* can be completely understood in terms of elementary arithmetic, in light of the first point below. Recall that the set of integers modulo n is

$$\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/n = \{cosetsofn\mathbb{Z}in\mathbb{Z}\} = \{x + n\mathbb{Z} : x \in \mathbb{Z}\}$$

2.4.1 Proposition: Let $G = \langle g \rangle$, of order n. Then G is isomorphic to $\mathbb{Z}/n$ with addition, by the map

$$f(g^i) = i + n\mathbb{Z} \in \mathbb{Z}/n$$

Proof: The main point is the well-definedness of the map. That is, that $g^i = g^j$ implies $i = j \bmod n$, for $i, j \in \mathbb{Z}$. Suppose, without loss of generality, that $i < j$. Then $g^{j-i} = e$. Let

$$j - i = q \cdot n + r$$

[11] This is one of too many uses of this term, but it is irretrievably standard.

[12] The term *invariant* surely comes closer to suggesting the intent, but is unfortunately archaic.

with $0 \leq r < n$. Then

$$e = e \cdot e = g^{j-i-qn} = g^r$$

and by the minimality of n we have $r = 0$. Thus, $n|j-i$, proving well-definedness of the map. The surjectivity and injectivity are then easy. The assertion that f is a homomorphism is just the well-definedness of addition modulo n together with properties of exponents:

$$f(g^i) + f(g^j) = (i + n\mathbb{Z}) + (j + n\mathbb{Z}) = (i + j) + n\mathbb{Z} = f(g^{i+j}) = f(g^i \cdot g^j)$$

This demonstrates the isomorphism. ///

2.4.2 Corollary: Up to isomorphism, there is only one finite cyclic group of a given order. ///

The following facts are immediate corollaries of the proposition and elementary properties of $\mathbb{Z}/n$.

- The *distinct* subgroups of G are exactly the subgroups $\langle g^d \rangle$ for all *divisors* d of N.
- For $d|N$ the order of the subgroup $\langle g^d \rangle$ is the order of g^d, which is N/d.
- The order of g^k with arbitrary integer $k \neq 0$ is $N/\gcd(k, N)$.
- For any integer n we have

$$\langle g^n \rangle = \langle g^{\gcd(n,N)} \rangle$$

- The distinct generators of G are the elements g^r where $1 \leq r < N$ and $\gcd(r, N) = 1$. Thus, there are $\varphi(N)$ of them, where φ is Euler's phi function.
- The number of elements of order n in a finite cyclic group of order N is 0 unless $n|N$, in which case it is N/n.

2.4.3 Proposition: A homomorphic image of a finite cyclic group is finite cyclic.

Proof: The image of a generator is a generator for the image. ///

Using the isomorphism of a cyclic group to some $\mathbb{Z}/n$, it is possible to reach definitive conclusions about the solvability of the equation $x^r = y$.

2.4.4 Theorem: Let G be a cyclic group of order n with generator g. Fix an integer r, and define

$$f : G \longrightarrow G$$

by

$$f(x) = x^r$$

This map f is a group homomorphism of G to itself. If $\gcd(r, n) = 1$, then f is an *isomorphism*, and in that case every $y \in G$ has a unique r^{th} root. More generally,

$$\text{order of kernel of } f = \gcd(r, n)$$

$$\text{order of image of } f = n/\gcd(r, n)$$

If an element y has an rth root, then it has exactly $\gcd(r, n)$ of them. There are exactly $n/\gcd(r, n)$ r^{th} powers in G.

Proof: Since G is abelian the map f is a homomorphism. Use the fact that G is isomorphic to $\mathbb{Z}/n$. Converting to the additive notation for $\mathbb{Z}/n$-with-addition, f is

$$f(x) = r \cdot x$$

If $\gcd(r,n) = 1$ then there is a multiplicative inverse r^{-1} to $r \bmod n$. Thus, the function

$$g(x) = r^{-1} \cdot x$$

gives an inverse function to f, so f is an isomorphism.

For arbitrary r, consider the equation

$$r \cdot x = y \bmod n$$

for given y. This condition is

$$n|(rx - y)$$

Let $d = \gcd(r,n)$. Then certainly it is *necessary* that $d|y$ or this is impossible. On the other hand, suppose that $d|y$. Write $y = dy'$ with integer y'. We want to solve

$$r \cdot x = dy' \bmod n$$

Dividing through by the common divisor d, this congruence is

$$\frac{r}{d} \cdot x = y' \bmod \frac{n}{d}$$

The removal of the common divisor makes r/d prime to n/d, so there is an inverse $(r/d)^{-1}$ to $r/d \bmod n/d$, and

$$x = (r/d)^{-1} \cdot y' \bmod (n/d)$$

That is, any integer x meeting this condition is a solution to the original congruence. Letting x_0 be one such solution, the integers

$$x_0,\ x_0 + \frac{n}{d},\ x_0 + 2 \cdot \frac{n}{d},\ x_0 + 3 \cdot \frac{n}{d},\ \dots x_0 + (d-1) \cdot \frac{n}{d}$$

are also solutions, and are distinct mod n. That is, we have d distinct solutions mod n.

The kernel of f is the collection of x so that $rx = 0 \bmod n$. Taking out the common denominator $d = \gcd(r,n)$, this is $(r/d)x = 0 \bmod n/d$, or $(n/d)|(r/d)x$. Since r/d and n/d have no common factor, n/d divides x. Thus, mod n, there are d different solutions x. That is, the kernel of f has d elements. ///

2.5 *Quotient groups*

Let G be a group and H a subgroup. The **quotient set** G/H of G by H is the set of H-cosets

$$G/H = \{xH : x \in G\}$$

in G. In general, there is *no* natural group structure on this set. [13] But if H is *normal*, then we define a group operation $*$ on G/H by

$$xH * yH = (xy)H$$

Granting in advance that this works out, the **quotient map** $q : G \longrightarrow G/N$ defined by

$$q(g) = gN$$

will be a group homomorphism.

[13] The key word is *natural*: of course any set can have several group structures put on it, but, reasonably enough, we are interested in group structures on G/H that have some connection with the original group structure on G.

Of course, the same symbols can be written for non-normal H, *but will not give a well-defined operation.* That is, for well-definedness, one must verify that the operation does not depend upon the choice of coset representatives x, y in this formula. That is, one must show that if

$$xH = x'H \quad \text{and} \quad yH = y'H$$

then

$$(xy)H = (x'y')H$$

If H is *normal*, then $xH = Hx$ for all $x \in G$. Then, literally, as sets,

$$xH \cdot yH = x \cdot Hy \cdot H = x \cdot yH \cdot H = (xy)H \cdot H = (xy)H$$

That is, we can more directly define the group operation $*$ as

$$xH * yH = xH \cdot yH$$

2.5.1 Remark: If H is *not* normal, take $x \in G$ such that $Hx \not\subset xH$. That is, there is $h \in H$ such that $hx \notin xH$. Then $hxH \neq H$, and, if the same definition were to work, supposedly

$$hH * xH = (hx)H \neq xH$$

But, on the other hand, since $hH = eH$,

$$hH * xH = eH * xH = (ex)H = xH$$

That is, if H is not normal, this apparent definition is in fact not well-defined.

2.5.2 Proposition: *(Isomorphism Theorem)* Let $f : G \longrightarrow H$ be a *surjective* group homomorphism. Let $K = \ker f$. Then the map $\bar{f} : G/K \longrightarrow H$ by

$$\bar{f}(gK) = f(g)$$

is well-defined and is an isomorphism.

Proof: If $g'K = gK$, then $g' = gk$ with $k \in K$, and

$$f(g') = f(gk) = f(g)\, f(k) = f(g)\, e = f(g)$$

so the map $\bar{f}$ is well-defined. It is surjective because f is. For injectivity, if $\bar{f}(gK) = \bar{f}(g'K)$, then $f(g) = f(g')$, and

$$e_H = f(g)^{-1} \cdot f(g') = f(g^{-1}) \cdot f(g) = f(g^{-1}g')$$

Thus, $g^{-1}g' \in K$, so $g' \in gK$, and $g'K = gK$. ///

In summary, the normal subgroups of a group are exactly the kernels of surjective homomorphisms.

As an instance of a counting principle, we have

2.5.3 Corollary: Let $f : G \longrightarrow H$ be a surjective homomorphism of finite groups. Let Y be a subgroup of H. Let

$$X = f^{-1}(Y) = \{x \in G : f(x) \in Y\}$$

be the **inverse image** of Y in G. Then

$$|X| = |\ker f| \cdot |Y|$$

Proof: By the isomorphism theorem, without loss of generality $Y = G/N$ where $N = \ker f$ is a normal subgroup in G. The quotient group is the set of cosets gN. Thus,

$$f^{-1}(Y) = \{xN : f(x) \in Y\}$$

That is, the inverse image is a disjoint union of cosets of N, and the number of cosets in the inverse image is $|Y|$. We proved earlier that X is a subgroup of G. ///

A variant of the previous corollary gives

2.5.4 Corollary: Given a normal subgroup N of a group G, and given any other subgroup H of G, let $q : G \longrightarrow G/N$ be the quotient map. Then

$$H \cdot N = \{hn : h \in H, n \in N\} = q^{-1}(q(H))$$

is a subgroup of G. If G is finite, the order of this group is

$$|H \cdot N| = \frac{|H| \cdot |N|}{|H \cap N|}$$

Further,

$$q(H) \approx H/(H \cap N)$$

Proof: By definition the inverse image $q^{-1}(q(H))$ is

$$\{g \in G : q(g) \in q(H)\} = \{g \in G : gN = hN \text{ for some } h \in H\}$$

$$= \{g \in G : g \in hN \text{ for some } h \in H\} = \{g \in G : g \in H \cdot N\} = H \cdot N$$

The previous corollary already showed that the inverse image of a subgroup is a subgroup. And if $hN = h'N$, then $N = h^{-1}h'N$, and $h^{-1}h' \in N$. Yet certainly $h^{-1}h' \in H$, so $h^{-1}h' \in H \cap N$. And, on the other hand, if $h^{-1}h' \in H \cap N$ then $hN = h'N$. Since $q(h) = hN$, this proves the isomorphism. From above, the inverse image $H \cdot N = q^{-1}(q(H))$ has cardinality

$$\operatorname{card} H \cdot N = |\ker q| \cdot |q(H)| = |N| \cdot |H/(H \cap N)| = \frac{|N| \cdot |H|}{|H \cap N|}$$

giving the counting assertion. ///

2.6 *Groups acting on sets*

Let G be a group and S a set. A map $G \times S \longrightarrow S$, denoted by juxaposition

$$g \times s \longrightarrow gs$$

is an **action** of the group on the set if
• $es = s$ for all $s \in S$
• *(Associativity)* $(gh)s = g(hs)$ for all $g, h \in G$ and $s \in S$.

These conditions assure that, for example, $gs = t$ for $s, t \in S$ and $g \in G$ implies that $g^{-1}t = s$. Indeed,

$$g^{-1}t = g^{-1}(gs) = (g^{-1}g)s = es = s$$

Sometimes a set with an action of a group G on it is called a G-**set**.

The action of G on a set is **transitive** if, for all $s, t \in S$, there is $g \in G$ such that $gs = t$. This definition admits obvious equivalent variants: for example, the seemingly weaker condition that there is $s_o \in S$ such that for every $t \in S$ there is $g \in G$ such that $gs_o = t$ implies transitivity. Indeed, given $s, t \in S$, let $g_s s_o = s$ and $g_t s_o = t$. Then

$$(g_t g_s^{-1})s = g_t(g_s^{-1}s) = g_t(s_o) = t$$

For G acting on a set S, a subset T of S such that $g(T) \subset T$ is G-**stable**.

2.6.1 Proposition: For a group G acting on a set S, and for a G-stable subset T of S, in fact $g(T) = T$ for all $g \in G$.

Proof: We have

$$T = eT = (gg^{-1})T = g(g^{-1}(T)) \subset g(T) \subset T$$

Thus, all the inclusions must be equalities. ///

A single element $s_o \in S$ such that $gs_o = s_o$ for *all* $g \in G$ is G-**fixed**. Given an element $s_o \in S$, the **stabilizer** of s_o in G, often denoted G_{s_o}, is

$$G_{s_o} = \{g \in G : gs_o = s_o\}$$

More generally, for a subset T of S, the **stabilizer** of T in G is

$$\text{stabilizer of } T \text{ in } G = \{g \in G : g(T) = T\}$$

The **point-wise fixer** or **istropy subgroup** of a subset T is

$$\text{isotropy subgroup of } T = \text{point-wise fixer of } T \text{ in } G = \{g \in G : gt = t \text{ for all } t \in T\}$$

For a subgroup H of G, the **fixed points** S^H of H on S are the elements of the set

$$\text{fixed point set of } H = \{s \in S : hs = s \text{ for all } h \in H\}$$

2.6.2 Remark: In contrast to the situation of the previous proposition, if we attempt to define the stabilizer of a subset by the weaker condition $g(T) \subset T$, the following proposition can fail (for infinite sets S).

2.6.3 Proposition: Let G act on a set S, and let T be a subset of S. Then both the stabilizer and point-wise fixer of T in G are *subgroups* of G.

Proof: We only prove that the stabilizer of T is stable under inverses. Suppose $gT = T$. Then

$$g^{-1}T = g^{-1}(g(T)) = (g^{-1}g)(T) = e(T) = T$$

since $g(T) = T$. ///

With an action of G on the set S, a G-**orbit** in S is a *non-empty* G-stable subset of S, on which G is *transitive*.

2.6.4 Proposition: Let G act on a set S. For any element s_o in an orbit O of G on S,

$$O = G \cdot s_o = \{gs_o : g \in G\}$$

Conversely, for any $s_o \in S$, the set $G \cdot s_o$ is a G-orbit on S.

Proof: Since an orbit O is required to be non-empty, O contains an element s_o. Since O is G-stable, certainly $gs_o \in S$ for all $g \in G$. Since G is transitive on O, the collection of all images gs_o of s_o by elements $g \in G$ must be the whole orbit O. On the other hand, any set

$$Gs_o = \{gs_o : g \in G\}$$

is G-stable, since $h(gs_o) = (hg)s_o$. And certainly G is transitive on such a set. ///

Now we come to some consequences for counting problems. [14]

2.6.5 Proposition: Let G act transitively on a (non-empty) set S, and fix $s \in S$. Then S is in bijection with the set G/G_s of cosets gG_s of the isotropy group G_s of s in G, by

$$gs \longleftrightarrow gG_s$$

Thus,

$$\mathrm{card}\, S = [G : G_s]$$

Proof: If $hG_s = gG_s$, then there is $x \in G_s$ such that $h = gx$, and $hs = gxs = gs$. On the other hand, if $hs = gs$, then $g^{-1}hs = s$, so $g^{-1}h \in G_s$, and then $h \in gG_s$. ///

2.6.6 Corollary: *(Counting formula)* Let G be a finite group acting on a finite set S. Let X be the set of G-orbits in S. For $O \in X$ let $s_O \in O$. And

$$\mathrm{card}\, S = \sum_{O \in X} \mathrm{card}\, O = \sum_{O \in X} [G : G_{s_O}]$$

Proof: The set S is a disjoint union of the G-orbits in it, so the cardinality of S is the sum of the cardinalities of the orbits. The cardinality of each orbit is the index of the isotropy group of a chosen element in it, by the previous proposition. ///

Two fundamental examples of natural group actions are the following.

2.6.7 Example: A group G acts on itself (as a set) by **conjugation**: [15] for $g, x \in G$,

$$\text{conjugate of } x \text{ by } g = gxg^{-1}$$

It is easy to verify that for fixed $g \in G$, the map

$$x \longrightarrow gxg^{-1}$$

is an isomorphism of G to itself. For x and y elements of G in the same G-orbit under this action, say that x and y **are conjugate**. The orbits of G on itself with the conjugation action are **conjugacy classes** (of elements). The **center** of a group G is the set of elements z whose orbit under conjugation is just $\{z\}$. That is,

$$\text{center of } G = \{z \in G : gz = zg \text{ for all } g \in G\}$$

Either directly or from general principles (above), the center Z of a group G is a *subgroup* of G. Further, it is *normal*:

$$gZg^{-1} = \{gzg^{-1} : z \in Z\} = \{z : z \in Z\} = Z$$

[14] Yes, these look boring and innocent, in this abstraction.

[15] It is obviously not wise to use the notation gh for ghg^{-1}.

And of course the center is itself an *abelian* group.

2.6.8 Example: For a subgroup H of G and for $g \in G$, the **conjugate** subgroup gHg^{-1} is

$$gHg^{-1} = \{ghg^{-1} : h \in H\}$$

Thus, G acts on the set of its own subgroups by conjugation. [16] As with the element-wise conjugation action, for H and K subgroups of G in the same G-orbit under this action, say that H and K **are conjugate**. The orbits of G on its subgroups with the conjugation action are **conjugacy classes** (of subgroups). The *fixed points* of G under the conjugation action on subgroups are just the *normal* subgroups. On the other hand, for a given subgroup H, the isotropy subgroup in G for the conjugation action is called the **normalizer** of H in G:

$$\text{normalizer of } H \text{ in } G = \{g \in G : gHg^{-1} = H\}$$

Either directly or from more general principles (above), the normalizer of H in G is a subgroup of G (containing H).

2.7 *The Sylow theorem*

There is not much that one can say about the subgroups of an arbitrary finite group. Lagrange's theorem is the simplest very general assertion. Sylow's theorem is perhaps the strongest and most useful relatively elementary result limiting the possibilities for subgroups and, therefore, for finite groups.

Let p be a prime. A p-**group** is a finite group whose order is a power of the prime p. Let G be a finite group. Let p^e be the largest power of p dividing the order of G. A p-**Sylow** subgroup (if it exists) is a subgroup of G of order p^e.

2.7.1 Remark: By Lagrange's theorem, no larger power of p can divide the order of *any* subgroup of G.

2.7.2 Theorem: Let p be a prime. Let G be a finite group. Let p^e be the largest power of p dividing the order of G. Then
- G has p-Sylow subgroups.
- Every subgroup of G with order a power of p lies inside a p-Sylow subgroup of G.
- The number n_p of p-Sylow subgroups satisfies

$$n_p | \text{order}(G) \qquad n_p = 1 \bmod p$$

- Any two p-Sylow subgroups P and Q are **conjugate**. [17]
- A group of order p^n has a non-trivial center.

It is convenient to prove a much weaker and simpler result first, which also illustrates a style of induction via subgroups and quotients:

2.7.3 Lemma: Let A be a finite *abelian* group, and let p be a prime dividing the order of A. Then there is an element a of A of order exactly p. Thus, there exists a subgroup of A of order p.

Proof: (of lemma) Use induction on the order of A. If the order is p exactly, then any non-identity element is of order p. Since a prime divides its order, A is not the trivial group, so we can choose a non-identity

[16] And, again, it is manifestly unwise to write gH for gH^{-1}.

[17] This property is the sharpest and most surprising assertion here.

element g of A. By Lagrange, the order n of g divides the order of A. If p divides n, then $g^{n/p}$ is of order exactly p and we're done. So suppose that p does *not* divide the order of g. Then consider the quotient

$$q(A) = B = A/\langle g\rangle$$

of A by the cyclic subgroup generated by g. The order of B is still divisible by p (since $|<\langle g\rangle|$ is not), so by induction on order there is an element y in B of order exactly p. Let x be any element in A which maps to y under the quotient map $q : A \longrightarrow B$. Let N be the order of x. The prime p divides N, or else write $N = \ell p + r$ with $0 < r < p$, and

$$e_B = q(e_A) = q(x^N) = y^N = y^{\ell p + r} = y^r \neq e_B$$

contradiction. Then $x^{N/p}$ has order exactly p, and the cyclic subgroup generated by $x^{N/p}$ has order p.
///

Proof: Now prove the theorem. First, we prove *existence* of p-Sylow subgroups by induction on the exponent e of the power p^e of p dividing the order of G. (Really, the induction uses subgroups and quotient groups of G.) If $e = 0$ the p-Sylow subgroup is the trivial subgroup, and there is nothing to prove. For fixed $e > 1$, do induction on the order of the group G. If any proper subgroup H of G has order divisible by p^e, then invoke the theorem for H, and a p-Sylow subgroup of H is one for G. So suppose that *no* proper subgroup of G has order divisible by p^e. Then for any subgroup H of G the prime p divides $[G : H]$. By the *counting formula* above, using the conjugation action of G on itself,

$$\mathrm{card}\, G = \sum_x [G : G_x]$$

where x is summed over (irredundant) representatives for conjugacy classes. Let Z be the center of G. Then Z consists of G-orbits each with a single element. We rewrite the counting formula as

$$\mathrm{card}\, G = \mathrm{card}\, Z + \sum_x^{\mathrm{non-central}} [G : G_x]$$

where now x is summed over representatives *not* in the center. For non-central x the isotropy group G_x is a proper subgroup, so by assumption, p divides $[G : G_x]$ for all x. Since p divides the order of G, we conclude from the counting formula that p divides the order of the center Z (but p^e does not divide the order of Z). Using the lemma above, let A be a subgroup of Z of order p. Since A is inside the center it is still normal. Consider the quotient group $H = G/A$, with quotient map $q : A \longrightarrow B$. The power of p dividing the order of H is p^{e-1}, strictly smaller than p^e. By induction, let Q be a p-Sylow subgroup of H, and let $P = q^{-1}(Q)$ be the inverse image of Q under the quotient map q. Then

$$|P| = |q^{-1}(Q)| = |\ker q| \cdot |Q| = p \cdot p^{e-1} = p^e$$

from the counting corollary of the isomorphism theorem (above). Thus, G does have a p-Sylow theorem after all.

If it happens that $|G| = p^e$, looking at that same formula

$$\mathrm{card}\, G = \mathrm{card}\, Z + \sum_x^{\mathrm{non-central}} [G : G_x]$$

the left-hand side is p^e, and all the summands corresponding to non-central conjugacy classes are divisible by p, so the order of the center is divisible by p. That is, p-power-order groups have non-trivial centers.

Let X be *any* G-conjugation stable set of p-Sylow subgroups. Fix a p-power-order subgroup Q not necessarily in X, and let Q act on the set X by conjugation. The counting formula gives

$$\operatorname{card} X = \sum_x [Q : Q_x]$$

where x runs over representatives for Q-conjugacy classes in X. If Q normalized a p-Sylow subgroup x not containing Q, then

$$H = Q \cdot x$$

would be a subgroup of order

$$|Q \cdot x| = \frac{|Q| \cdot |x|}{|Q \cap x|} > |x|$$

and would be a power of p, contradicting the maximality of x. Thus, the only p-Sylow subgroups normalized by any p-power-order subgroup Q are those containing Q. Thus, except for x containing Q, all the indices $[Q : Q_x]$ are divisible by p. Thus,

$$|X| = |\{x \in X : Q \subset x\}| + \sum_x^{Q \not\subset x} [Q : Q_x]$$

In the case that Q itself is a p-Sylow subgroup, and X is *all* p-Sylow subgroups in G,

$$|\{x \in X : Q \subset x\}| = |\{Q\}| = 1$$

so the number of *all* p-Sylow subgroups is 1 mod p.

Next, let X consist of a single G-conjugacy class of p-Sylow subgroups. Fix $x \in X$. Since X is a single orbit,

$$|X| = [G : G_x]$$

and the latter index is *not* divisible by p, since the normalizer G_x of x contains x. Let a p-power-subgroup Q act by conjugation on X. In the counting formula

$$|X| = |\{x \in X : Q \subset x\}| + \sum_x^{Q \not\subset x} [Q : Q_x]$$

all the indices $[Q : Q_x]$ are divisible by p, but $|X|$ is *not*, so

$$|\{x \in X : Q \subset x\}| \neq 0$$

That is, given a p-power-order subgroup Q, *every* G-conjugacy class of p-Sylow subgroups contains an x containing Q. This is only possible if there is a *unique* G-conjugacy class of p-Sylow subgroups. That is, the conjugation action of G is *transitive* on p-Sylow subgroups.

Further, since Q was not necessarily maximal in this last discussion, we have shown that every p-power-order subgroup of G lies inside at least one p-Sylow subgroup.

And, fixing a single p-Sylow subgroup x, using the transitivity, the number of *all* p-Sylow subgroups is

$$\text{number } p\text{-Sylow subgroups} = [G : G_x] = |G|/|G_x|$$

This proves the divisibility property. ///

2.7.4 Remark: For general integers d dividing the order of a finite group G, it is seldom the case that there is a subgroup of G of order d. By contrast, if G is *cyclic* there is a *unique* subgroup for every divisor of its order. If G is *abelian* there is *at least one* subgroup for every divisor of its order.

2.7.5 Remark: About the proof of the Sylow theorem: once one knows that the proof uses the conjugation action on elements and on subgroups, there are not so many possible directions the proof could go. Knowing these limitations on the proof methodology, one could hit on a correct proof after a relatively small amount of trial and error.

2.8 *Trying to classify finite groups, part I*

Lagrange's theorem and the Sylow theorem allow us to make non-trivial progress on the project of classifying finite groups whose orders have relatively few prime factors. That is, we can prove that there are not many non-isomorphic groups of such orders, sometimes a single isomorphism class for a given order. This sort of result, proving that an abstraction miraculously allows fewer instances than one might have imagined, is often a happy and useful result.

Groups of prime order: Let p be a prime, and suppose that G is a group with $|G| = p$. Then by Lagrange's theorem there are no proper subgroups of G. Thus, picking any element g of G other than the identity, the (cyclic) subgroup $\langle g \rangle$ generated by g is necessarily the whole group G. That is, for such groups G, choice of a non-identity element g yields

$$G = \langle g \rangle \approx \mathbb{Z}/p$$

Groups of order pq, part I: Let $p < q$ be primes, and suppose that G is a group with $|G| = pq$. Sylow's theorem assures that there exist subgroups P and Q of orders p and q, respectively. By Lagrange, the order of $P \cap Q$ divides both p and q, so is necessarily 1. Thus,

$$P \cap Q = \{e\}$$

Further, the number n_q of q-Sylow subgroups must be 1 mod q and also divide the order pq of the group. Since $q = 0 \bmod q$, the only possibilities (since p is prime) are that either $n_q = p$ or $n_q = 1$. But $p < q$ precludes the possibility that $n_q = p$, so $n_q = 1$. That is, with $p < q$, the q-Sylow subgroup is necessarily *normal*.

The same argument, apart from the final conclusion invoking $p < q$, shows that the number n_p of p-Sylow subgroups is either $n_p = 1$ or $n_p = q$, and (by Sylow) is $n_p = 1 \bmod p$. But now $p < q$ does *not* yield $n_p = 1$. There are two cases, $q = 1 \bmod p$ and otherwise.

If $q \neq 1 \bmod p$, then we *can* reach the conclusion that $n_p = 1$, that is, that the p-Sylow subgroup is also normal. Thus, for $p < q$ and $q \neq 1 \bmod p$, we have a normal p-Sylow group P and a normal q-Sylow subgroup Q. Again, $P \cap Q = \{e\}$ from Lagrange.

How to reconstruct G from such facts about its subgroups?

We need to enrich our vocabulary: given two groups G and H, the **(direct) product** group $G \times H$ is the cartesian product with the operation

$$(g, h) \cdot (g', h') = (gg', hh')$$

(It is easy to verify the group properties.) For G and H abelian, with group operations written as addition, often the direct product is written instead as a **(direct) sum** [18]

$$G \oplus H$$

[18] Eventually we will make some important distinctions between direct sums and direct products, but there is no need to do so just now.

2.8.1 Proposition: Let A and B be normal [19] subgroups of a group G, such that $A \cap B = \{e\}$. Then

$$f : A \times B \longrightarrow A \cdot B = \{ab : a \in A, b \in B\}$$

by

$$f(a, b) = ab$$

is an isomorphism. The subgroup $A \cdot B \approx A \times B$ is a normal subgroup of G. In particular, $ab = ba$ for all $a \in A$ and $b \in B$.

Proof: The trick is to consider **commutator** expressions

$$aba^{-1}b^{-1} = aba^{-1} \cdot b^{-1} = a \cdot ba^{-1}b^{-1}$$

for $a \in A$ and $b \in B$. Since B is normal, the second expression is in B. Since A is normal, the third expression is in A. Thus, the commutator $aba^{-1}b^{-1}$ is in $A \cap B$, which is $\{e\}$. [20] Thus, right multiplying by b

$$aba^{-1} = b$$

or, right multiplying further by a,

$$ab = ba$$

The fact that $ab = ba$ for all $a \in A$ and all $\underline{in}B$ allows one to easily show that f is a group homomorphism. Its kernel is trivial, since $ab = e$ implies

$$a = b^{-1} \in A \cap B = \{e\}$$

Thus, the map is injective, from earlier discussions. Now $|A \times B| = pq$, and the map is injective, so $f(A \times B)$ is a subgroup of G with order pq. Thus, the image is all of G. That is, f is an isomorphism. ///

2.8.2 Proposition: Let A and B by cyclic groups of relatively prime orders m and n. Then $A \times B$ is cyclic of order mn. In particular, for a a generator for A and b a generator for B, (a, b) is a generator for the product.

Proof: Let N be the least positive integer such that $N(a, b) = (e_A, e_B)$. Then $Na = e_A$, so $|a|$ divides N. Similarly, $|b|$ divides N. Since $|a|$ and $|b|$ are relatively prime, this implies that their product divides N. ///

2.8.3 Corollary: For $|G| = pq$ with $p < q$ and $q \neq 1 \bmod p$, G is cyclic of order pq. Hence, in particular, there is only *one* isomorphism class of groups of such orders pq. ///

2.8.4 Remark: Even without the condition $q \neq 1 \bmod p$, we do have the cyclic group $\mathbb{Z}/pq$ of order pq, but without that condition we cannot prove that there are no *other* groups of order pq. [21] We'll delay treatment of $|G| = pq$ with primes $p < q$ and $q = 1 \bmod p$ till after some simpler examples are treated.

2.8.5 Example: Groups of order $15 = 3 \cdot 5$, or order $35 = 5 \cdot 7$, of order $65 = 5 \cdot 13$, etc., are necessarily cyclic of that order. By contrast, we reach no such conclusion about groups of order $6 = 2 \cdot 4$, $21 = 3 \cdot 7$, $55 = 5 \cdot 11$, etc. [22]

---

[19] Unless at least one of the subgroups is normal, the set $A \cdot B$ may not even be a subgroup, much less normal.

[20] By Lagrange, again. Very soon we will tire of explicit invocation of Lagrange's theorem, and let it go without saying.

[21] And, indeed, there *are* non-cyclic groups of those orders.

[22] And, again, there *are* non-cyclic groups of such orders.

Groups G of order pqr with distinct primes p, q, r: By Sylow, there is a p-Sylow subgroup P, a q-Sylow subgroup Q, and an r-Sylow subgroup R. Without any further assumptions, we cannot conclude anything about the normality of any of these Sylow subgroups, by contrast to the case where the order was pq, wherein the Sylow subgroup for the larger of the two primes was invariably normal.

One set of hypotheses which allows a simple conclusion is

$$\begin{array}{lll} q \neq 1 \bmod p & r \neq 1 \bmod p & qr \neq 1 \bmod p \\ p \neq 1 \bmod q & r \neq 1 \bmod q & pr \neq 1 \bmod q \\ p \neq 1 \bmod r & q \neq 1 \bmod r & pq \neq 1 \bmod r \end{array}$$

These conditions would suffice to prove that all of P, Q, and R are normal. Then the little propositions above prove that $P \cdot Q$ is a normal cyclic subgroup of order pq, and then (since still pq and r are relatively prime) that $(PQ) \cdot R$ is a cyclic subgroup of order pqr, so must be the whole group G. That is, G is cyclic of order pqr.

Groups of order pq, part II: Let $p < q$ be primes, and now treat the case that $q = 1 \bmod p$, so that a group G of order pq need *not* be cyclic. Still, we know that the q-Sylow subgroup Q is *normal*. Thus, for each x in a fixed p-Sylow subgroup P, we have a map $a_x : Q \longrightarrow Q$ defined by

$$a_x(y) = xyx^{-1}$$

Once the normality of Q assures that this really does map back to Q, it is visibly an isomorphism of Q to itself. This introduces:

An isomorphism of a group to itself is an **automorphism**. [23] The **group of automorphisms** of a group G is

$$\mathrm{Aut}(G) = \mathrm{Aut}G = \{\text{isomorphisms } G \longrightarrow G\}$$

It is easy to check that $\mathrm{Aut}(G)$ is indeed a group, with operation being the composition of maps, and identity being the identity map 1_G defined by [24]

$$1_G(g) = g \quad (\text{for any } g \text{ in } G)$$

In general it is a non-trivial matter to determine in tangible terms the automorphism group of a given group, but we have a simple case:

2.8.6 Proposition:

$$\mathrm{Aut}(\mathbb{Z}/n) \approx (\mathbb{Z}/n)^\times$$

by defining, for each $z \in (\mathbb{Z}/n)^\times$, and for $x \in \mathbb{Z}/n$,

$$f_z(x) = zx$$

On the other hand, given an automorphism f, taking $z = f(1)$ gives $f_z = f$.

Proof: For z multiplicatively invertible mod n, since the addition and multiplication in $\mathbb{Z}/n$ enjoy a distributive property, f_z is an automorphism of $\mathbb{Z}/n$ to itself. On the other hand, given an automorphism f of $\mathbb{Z}/n$, let $z = f(1)$. Then, indeed, identifying x in $\mathbb{Z}/n$ with an ordinary integer,

$$f(x) = f(x \cdot 1) = f(\underbrace{1 + \ldots + 1}_{x}) = \underbrace{f(1) + \ldots + f(1)}_{x} = \underbrace{z + \ldots + z}_{x} = zx$$

[23] A homomorphism that is not necessarily an isomorphism of a group to itself is an *endomorphism*.

[24] This should be expected.

That is, *every* automorphism is of this form. ///

Given two groups H and N, with a group homomorphism

$$f : H \longrightarrow \mathrm{Aut}(N) \quad \text{denoted} \quad h \longrightarrow f_h$$

the **semi-direct product** group

$$H \times_f N$$

is the set $H \times N$ with group operation intending to express the idea that

$$hnh^{-1} = f_h(n)$$

But since H and N are not literally subgroups of anything yet, we must say, instead, that we want

$$(h, e_N)(e_H, n)(h^{-1}, e_N) = (e_H, f_h(n))$$

After some experimentation, one might decide upon the definition of the operation

$$(h, n) \cdot (h', n') = (hh', f_{h'^{-1}}(n)\, n')$$

Of course, when f is the trivial homomorphism (sending everything to the identity) the semi-direct product is simply the direct product of the two groups.

2.8.7 Proposition: With a group homomorphism $f : H \longrightarrow \mathrm{Aut}(N)$, the semi-direct product $H \times_f N$ is a group. The maps $h \longrightarrow (h, e_N)$ and $n \longrightarrow (e_H, n)$ inject H and N, respectively, and the image of N is normal.

Proof: [25] The most annoying part of the argument would be proof of associativity. On one hand,

$$((h, n)(h', n'))\ (h'', n'') = (hh', f_{h'^{-1}}(n)n')\,(h'', n'') = (hh'h'', f_{h''^{-1}}\,(f_{h'^{-1}}(n)n')\,n'')$$

The H-component is uninteresting, so we only look at the N-component:

$$f_{h''^{-1}}\,(f_{h'^{-1}}(n)n')\,n'' = f_{h''^{-1}} \circ f_{h'^{-1}}(n) \cdot f_{h''^{-1}}(n') \cdot n'' = f_{(h'h'')^{-1}}(n) \cdot f_{h''^{-1}}(n') \cdot n''$$

which is the N-component which would arise from

$$(h, n)\ ((h', n')(h'', n''))$$

This proves the associativity. The other assertions are simpler. ///

Thus, in the $|G| = pq$ situation, because Q is normal, we have a group homomorphism

$$\mathbb{Z}/p \approx P \longrightarrow \mathrm{Aut}(Q) \approx (\mathbb{Z}/q)^\times$$

The latter is of order $q-1$, so unless $p|(q-1)$ this homomorphism must have trivial image, that is, P and Q commute, giving yet another approach to the case that $q \neq 1 \bmod p$. But for $p|(q-1)$ there is at least one non-trivial homomorphism to the automorphism group: $(\mathbb{Z}/q)^\times$ is an abelian group of order divisible by p, so there exists [26] an element z of order p. Then take $f : \mathbb{Z}/p \longrightarrow \mathrm{Aut}(\mathbb{Z}/q)$ by

$$f(x)(y) = z^x \cdot y \in \mathbb{Z}/q$$

[25] There is nothing surprising in this argument. It amounts to checking what must be checked, and there is no obstacle other than bookkeeping. It is surely best to go through it oneself rather than watch someone else do it, but we write it out here just to prove that it is possible to force oneself to carry out some of the details.

[26] Existence follows with or without use of the fact that there are primitive roots modulo primes. For small numerical examples this cyclicness can be verified directly, without necessarily appealing to any theorem that guarantees it.

This gives a semi-direct product of $\mathbb{Z}/p$ and $\mathbb{Z}/q$ which cannot be abelian, since elements of the copy P of $\mathbb{Z}/p$ and the copy Q of $\mathbb{Z}/q$ do not all commute with each other. That is, there is at least one non-abelian [27] group of order pq if $q = 1 \bmod p$.

How many different semi-direct products are there? [28] Now we must use the non-trivial fact that $(\mathbb{Z}/q)^\times$ is *cyclic* for q prime. [29] That is, granting this cyclicness, there are *exactly* $p-1$ elements in $(\mathbb{Z}/q)^\times$ of order p. Thus, given a fixed element z of order p in $(\mathbb{Z}/q)^\times$, any other element of order p is a power of z.

Luckily, this means that, given a choice of isomorphism $i : \mathbb{Z}/p \approx P$ to the p-Sylow group P, and given non-trivial $f : P \longrightarrow \mathrm{Aut}(Q)$, whatever the image $f(i(1))$ may be, we can alter the choice of i to achieve the effect that

$$f(i(1)) = z$$

Specifically, if at the outset

$$f(i(1)) = z'$$

with some other element z' of order p, use the cyclicness to find an integer ℓ (in the range $1, 2, \ldots, p-1$) such that

$$z' = z^\ell$$

Since ℓ is prime to p, it has an inverse modulo $k \bmod p$. Then

$$f(i(k)) = f(k \cdot i(1)) = f(i(1))^k = (z')^k = (z^\ell)^k = z$$

since $\ell k = 1 \bmod p$ and z has order p.

In summary, with primes $q = 1 \bmod p$, up to isomorphism there is a *unique* non-abelian group of order pq, and it is a semi-direct product of $\mathbb{Z}/p$ and $\mathbb{Z}/q$. [30]

Groups of order p^2, with prime p: A different aspect of the argument of the Sylow theorem is that a p-power-order group G necessarily has a non-trivial center Z. If $Z = G$ we have proven that G is abelian. Suppose Z is proper. Then [31] it is of order p, thus [32] necessarily *cyclic*, with generator z. Let x be any other group element *not* in Z. It cannot be that the order of x is p^2, or else $G = \langle x \rangle$ and $G = Z$, contrary to hypothesis. Thus, [33] the order of x is p, and [34]

$$\langle x \rangle \cap \langle z \rangle = \{e\}$$

Abstracting the situation just slightly:

2.8.8 Proposition: [35] Let G be a finite group with center Z and a subgroup A of Z. Let B be another abelian subgroup of G, such that $A \cap B = \{e\}$ and $A \cdot B = G$. Then the map

$$f : A \times B \longrightarrow A \cdot B$$

[27] So surely non-cyclic.

[28] As usual in this context, *different* means *non-isomorphic*.

[29] This is the existence of *primitive roots* modulo primes.

[30] The argument that showed that seemingly different choices yield isomorphic groups is an ad hoc example of a wider problem, of classification up to isomorphism of *group extensions*.

[31] Lagrange

[32] Lagrange

[33] Lagrange

[34] Lagrange

[35] This is yet another of an endless stream of variations on a theme.

by

$$a \times b \longrightarrow ab$$

is an isomorphism, and $A \cdot B$ is abelian.

Proof: If $a \times b$ were in the kernel of f, then $ab = e$, and

$$a = b^{-1} \in A \cap B = \{e\}$$

And

$$f((a,b)\cdot(a',b')) = f(aa',bb') = aa'bb'$$

while

$$f(a,b)\cdot f(a',b') = ab \cdot a'b' = aa'bb'$$

because $ba' = a'b$, because elements of A commute with everything. That is, f is a homomorphism. Since both A and B are abelian, certainly the product is. ///

That is, any group G of order p^2 (with p prime) is abelian. So our supposition that the center of such G is of order only p is false.

Starting over, but knowing that G of order p^2 is abelian, if there is no element of order p^2 in G (so G is not cyclic), then in any case there is an element z of order p. [36] And take x not in $\langle z\rangle$. Necessarily x is of order p. By the same clichéd sort of argument as in the last proposition, $\langle x\rangle \cap \langle z\rangle = \{e\}$ and

$$\mathbb{Z}/p \times \mathbb{Z}/p \approx \langle x\rangle \times \langle z\rangle \approx \langle x\rangle \cdot \langle z\rangle = G$$

That is, *non-cyclic* groups of order p^2 are isomorphic to $\mathbb{Z}/p \times \mathbb{Z}/p$.

Automorphisms of groups of order q^2: Anticipating that we'll look at groups of order pq^2 with normal subgroups of order q^2, to understand semi-direct products $P \times_f Q$ with P of order p and Q of order q^2 we must have *some* understanding of the automorphism groups of groups of order q^2, since $f : P \longrightarrow \mathrm{Aut}\, Q$ determines the group structure. For the moment we will focus on merely the *order* of these automorphism groups. [37]

For Q cyclic of order q^2, we know that $Q \approx \mathbb{Z}/q^2$, and from above

$$\mathrm{Aut}\, Q \approx \mathrm{Aut}(\mathbb{Z}/q^2) \approx (\mathbb{Z}/q^2)^\times$$

In particular, the *order* is [38]

$$|\mathrm{Aut}\, Q| = \mathrm{card}\,(\mathbb{Z}/q^2)^\times = \varphi(q^2) = q(q-1)$$

This is easy.

For Q non-cyclic of order q^2, we saw that

$$Q \approx \mathbb{Z}/q \oplus \mathbb{Z}/q$$

[36] For example, this follows from the lemma preparatory to the Sylow theorem.

[37] After some further preparation concerning finite fields and linear algebra we can say more definitive structural things.

[38] Using Euler's totient function φ.

where we write direct sum to emphasize the abelian-ness of Q. [39] For the moment we only aim to *count* these automorphisms. Observe that [40]

$$(\bar{x}, \bar{y}) = x \cdot (\bar{1}, \bar{0}) + y \cdot (\bar{0}, \bar{1})$$

for any $x, y \in \mathbb{Z}$, where for the moment the bars denotes residue classes modulo q. Thus, for any automorphism α of Q

$$\alpha(\bar{x}, \bar{y}) = x \cdot \alpha(\bar{1}, \bar{0}) + y \cdot \alpha(\bar{0}, \bar{1})$$

where multiplication by an integer is repeated addition. Thus, the images of $(\bar{1}, \bar{0})$ and $(\bar{0}, \bar{1})$ determine α completely. And, similarly, *any* choice of the two images gives a group homomorphism of Q to itself. The only issue is to avoid having a proper kernel. To achieve this, $\alpha(\bar{1}, \bar{0})$ certainly must not be $e \in Q$, so there remain $q^2 - 1$ possible choices for the image of $(\bar{1}, \bar{0})$. Slightly more subtly, $\alpha(\bar{0}, \bar{1})$ must not lie in the cyclic subgroup generated by $\alpha(\bar{1}, \bar{0})$, which excludes exactly q possibilities, leaving $q^2 - q$ possibilities for $\alpha(\bar{0}, \bar{1})$ for each choice of $\alpha(\bar{1}, \bar{0})$. Thus, altogether,

$$\text{card Aut}(\mathbb{Z}/q \oplus \mathbb{Z}/q) = (q^2 - 1)(q^2 - q)$$

We will pursue this later.

Groups of order pq^2: As in the simpler examples, the game is to find some mild hypotheses that combine with the Sylow theorem to limit the possibilities for the arrangement of Sylow subgroups, and then to look at direct product or semi-direct product structures that can arise. Let G be a group of order pq^2 with p, q distinct primes.

As a preliminary remark, if G is assumed abelian, then G is necessarily the direct product of a p-Sylow subgroup and a q-Sylow subgroup (both of which are necessarily norml), and by the classification of order q^2 groups this gives possibilities

$$G = P \cdot Q \approx P \times Q \approx \mathbb{Z}/p \times Q \approx \begin{cases} \mathbb{Z}/p \oplus \mathbb{Z}/q^2 & \approx & \mathbb{Z}/pq^2 \\ \mathbb{Z}/p \oplus \mathbb{Z}/q \oplus \mathbb{Z}/q & \approx & \mathbb{Z}/q \oplus \mathbb{Z}/pq \end{cases}$$

in the two cases for Q, writing direct sums to emphasize the abelian-ness. So now we consider non-abelian possibilities, and/or hypotheses which force a return to the abelian situation.

The first and simplest case is that neither $p = 1 \bmod q$ nor $q^2 = 1 \bmod p$. Then, by Sylow, there is a unique q-Sylow subgroup Q and a unique p-Sylow subgroup P, both necessarily normal. We just saw that the group Q of order q^2 is necessarily [41] abelian. Since both subgroups are normal, elements of Q commute with elements of P. [42] This returns us to the abelian case, above.

A second case is that $p|(q-1)$. This implies that $p < q$. The number n_q of q-Sylow subgroups is $1 \bmod q$ and divides pq^2, so is either 1 or p, but $p < q$, so necessarily $n_q = 1$. That is, the q-Sylow subgroup Q is normal. But this does not follow for the p-Sylow subgroup, since now $p|(q-1)$. The Sylow theorem would seemingly allow the number n_p of p-Sylow subgroups to be 1, q, or q^2. Thus, we should consider the possible semi-direct products

$$\mathbb{Z}/p \times_f Q$$

[39] Thus, in fact, Q is a two-dimensional vector space over the finite field $\mathbb{Z}/q$. We will more systematically pursue this viewpoint shortly.

[40] Yes, this is linear algebra.

[41] As a different sort of corollary of Sylow.

[42] The earlier argument is worth repeating: for a in one and b in another of two normal subgroups with trivial intersection, $(aba^{-1})b^{-1} = a(ba^{-1})b^{-1}$ must lie in both, so is e. Then $ab = ba$.

for

$$f : \mathbb{Z}/p \longrightarrow \operatorname{Aut} Q$$

If f is the trivial homomorphism, then we obtain a direct product, returning to the abelian case. For $Q \approx \mathbb{Z}/q^2$ its automorphism group has order $q(q-1)$, which is divisible by p (by hypothesis), so [43] has an element of order p. That is, there does exist a non-trivial homomorphism

$$f : P \approx \mathbb{Z}/p \longrightarrow \operatorname{Aut} \mathbb{Z}/q^2$$

That is, there do exist non-abelian semi-direct products

$$\mathbb{Z}/p \times_f \mathbb{Z}/q^2$$

Distinguishing the isomorphism classes among these is similar to the case of groups of order pq with $p|(q-1)$, and we would find just a single non-abelian isomorphism class. For $\mathbb{Q} \approx \mathbb{Z}/q \oplus \mathbb{Z}/q$, we saw above that

$$|\operatorname{Aut}(\mathbb{Z}/q \oplus \mathbb{Z}/q)| = (q^2-1)(q^2-q) = (q-1)^2\, q\, (q+1)$$

Thus, there is at least one non-abelian semi-direct product

$$\mathbb{Z}/p \times_f (\mathbb{Z}/q \oplus \mathbb{Z}/q)$$

Attempting to count the isomorphism classes would require that we have more information on the automorphism group, which we'll obtain a little later.

A third case is that $p|(q+1)$. This again implies that $p < q$, except in the case that $q = 2$ and $p = 3$, which we'll ignore for the moment. The number n_q of q-Sylow subgroups is 1 mod q and divides pq^2, so is either 1 or p, but $p < q$, so necessarily $n_q = 1$. That is, the q-Sylow subgroup Q is normal. But this does not follow for the p-Sylow subgroup, since now $p|(q+1)$. The Sylow theorem would seemingly allow the number n_p of p-Sylow subgroups to be 1 or q^2. Thus, we should consider the possible semi-direct products

$$\mathbb{Z}/p \times_f Q$$

for

$$f : \mathbb{Z}/p \longrightarrow \operatorname{Aut} Q$$

If f is the trivial homomorphism, then we obtain a direct product, returning to the abelian case. For $Q \approx \mathbb{Z}/q^2$ its automorphism group has order $q(q-1)$, which is not divisible by p, since $p|(q+1)$. That is, for $Q \approx \mathbb{Z}/q^2$ and $p|(q+1)$ and $q > 2$, there is *no* non-trivial homomorphism

$$f : P \approx \mathbb{Z}/p \longrightarrow \operatorname{Aut} \mathbb{Z}/q^2$$

That is, in this case there is *no* non-abelian semi-direct product

$$\mathbb{Z}/p \times_f \mathbb{Z}/q^2$$

For $\mathbb{Q} \approx \mathbb{Z}/q \oplus \mathbb{Z}/q$, we saw above that

$$|\operatorname{Aut}(\mathbb{Z}/q \oplus \mathbb{Z}/q)| = (q^2-1)(q^2-q) = (q-1)^2\, q\, (q+1)$$

Thus, there is at least one non-abelian semi-direct product

$$\mathbb{Z}/p \times_f (\mathbb{Z}/q \oplus \mathbb{Z}/q)$$

[43] By the lemma preceding the Sylow theorem, for example. In fact, all primes q the group $(\mathbb{Z}/q^2)^\times$ is *cyclic*, so will have *exactly one* subgroup of order p.

Again, attempting to count the isomorphism classes would require that we have more information on the automorphism group.

A fourth case is that $q|(p-1)$. Thus, by the same arguments as above, the p-Sylow subgroup P is normal, but the q-Sylow subgroup Q might not be. There are non-trivial homomorphisms in both cases

$$f : \begin{cases} \mathbb{Z}/p \oplus \mathbb{Z}/q^2 & \approx & \mathbb{Z}/pq^2 \\ \mathbb{Z}/p \oplus \mathbb{Z}/q \oplus \mathbb{Z}/q & \approx & \mathbb{Z}/q \oplus \mathbb{Z}/pq \end{cases} \longrightarrow \mathrm{Aut}\,\mathbb{Z}/p \approx (\mathbb{Z}/p)^\times$$

so either type of q-Sylow subgroup Q of order q^2 can give non-trivial automorphisms of the normal p-Sylow group P. Again, determining isomorphism classes in the first case is not hard, but in the second requires more information about the structure of $\mathrm{Aut}(\mathbb{Z}/q \oplus \mathbb{Z}/q)$.

2.8.9 Remark: The above discussion is fast approaching the limit of what we can deduce about finite groups based only on the prime factorization of their order. The cases of prime order and order pq with distinct primes are frequently genuinely useful, the others less so.

2.9 *Worked examples*

2.9.1 Example: Let G, H be finite groups with relatively prime orders. Show that any group homomorphism $f : G \longrightarrow H$ is necessarily trivial (that is, sends every element of G to the identity in H.)

The isomorphism theorem implies that

$$|G| = |\ker f| \cdot |f(G)|$$

In particular, $|f(G)|$ divides $|G|$. Since $f(G)$ is a subgroup of H, its order must also divide $|H|$. These two orders are relatively prime, so $|f(G)| = 1$.

2.9.2 Example: Let m and n be integers. Give a formula for an isomorphism of abelian groups

$$\frac{\mathbb{Z}}{m} \oplus \frac{\mathbb{Z}}{n} \longrightarrow \frac{\mathbb{Z}}{\gcd(m,n)} \oplus \frac{\mathbb{Z}}{\mathrm{lcm}(m,n)}$$

Let r, s be integers such that $rm + sn = \gcd(m,n)$. Let $m' = m/\gcd(m,n)$ and $n' = n/\gcd(m,n)$. Then $rm' + sn' = 1$. We claim that

$$f(a + m\mathbb{Z}, b + n\mathbb{Z}) = ((a-b) + \gcd(m,n)\mathbb{Z},\ (b \cdot rm' + a \cdot sn') + \mathrm{lcm}(m,n)\mathbb{Z})$$

is such an isomorphism. To see that it is well-defined, observe that

$$(a + m\mathbb{Z}) - (b + n\mathbb{Z}) = (a-b) + \gcd(m,n)\mathbb{Z}$$

since

$$m\mathbb{Z} + n\mathbb{Z} = \gcd(m,n)\mathbb{Z}$$

which itself follows from the facts that

$$\gcd(m,n) = rm + sn \in m\mathbb{Z} + n\mathbb{Z}$$

and (by definition) $m\mathbb{Z} \subset \gcd(m,n)\mathbb{Z}$ and $n\mathbb{Z} \subset \gcd(m,n)\mathbb{Z}$. And, similarly

$$sn' \cdot m\mathbb{Z} + rm' \cdot n\mathbb{Z} = \mathrm{lcm}(m,n)\mathbb{Z}$$

so the second component of the map is also well-defined.

Now since these things are finite, it suffices to show that the kernel is trivial. That is, suppose $b = a + k\text{gcd}(m,n)$ for some integer k, and consider

$$b \cdot rm' + a \cdot sn'$$

The latter is

$$(a + k\text{gcd}(m,n))rm' + a \cdot sn' = a \cdot rm' + a \cdot sn' = a \bmod m$$

since $\text{gcd}(m,n)m' = m$ and $rm' + sn' = 1$. Symmetrically, it is b mod n. Thus, if it is 0 mod $\text{lcm}(m,n)$, $a = 0 \bmod m$ and $b = 0 \bmod n$. This proves that the kernel is trivial, so the map is injective, and, because of finiteness, surjective as well.

2.9.3 Remark: I leave you the fun of guessing where the $a - b$ expression (above) comes from.

2.9.4 Example: Show that every group of order $5 \cdot 13$ is cyclic.

Invoke the Sylow theorem: the number of 5-Sylow subgroups is 1 mod 5 and also divides the order $5 \cdot 13$, so must be 1 (since 13 is not 1 mod 5). Thus, the 5-Sylow subgroup is normal. Similarly, even more easily, the 13-Sylow subgroup is normal. The intersection of the two is trivial, by Lagrange. Thus, we have two normal subgroups with trivial intersection and the product of whose orders is the order of the whole group, and conclude that the whole group is isomorphic to the (direct) product of the two, namely $\mathbb{Z}/5 \oplus \mathbb{Z}/13$. Further, this is isomorphic to $\mathbb{Z}/65$.

2.9.5 Example: Show that every group of order $5 \cdot 7^2$ is abelian.

From the classification of groups of prime-squared order, we know that there are only two (isomorphism classes of) groups of order 7^2, $\mathbb{Z}/49$ and $\mathbb{Z}/7 \oplus \mathbb{Z}/7$. From the Sylow theorem, since the number of 7-Sylow subgroups is 1 mod 7 and also divides the group order, the 7-Sylow subgroup is normal. For the same reason the 5-Sylow subgroup is normal. The intersection of the two is trivial (Lagrange). Thus, again, we have two normal subgroups with trivial intersection the product of whose orders is the group order, so the group is the direct product. Since the factor groups are abelian, so is the whole.

2.9.6 Example: Exhibit a non-abelian group of order $3 \cdot 7$.

We can construct this as a semi-direct product, since there exists a non-trivial homomorphism of $\mathbb{Z}/3$ to $\text{Aut}(\mathbb{Z}/7)$, since the latter automorphism group is isomorphic to $(\mathbb{Z}/7)^\times$, of order 6. Note that we are assured of the *existence* of a subgroup of order 3 of the latter, whether or not we demonstrate an explicit element.

2.9.7 Example: Exhibit a non-abelian group of order $5 \cdot 19^2$.

We can construct this as a semi-direct product, since there exists a non-trivial homomorphism of $\mathbb{Z}/5$ to $\text{Aut}(\mathbb{Z}/19 \oplus \mathbb{Z}/19)$, since the latter automorphism group has order $(19^2 - 1)(19^2 - 19)$, which is divisible by 5. Note that we are assured of the *existence* of a subgroup of order 5 of the latter, whether or not we demonstrate an explicit element.

2.9.8 Example: Show that every group of order $3 \cdot 5 \cdot 17$ is cyclic.

Again, the usual divisibility trick from the Sylow theorem proves that the 17-group is normal. Further, since neither 3 nor 5 divides $17 - 1 = |\text{Aut}(\mathbb{Z}/17)|$, the 17-group is *central*. But, since $3 \cdot 17 = 1 \bmod 5$, and $5 \cdot 17 = 1 \bmod 3$, we cannot immediately reach the same sort of conclusion about the 3-group and 5-group. But if *both* the 3-group and 5-group were *not* normal, then we'd have at least

$$1 + (17 - 1) + (5 - 1) \cdot 3 \cdot 17 + (3 - 1) \cdot 5 \cdot 17 = 391 > 3 \cdot 5 \cdot 17 = 255$$

elements in the group. So at least one of the two is normal. If the 5-group is normal, then the 3-group acts trivially on it by automorphisms, since 3 does not divide $5 - 1 = |\text{Aut}(\mathbb{Z}/5)|$. Then we'd have a *central*

subgroup of order $5 \cdot 17$ group, and the whole group is abelian, so is cyclic by the type of arguments given earlier. Or, if the 3-group is normal, then for the same reason it is is central, so we have a central (cyclic) group of order $3 \cdot 17$, and again the whole group is cyclic.

2.9.9 Example: Do there exist 4 primes p, q, r, s such that every group of order $pqrs$ is necessarily abelian?

We want to arrange that all of the p, q, r, s Sylow subgroups P, Q, R, S are normal. Then, because the primes are distinct, still

$$P \cap Q = \{e\}$$

$$P \cdot Q \cap R = \{e\}$$

$$P \cdot Q \cdot R \cap S = \{e\}$$

(and all other combinations) so these subgroups commute with each other. And then, as usual, the whole group is the direct product of these Sylow subgroups.

One way to arrange that all the Sylow subgroups are normal is that, mod p, none of q, r, s, qr, qs, rs, qrs is 1, and symmetrically for the other primes. Further, with none of q, r, s dividing $p - 1$ the p-group is *central*. For example, after some trial and error, plausible $p < q < r < s$ has $p = 17$. Take $q, r, s \bmod 11 = 2, 3, 5$ respectively. Take $q = 13$, so $p = -2 \bmod 13$, and require $r, s = 2, 5 \bmod q$. Then $r = 3 \bmod 11$ and $r = 2 \bmod 13$ is $80 \bmod 143$, and 223 is the first prime in this class. With $s = 5 \bmod 223$, none of the 7 quantities is 1 mod r.. Then $s = 5 \bmod 11 \cdot 13 \cdot 223$ and the first prime of this form is

$$s = 5 + 6 \cdot 11 \cdot 13 \cdot 223 = 191339$$

By this point, we know that the p, q, and r-sylow groups are central, so the whole thing is cyclic.

Exercises

2.1 Classify groups of order 7 or less, up to isomorphism.

2.2 Find two different non-abelian groups of order 8.

2.3 Classify groups of order 9 and 10.

2.4 Classify groups of order 12.

2.5 Classify groups of order 21.

2.6 Classify groups of order 27.

2.7 Classify groups of order 30.

2.8 Classify groups of order 77.

2.9 Let G be a group with just two subgroups, G and $\{e\}$. Prove that either $G = \{e\}$ or G is cyclic of prime order.

2.10 Let N be a normal subgroup of a group G, and let H be a subgroup of G such that $G = H \cdot N$, that is, such that the collection of all products $h \cdot n$ with $h \in H$ and $n \in N$ is the whole group G. Show that $G/N \approx H/(H \cap N)$.

2.11 (*Cayley's theorem*) Show that every finite group is isomorphic to a subgroup of a permutation group. (*Hint*: let G act on the *set* G by left multiplication.)

2.12 Let G be a group in which $g^2 = 1$ for every $g \in G$. Show that G is abelian.

2.13 Let H be a subgroup of index 2 in a finite group G. Show that H is normal.

2.14 Let p be the smallest prime dividing the order of a finite group G. Let H be a subgroup of index p in G. Show that H is normal.

2.15 Let H be a subgroup of finite index in a (not necessarily finite) group G. Show that there is a *normal* subgroup N of G such that $N \subset H$ and N is of finite index in G.

2.16 Find the automorphism group $\mathrm{Aut}\mathbb{Z}/n$ of the additive group $\mathbb{Z}/n$.

3. The players: rings, fields, etc.

3.1 Rings, fields
3.2 Ring homomorphisms
3.3 Vector spaces, modules, algebras
3.4 Polynomial rings I

Here we introduce some basic terminology, and give a sample of a modern construction of a *universal object*, namely a polynomial ring in one variable.

3.1 *Rings, fields*

The idea of **ring** generalizes the idea of *collection of numbers*, among other things, so maybe it is a little more intuitive than the idea of **group**. A **ring** R is a set with two operations, $+$ and $\cdot$, and with a special element 0 (**additive identity**) with most of the usual properties we expect or demand of *addition* and *multiplication*:

• R with its addition and with 0 is an abelian group. [44]

• The multiplication is *associative*: $a(bc) = (ab)c$ for all $a, b, c \in R$.

• The multiplication and addition have left and right **distributive** properties: $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$ for all $a, b, c \in R$.

Often the multiplication is written just as juxtaposition

$$ab = a \cdot b$$

Very often, a particular ring has some additional special features or properties:

[44] This is a compressed way to say that 0 behaves as an additive identity, that there are additive inverses, and that addition is associative.

• If there is an element 1 in a ring with $1 \cdot a = a \cdot 1$ for all $a \in R$, then 1 is said to be **the (multiplicative) identity** or **the unit** [45] in the ring, and the ring is said to **have an identity** or **have a unit** or be a **ring with unit**. [46]

• If $ab = ba$ for all a, b in a ring R, then the ring is a **commutative ring**. That is, a ring is called *commutative* if and only if the *multiplication* is commutative.

• In a ring R with 1, for a given element $a \in R$, if there is $a^{-1} \in R$ so that $a \cdot a^{-1} = 1$ and $a^{-1} \cdot a = 1$, then a^{-1} is said to be a **multiplicative inverse** for a. If $a \in R$ *has* a multiplicative inverse, then a is called **a unit** [47] in R. The collection of all units in a ring R is denoted $R^\times$ and is called **the group of units in** R. [48]

• A commutative ring in which every nonzero element is a *unit* is a **field**.

• A not-necessarily commutative ring in which every nonzero element is a unit is a **division ring**.

• In a ring R an element r so that $r \cdot s = 0$ or $s \cdot r = 0$ for some nonzero $s \in R$ is called a **zero divisor**. [49] A commutative ring *without* nonzero zero-divisors is an **integral domain**.

• A commutative ring R has the **cancellation property** if, for any $r \neq 0$ in R, if $rx = ry$ for $x, y \in R$, then $x = y$.

If we take a ring R with 0 and with its addition, forgetting the multiplication in R, then we get an abelian group, called **the additive group of** R. And the group of units $R^\times$ is a (possibly non-abelian) group.

3.1.1 Example: The integers $\mathbb{Z}$ with usual addition and multiplication form a ring. This ring is certainly *commutative* and has a multiplicative identity 1. The group of units $\mathbb{Z}^\times$ is just $\{\pm 1\}$. This ring is an integral domain. The *even* integers $2\mathbb{Z}$ with the usual addition and multiplication form a commutative ring *without* unit. Just as this example suggests, sometimes the lack of a unit in a ring is somewhat artificial, because there is a larger ring it sits inside which *does* have a unit. There are no units in $2\mathbb{Z}$.

3.1.2 Example: The integers mod m, denoted $\mathbb{Z}/m$, form a commutative ring with identity. It is not hard to verify that addition and multiplication are well-defined. *As the notation suggests,* the group of units is $\mathbb{Z}/m^\times$. [50]

[45] Sometimes the word *unity* is used in place of *unit* for the special element 1, but this cannot be relied upon, and in any case does not fully succeed in disambiguating the terminology.

[46] We also demand that $1 \neq 0$ in a ring, if there is a 1.

[47] Yes, this usage is partly in conflict with the terminology for a special element 1.

[48] It is almost immediate that $R^\times$ truly is a group.

[49] The question of whether or not 0 should by convention be counted as a zero divisor has no clear answer.

[50] Yes, we used the group-of-units notation in this case before we had introduced the terminology.

3.1.3 Example: The ring $\mathbb{Z}/p$ of integers mod p is a *field* for p prime, since all non-zero residue classes have multiplicative inverses. [51] The group of units is $(\mathbb{Z}/p)^\times$. For n non-prime, $\mathbb{Z}/n$ is definitely not a field, because a proper factorization $n = ab$ exhibits non-zero zero divisors.

3.1.4 Example: Generally, a finite field with q elements is denoted $\mathbb{F}_q$. We will see later that, up to isomorphism, there is at most one finite field with a given number of elements, and, in fact, none unless that number is the power of a prime.

3.1.5 Example: The collection of n-by-n matrices (for fixed n) with entries in a ring R is a ring, with the usual matrix addition and multiplication. [52] Except for the silly case $n = 1$, rings of matrices over commutative rings R are *non-commutative*. The group of units, meaning matrices with an inverse of the same form, is the group $GL(n, R)$, the **general linear group** of size n over R.

3.1.6 Example: The rational numbers $\mathbb{Q}$, the real numbers $\mathbb{R}$, and the complex numbers $\mathbb{C}$ are all examples of *fields*, because all their nonzero elements have multiplicative inverses. The integers $\mathbb{Z}$ do not form a field.

There are some things about the behavior of rings which we might accidentally take for granted.

Let R be a ring.
• *Uniqueness of* 0 *additive identity:* From the analogous discussion at the beginning of group theory, we know that there is exactly one element $z = 0$ with the property that $r + z = r$ for all r in R. And there is exactly one additive inverse to any $r \in R$. And for $r \in R$, we have $-(-r) = r$. Similarly, if R has a unit 1, then, using the group $R^\times$, we deduce uniqueness of 1, and uniqueness of multiplicative inverses.

The following items are slightly subtler than the things above, involving the interaction of the multiplication and addition. Still, there are no surprises. [53]

Let R be a ring.
• For any $r \in R$, $0 \cdot r = r \cdot 0 = 0$. [54]
• Suppose that there is a 1 in R. Let -1 be the additive inverse of 1. Then for any $r \in R$ we have $(-1) \cdot r = r \cdot (-1) = -r$, where as usual $-r$ denotes the additive inverse of r.
• Let $-x, -y$ be the additive inverses of $x, y \in R$. Then $(-x) \cdot (-y) = xy$.

Proof: Let $r \in R$. Then

$$\begin{aligned} 0 \cdot r &= (0+0) \cdot r \qquad \text{(since } 0+0=0\text{)} \\ &= 0 \cdot r + 0 \cdot r \qquad \text{(distributivity)} \end{aligned}$$

Then, adding $-(0 \cdot r)$ to both sides, we have

$$0 = 0 \cdot r - 0 \cdot r = 0 \cdot r + 0 \cdot r - 0 \cdot r = 0 \cdot r + 0 = 0 \cdot r$$

That is, $0 \cdot r$. The proof that $r \cdot 0 = 0$ is identical.

To show that $(-1)r$ is the additive inverse of r, which by now we know is unique, we check that

$$r + (-1)r = 0$$

We have

$$r + (-1)r = 1 \cdot r + (-1) \cdot r = (1-1) \cdot r = 0 \cdot r = 0$$

[51] Again, for a residue class represented by x relatively prime to p, there are integers r, s such that $rx + yp = \gcd(x, p) = 1$, and then the residue class of r is a multiplicative inverse to the residue class of x.

[52] Verification of the ring axioms is not terrifically interesting, but is worth doing once.

[53] No surprises except perhaps that these things do follow from the innocent-seeming ring axioms.

[54] One can easily take the viewpoint that this universal assertion has very little *semantic content*.

using the result we just $0 \cdot r = 0$.

To show that $(-x)(-y) = xy$, prove that $(-x)(-y) = -(-(xy))$, since $-(-r) = r$. We claim that $-(xy) = (-x)y$: this follows from

$$(-x)y + xy = (-x + x)y = 0 \cdot y = 0$$

Thus, we want to show

$$(-x)(-y) + (-x)y = 0$$

Indeed,

$$(-x)(-y) + (-x)y = (-x)(-y + y) = (-x) \cdot 0 = 0$$

using $r \cdot 0 = 0$ verified above. Thus, $(-x)(-y) = xy$. ///

An **idempotent** element of a ring R is an element e such that

$$e^2 = e$$

A **nilpotent** element is an element z such that for some positive integer n

$$z^n = 0_R$$

3.2 *Ring homomorphisms*

Ring homomorphisms are maps from one ring to another which respect the ring structures.

Precisely, a **ring homomorphism** $f : R \to S$ from one ring R to another ring S is a map such that for all r, r' in R

$$\begin{array}{rcl} f(r + r') & = & f(r) + f(r') \\ f(rr') & = & f(r)\ f(r') \end{array}$$

That is, f *preserves* or *respects* both addition and multiplication. [55] A ring homomorphism which has a two-sided inverse homomorphism is an **isomorphism**. If a ring homomorphism is a bijection, it is an isomorphism. [56]

The **kernel** of a ring homomorphism $f : R \to S$ is

$$\ker f = \{r \in R : f(r) = 0\}$$

3.2.1 Example: The most basic worthwhile example of a ring homomorphism is

$$f : \mathbb{Z} \longrightarrow \mathbb{Z}/n$$

[55] We do not make an attempt to use different notations for the addition and multiplication in the two different rings R and S in this definition, or in subsequent discussions. Context should suffice to distinguish the two operations.

[56] Since a bijective ring homomorphism has an inverse map which is a ring homomorphism, one could *define* an isomorphism to be a bijective homomorphism. However, in some other scenarios bijectivity of certain types of homomorphisms is *not* sufficient to assure that there is an inverse map of the same sort. The easiest example of such failure may be among continuous maps among topological spaces. For example, let $X = \{0, 1\}$ with the *indiscrete topology*, in which only the whole set and the empty set are open. Let $Y = \{0, 1\}$ with the *discrete* topology, in which all subsets are open. Then the identity map $X \longrightarrow Y$ is continuous, but its inverse is not. That is, the map is a continuous bijection, but its inverse is not continuous.

given by

$$f(x) = x + n\mathbb{Z}$$

The assertion that this f is a ring homomorphism is the combination of the two assertions

$$(x + n\mathbb{Z}) + (y + n\mathbb{Z}) = (x + y) + n\mathbb{Z}$$

and

$$(x + n\mathbb{Z}) \cdot (y + n\mathbb{Z}) + n\mathbb{Z} = (x \cdot y) + n\mathbb{Z}$$

Even though it is slightly misleading, this homomorphism is called the **reduction mod m homomorphism**.

3.2.2 Proposition: Let $f : R \to S$ be a ring homomorphism. Let $0_R, 0_S$ be the additive identities in R, S, respectively. Then $f(0_R) = 0_S$.

Proof: This is a corollary of the analogous result for groups. ///

3.2.3 Proposition: Let $f : R \longrightarrow S$ be a *surjective* ring homomorphism. Suppose that R has a multiplicative identity 1_R. Then S has a multiplicative identity 1_S and

$$f(1_R) = 1_S$$

3.2.4 Remark: Notice that, unlike the discussion about the additive identity, now we need the further hypothesis of surjectivity.

Proof: Given $s \in S$, let $r \in R$ be such that $f(r) = s$. Then

$$f(1_R) \cdot s = f(1_R) \cdot f(r) = f(1_R \cdot r) = f(r) = s$$

Thus, $f(1_R)$ behaves like a unit in S. By the *uniqueness* of units, $f(1_R) = 1_S$. ///

3.2.5 Example: The image of a multiplicative identity 1_R under a ring homomorphism $f : R \to S$ is not necessarily the multiplicative identity 1_S of S. For example, define a ring homomorphism

$$f : \mathbb{Q} \to S$$

from the rational numbers $\mathbb{Q}$ to the ring S of 2-by-2 rational matrices by

$$f(x) = \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix}$$

Then the image of 1 is

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

which is not the the multiplicative identity

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

in S. As another example, let $R = \mathbb{Z}/3$ and $S = \mathbb{Z}/6$, and define $f : R \longrightarrow S$ by

$$f(r \bmod 3) = 4r \bmod 6$$

(This is well-defined, and is a homomorphism.) The essential feature is that

$$4 \cdot 4 = 4 \bmod 6$$

Then

$$f(x \cdot y) = 4(x \cdot y) = (4 \cdot 4)(x \cdot y) = (4x) \cdot (4y) = f(x) \cdot f(y)$$

But $f(1) = 4 \neq 1 \bmod 6$.

3.3 *Vector spaces, modules, algebras*

Let k be a field. A **k-vectorspace** V is an abelian group V (with operation written additively, referred to as **vector addition**) and a **scalar multiplication**

$$k \times V \longrightarrow V$$

written

$$\alpha \times v \longrightarrow \alpha \cdot v = \alpha v$$

such that, for $\alpha, \beta \in k$ and $v, v' \in V$,

$$\begin{array}{ll} \text{(Distributivity)} & \alpha \cdot (v + v') = \alpha \cdot v + \alpha \cdot v' \\ \text{(Distributivity)} & (\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v \\ \text{(Associativity)} & (\alpha \cdot \beta) \cdot v = \alpha \cdot (\beta \cdot v) \\ & 1 \cdot v = v \end{array}$$

3.3.1 Remark: The requirement that $1 \cdot v = v$ does not follow from the other requirements. [57] By contrast, the zero element 0 in a field does reliably annihilate any vectorspace element v:

$$0_V = -(0 \cdot v) + 0 \cdot v = -(0 \cdot v) + (0 + 0) \cdot v = -(0 \cdot v) + 0 \cdot v + 0 \cdot v = 0 \cdot v$$

A k-vector **subspace** W of a k-vectorspace V is an additive subgroup closed under scalar multiplication.

A **k-linear combination** of vectors $v_1, \ldots, v_n$ in a k-vectorspace V is any vector of the form

$$\alpha_1 v_1 + \ldots + \alpha_n v_n$$

with $\alpha_i \in k$ and $v_i \in V$. Vectors $v_1, \ldots, v_n$ are **linearly dependent** if there is a linear combination of them which is 0, yet not all coefficients are 0. They are **linearly independent** if they are not linearly *dependent*. [58]

The pedestrian example of a vector space is, for fixed natural number n, the collection k^n of ordered n-tuples of elements of k with component-wise vector addition and component-wise scalar multiplication.

A **k-linear map** $T : V \longrightarrow W$ from one k-vectorspace V to another W is a homomorphism of abelian groups $T : V \longrightarrow W$

$$T(v + v') = Tv + Tv'$$

also respecting the scalar multiplication: for $\alpha \in k$

$$T(\alpha \cdot v) = \alpha \cdot Tv$$

[57] Sometimes the requirement that $1 \cdot v = v$ is given an unfortunate name, such as *unitary* property (in conflict with other usage), or *unital* property, which conjures up no clear image. The point is that the terminology is unpredictable.

[58] We will certainly continue this discussion of elementary linear algebra shortly, discussing the usual standard notions.

The collection of all k-linear maps from V to W is denoted

$$\mathrm{Hom}_k(V,W) = \{ \text{ all } k\text{-linear maps from } V \text{ to } W\}$$

When $V = W$, write

$$\mathrm{End}_k(V,V) = \mathrm{End}_k(V)$$

This is the **ring of k-linear endomorphisms** of V.

The **kernel** $\ker T$ of a k-linear map $T : V \longrightarrow W$ is

$$\ker T = \{v \in V : Tv = 0\}$$

Let R be a ring. An **R-module** [59] M is an abelian group M (with operation written additively) and a **multiplication**

$$R \times M \longrightarrow M$$

written

$$r \times m \longrightarrow r \cdot m = rm$$

such that, for $r, r' \in R$ and $m, m' \in M$

$$\begin{array}{ll} \text{(Distributivity)} & r \cdot (m + m') = r \cdot m + e \cdot m' \\ \text{(Distributivity)} & (r + r') \cdot m = r \cdot m + r' \cdot m \\ \text{(Associativity)} & (r \cdot r') \cdot m = r \cdot (r' \cdot m) \end{array}$$

The notion of module-over-ring obviously subsumes the notion of vectorspace-over-field.

A **R-linear combination** of elements $m_1, \ldots, m_n$ in a R module M is any module element of the form

$$r_1 m_1 + \ldots + r_n m_n$$

with $r_i \in R$ and $m_i \in M$. [60]

We specifically do *not* universally require that $1_R \cdot m = m$ for all m in an R-module M when the ring R contains a unit 1_R. Nevertheless, on many occasions we *do* require this, but, therefore, must say so explicitly to be clear.

An R **submodule** N of an R-module M is an additive subgroup which is closed under scalar multiplication.

An **R-linear map** $T : M \longrightarrow N$ from one T-module M to another N is a homomorphism of abelian groups $T : M \longrightarrow N$

$$T(m + m') = Tm + Tm'$$

also respecting the scalar multiplication: for $r \in R$

$$T(r \cdot m) = r \cdot Tm$$

The collection of all R-linear maps from M to N is denoted

$$\mathrm{Hom}_R(M,N) = \{ \text{ all } R\text{-linear maps from } M \text{ to } N\}$$

[59] In some older sources the word was *modul*, which is now obsolete. And, in some older sources, *module* was used for what we now call the multiplicative identity 1, as well as other things whose present names are otherwise.

[60] While one should think of linear algebra over fields as a prototype for some of the phenomena concerning modules more generally, one should at the same time be prepared for deviation from the simpler reasonable expectations.

When $M = N$, write

$$\mathrm{End}_R(M, M) = \mathrm{End}_R(M)$$

This is the **ring of R-linear endomorphisms** of M.

The **kernel** $\ker T$ of an R-linear map $T : M \longrightarrow N$ is

$$\ker T = \{m \in M : Tm = 0\}$$

3.3.2 Example: *Abelian groups are $\mathbb{Z}$-modules*: for $a \in A$ in an abelian group A, define the scalar multiplication by integers by

$$n \cdot a = \begin{cases} 0_A & (\text{for } n = 0) \\ \underbrace{a + \ldots + a}_{n} & (\text{for } n > 0) \\ -(\underbrace{a + \ldots + a}_{|n|}) & (\text{for } n < 0) \end{cases}$$

Observe that a homomorphism of abelian groups is inevitably $\mathbb{Z}$-linear.

3.3.3 Example: A **(left) ideal** I in a ring R is an additive subgroup I of R which is also closed under left multiplication by R: for $i \in I$ and $r \in R$, $r \cdot i \in I$. It is immediate that the collection of left ideals in R is identical to the collection of R-submodules of R (with left multiplication).

Let R be a *commutative* ring. [61] Let A be a not-necessarily commutative ring which is a left R-module. If, in addition to the requirements of a module, we have the *associativity*

$$r \cdot (a \cdot b) = (r \cdot a) \cdot b$$

for $r \in R$ and $a, b \in A$, then say A is an R-**algebra**. Often additional requirements are imposed. [62]

A ring homomorphism $f : A \longrightarrow B$ of R-algebras is an R-**algebra homomorphism** if it is also an R-module homomorphism.

3.4 *Polynomial rings I*

We should not be content to speak of *indeterminate* x or *variable* x to construct polynomial rings. Instead, we describe in precise terms the fundamental property that a polynomial ring is meant to have, namely, in colloquial terms, the *indeterminate* can be replaced by *any value*, or that any value can be *substituted for* the indeterminate.

Fix a commutative ring R, and let A be a *commutative* R-algebra with a distinguished element a_o. Say that A, or, more properly, the *pair* (A, a_o), is a **free (commutative) algebra** on one **generator** a_o if, for every commutative R-algebra B and chosen element $b_o \in B$ there is a *unique* R-algebra homomorphism

$$f_{B,b_o} : A \longrightarrow B$$

such that

$$f(a_o) = b_o$$

[61] The requirement that R be commutative is not at all necessary to give a definition of R-algebra, but without that hypothesis it is much less clear what is best to offer as a first and supposedly general definition.

[62] One might require the commutativity $(ra)b = a(rb)$, for example. One might require that R have a unit 1 and that $1 \cdot a = a$ for all $a \in A$. However, not all useful examples meet these additional requirements.

This condition is an example of a **universal mapping property**, and the polynomial ring (once we show that it is this object) is thus a **universal object** with respect to this property.

3.4.1 Remark: In different words, a_o can be mapped *anywhere*, and specifying the image of a_o completely determines the homomorphism.

3.4.2 Remark: We are to imagine that $A = R[x]$, $a_o = x$, and that the R-algebra homomorphism *is* the substitution of b_o for x.

The following uniqueness result is typical of what can be said when an object is characterized by universal mapping properties.

3.4.3 Proposition: Up to isomorphism, there is *at most one* free commutative R-algebra on one generator. That is, given two such things (A, a_o) and (A', a'_o), there is a unique isomorphism

$$i : A \longrightarrow A'$$

sending a_o to a'_o and such that, given a commutative R-algebra B with distinguished element b_o, the corresponding maps (as above)

$$f_{B,b_o} : A \longrightarrow B$$

$$f'_{B,b_o} : A' \longrightarrow B$$

satisfy

$$f = f' \circ i$$

3.4.4 Remark: Despite the possible unpalatableness of the definition and the proposition, this setup does what we want, and the proposition asserts the *essential uniqueness* of what will turn out to be recognizable as the polynomial ring $R[x]$.

Proof: This proof is typical of proving that there is at most one thing characterized by a universal property. First, take $B = A$ and $b_o = a_o$. Then there is a *unique* R-algebra homomorphism $A \longrightarrow A$ taking a_o to a_o. Since the identity map on A does this, apparently *only* the identity has this property among all endomorphisms of A.

Next, let $B = A'$ and $b = a'_o$, and

$$f_{A',a'_o} : A \longrightarrow A' \quad (\text{with } a_o \longrightarrow a'_o)$$

the unique R-algebra homomorphism postulated. Reversing the roles of A and A', we have another *unique*

$$f'_{A,a_o} : A' \longrightarrow A \quad (\text{with } a'_o \longrightarrow a_o)$$

Consider $g = f' \circ f'$. It sends a_o to a_o, so, by our first observation, must be the identity map on A. Similarly, $f \circ f'$ is the identity map on A'. Thus, f and f' are mutual inverses, and A and A' are isomorphic, by a *unique* isomorphism, [63] and a_o is mapped to a'_o by this map. ///

This slick uniqueness argument does not prove existence. Indeed, there seems to be no comparably magical way to prove existence, *but* the uniqueness result assures us that, whatever pathetic *ad hoc* device we do hit upon to construct the free algebra, the thing we make is *inevitably* isomorphic (and by a *unique* isomorphism)

[63] The virtues of there being a *unique* isomorphism may not be apparent at the moment, but already played a role in the uniqueness proof, and do play significant roles later.

to what any *other* construction might yield. That is, the uniqueness result shows that particular choice of construction does not matter. [64]

How to construct the thing? On one hand, since the possible images $f(a_o)$ can be *anything* in another R-algebra B, a_o ought not satisfy any relations such as $a_o^3 = a_o$ since a homomorphism would carry such a relation forward into the R-algebra B, and we have no reason to believe that $b_o^3 = b_o$ for every element b_o of every R-algebra B. [65] On the other hand, since the image of a_o under an R-algebra homomorphism is intended to determine the homomorphism completely, the free algebra A should not contain more than R-linear combinations of powers of a_o.

For fixed commutative ring R with identity 1, let S be the set [66] of R-valued functions P on the set $\{0, 1, 2, \ldots\}$ such that, for each P, there is an index n such that for $i > n$ we have $P(i) = 0$. [67] Introduce an addition which is simply componentwise: for $P, Q \in S$,

$$(P + Q)(i) = P(i) + Q(i)$$

And there is the value-wise R-module structure with scalar multiplication

$$(r \cdot P)(i) = r \cdot P(i)$$

All this is obligatory, simply to have an R-module. We take the distinguished element to be

$$a_o = \text{ the function } P_1 \text{ such that } P_1(1) = 1 \text{ and } P_1(i) = 0 \text{ for } i \neq 1$$

A misleadingly glib way of attempting to define the multiplication is [68] as

$$(P \cdot Q)(i) = \sum_{j+k=i} P(j)\, Q(k)$$

using the idea that a function is completely described by its values. Thus, since R is commutative,

$$P \cdot Q = Q \cdot P$$

For P, Q, T in S, associativity

$$(P \cdot Q) \cdot T = P \cdot (Q \cdot T)$$

[64] An elementary example of a construction whose internals are eventually ignored in favor of operational properties is *ordered pair*: in elementary set theory, the ordered pair (a, b) is defined as $\{\{a\}, \{a, b\}\}$, and the expected properties are verified. After that, this set-theoretic definition is forgotten. And, indeed, one should probably not consider this to be *correct* in any sense of providing further information about what an ordered pair truly is. Rather, it is an *ad hoc* construction which thereafter entitles us to do certain things.

[65] While it is certainly true that we should doubt that this a_o satisfies any relations, in other situations specification of universal objects *can* entail unexpected relations. In others, the fact that there *are* no relations apart from obvious ones may be non-trivial to prove. An example of this is the Poincaré-Birkhoff-Witt theorem concerning universal enveloping algebras. We may give this as an example later.

[66] This construction presumes that sets and functions are legitimate primitive objects. Thus, we tolerate possibly artificial-seeming constructions for their validation, while clinging to the uniqueness result above to rest assured that any peculiarities of a construction do not harm the object we create.

[67] We would say that P is *eventually zero*. The intent here is that $P(i)$ is the coefficient of x^i in a polynomial.

[68] If one is prepared to describe polynomial multiplication by telling the coefficients of the product then perhaps this is not surprising. But traditional informal discussions of polynomials often to treat them more as strings of symbols, *expressions*, rather than giving them set-theoretic substance.

follows from rewriting the left-hand side into a symmetrical form

$$((P \cdot Q) \cdot T)(i) = \sum_{j+k=i} (P \cdot Q)(j)\, T(k)$$

$$= \sum_{j+k=i} \sum_{m+n=j} P(m)Q(n)T(k) = \sum_{m+n+k=i} P(m)Q(n)T(k)$$

Distributivity of the addition and multiplication in S follows from that in R:

$$(P \cdot (Q+T))(i) = \sum_{j+k=i} P(j) \cdot (Q+T)(k) = \sum_{j+k=i} (P(j)Q(k) + P(j)T(k))$$

$$= \sum_{j+k=i} P(j)Q(k) + \sum_{j+k=i} P(j)T(k) = (P \cdot Q)(i) + (P \cdot T)(i)$$

The associativity

$$r \cdot (P \cdot Q) = (rP) \cdot Q$$

is easy. Note that, by an easy induction

$$P_1^i(j) = \begin{cases} 1 & (\text{if } j = i) \\ 0 & (\text{if } j \neq i) \end{cases}$$

So far, we have managed to make a commutative R-algebra S with a distinguished element P_1. With the above-defined multiplication, we claim that

$$\sum_i r_i\, P_1^i$$

(with coefficients r_i in R) is 0 (that is, the zero function in S) if and only if all coefficients are 0. Indeed, the value of this function at j is r_j. Thus, as a consequence, if

$$\sum_i r_i\, P_1^i = \sum_j r'_j\, P_1^j$$

then subtract one side from the other, so see that $r_i = r'_i$ for all indices i. That is, there is only one way to express an element of S in this form.

Given another R-algebra B and element $b_o \in B$, we would like to define

$$f(\sum_i r_i P_1^i) = \sum_i r_i b_o^i$$

At least this is *well-defined*, since there is only one expression for elements of S as R-linear combinations of powers of P_1^i. The R-linearity of this f is easy. The fact that it respects the multiplication of S is perhaps less obvious, but not difficult: [69]

$$f((\sum_i r_i P_1^i) \cdot (\sum_j r'_j P_1^j)) = f(\sum_{i,j} r_i r'_j P_1^{i+j}) = \sum_{i,j} r_i r'_j b_o^{i+j} = (\sum_i r_i b_o^i) \cdot (\sum_j r'_j b_o^j))$$

Thus, this f is an R-algebra homomorphism which sends $a_o = P_1$ to b_o.

[69] The formulas for multiplication of these finite sums with many summands could be proven by induction if deemed necessary.

Finally, there is no *other* R-algebra homomorphism of S to B sending $a_o = P_1$ to b_o, since every element of S is expressible as $\sum r_i P_1^i$, and the R-algebra homomorphism property yields

$$f(\sum r_i P_1^i) = \sum_i f(r_i P_1^i) = \sum_i r_i\, f(P_1^i) = \sum_i r_i\, f(P_1)^i$$

There is no further choice possible. [70]

3.4.5 Remark: This tedious construction (or something equivalent to it) is necessary. The uniqueness result assures us that no matter which of the several choices we make for (this tiresome) construction, the resulting thing is the same.

Exercises

3.1 Let r be nilpotent in a commutative ring. Show that $1 + r$ is a unit.

3.2 Give an example of an integer n such that $\mathbb{Z}/n$ has at least 4 different idempotent elements.

3.3 Give an example of an integer n such that $\mathbb{Z}/n$ has at least 4 different idempotent elements.

3.4 Let $f \in k[x]$ for a field k. For indeterminates x, y, show that we have a Taylor-Maclaurin series expansion of the form

$$f(x+y) \;=\; f(x) + \sum_{i=1}^{n} f_i(x)\, y^i$$

for some polynomials $f_i(x)$. For k of characteristic 0, show that

$$f_i(x) \;=\; \left(\frac{\partial}{\partial x}\right)^i f(x)/i!$$

3.5 Show that a **local ring** R (that is, a ring having a unique *maximal* proper ideal) has no idempotent elements other than 0 and 1.

3.6 Let $p > 2$ be a prime. Show that for $\ell \geq 1 \geq \frac{p}{p-1}$ the power of p dividing $(p^\ell)^n$ is larger than or equal to the power of p dividing $n!$.

3.7 *The exponential map modulo p^n:* Let $p > 2$ be prime. Make sense of the map

$$E : p\mathbb{Z}/p^n \to 1 + p\mathbb{Z} \bmod p^n$$

defined by the dubious formula

$$E(px) = 1 + \frac{px}{1!} + \frac{(px)^2}{2!} + \frac{(px)^3}{3!} + \dots$$

(*Hint:* cancel powers of p before trying to make sense of the fractions. And only finitely-many of the summands are non-zero mod p^n, so this is a *finite* sum.)

3.8 With the exponential map of the previous exercise, show that $E(px + py) = E(px) \cdot E(py)$ modulo p^n, for $x, y \in \mathbb{Z}/p^{n-1}$. That is, prove that E is a group homomorphism from $p\mathbb{Z}/p^n\mathbb{Z}$ to the subgroup of $(\mathbb{Z}/p^n)^\times$ consisting of $a = 1 \bmod p$.

[70] One may paraphrase this by saying that if g were another such map, then $f - g$ evaluated on any such element of S is 0.

3.9 Prove that $(\mathbb{Z}/p^n)^\times$ is *cyclic* for $p > 2$ prime.

3.10 Figure out the correct analogue of the exponential map for $p = 2$.

3.11 Figure out the correct analogue of the exponential maps modulo primes for the Gaussian integers $\mathbb{Z}[i]$.

3.12 For which ideals I of $\mathbb{Z}[i]$ is the multiplicative group $\mathbb{Z}[i]/I^\times$ of the quotient ring $\mathbb{Z}[i]/I$ *cyclic*?

3.13 Show that there are no *proper* two-sided ideals in the ring R of 2-by-2 rational matrices.

3.14 (*Hamiltonian quaternions*) Define the **quaternions** $\mathfrak{H}$ to be an $\mathbb{R}$-algebra generated by $1 \in \mathbb{R}$ and by elements i, j, k such that $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, and $ki = j$. Define the *quaternion conjugation* $\alpha \longrightarrow \alpha^*$ by

$$(a + bi + cj + dk)^* = a - bi - cj - dk$$

Show that $*$ is an *anti-automorphism*, meaning that

$$(\alpha \cdot \beta)^* = \beta^* \cdot \alpha^*$$

for quaternions α, β. Show that $\mathfrak{H}$ is a *division ring*.

3.15 Provide a *construction* of the quaternions, by showing that

$$a + bi + cj + dk \longrightarrow \begin{pmatrix} a + bi & c + di \\ c - di & a - bi \end{pmatrix}$$

is a ring homomorphism from the quaternions to a subring of the 2-by-2 complex matrices.

4. Commutative rings I

4.1 Divisibility and ideals
4.2 Polynomials in one variable over a field
4.3 Ideals
4.4 Ideals and quotient rings
4.5 Maximal ideals and fields
4.6 Prime ideals and integral domains
4.7 Fermat-Euler on sums of two squares
4.8 Worked examples

Throughout this section the rings in question will be commutative, and will have a unit 1.

4.1 *Divisibility and ideals*

Many of the primitive ideas about divisibility we bring from the ordinary integers $\mathbb{Z}$, though few of the conclusions are as simple in any generality.

Let R be a commutative [71] ring with unit [72] 1. Let $\mathbb{R}^\times$ be the group of units in R.

Say d **divides** m, equivalently, that m is a **multiple** of d, if there exists a $q \in R$ such that $m = qd$. Write $d|m$ if d divides m. It is easy to prove, from the definition, that if $d|x$ and $d|y$ then $d|(ax + by)$ for any $x, y, a, b \in R$: let $x = rd$ and $y = sd$, and

$$ax + by = a(rd) + b(sd) = d \cdot (ar + bs)$$

A ring element d is a **common divisor** of ring elements $n_1, \ldots, n_m$ if d divides each n_i. A ring element N is a **common multiple** of ring elements $n_1, \ldots, n_m$ if N is a multiple of each.

[71] Divisibility and ideals can certainly be discussed without the assumption of commutativity, but the peripheral complications obscure simpler issues.

[72] And, certainly, one can contemplate divisibility in rings without units, but this leads to needlessly counterintuitive situations.

A divisor d of n is **proper** if it is not a unit multiple of n and is not a unit itself. A ring element is **irreducible** if it has no proper factors. A ring element p is **prime** if $p|ab$ implies $p|a$ or $p|b$ and p is not a unit and is not 0. [73] If two prime elements p and p' are related by $p = up'$ with a unit u, say that p and p' are **associate**. We view associate primes as being essentially identical. [74] Recall that an **integral domain** [75] is a commutative ring in which $cd = 0$ implies either c or d is 0. [76]

4.1.1 Proposition: Prime elements of an integral domain R (with 1) are irreducible. [77]

Proof: Let p be a prime element of R, and suppose that $p = ab$. Then $p|ab$, so $p|a$ or $p|b$. Suppose $a = a'p$. Then $p = p \cdot a'b$, and $p \cdot (1 - a'b) = 0$. Since the ring is an integral domain, either $p = 0$ or $a'b = 1$, but $p \neq 0$. Thus, $a'b = 1$, and b is a unit. This proves that any factorization of the prime p is non-proper. ///

An integral domain R with 1 is a **unique factorization domain** (**UFD**) if every element $r \in R$ has a unique (up to ordering of factors and changing primes by units) expression

$$r = up_1 \dots p_\ell$$

with unit u and primes p_i.

4.1.2 Remark: The ordinary integers are the primary example of a UFD. The second important example is the ring of polynomials in one variable over a field, treated in the next section.

4.2 *Polynomials in one variable over a field*

We will prove that the ring $k[x]$ of polynomials in one variable with coefficients in a field k is *Euclidean*, and thus has unique factorization. This example is comparable in importance to $\mathbb{Z}$ and its elementary properties.

As usual, the **degree** of a polynomial $\sum_i c_i x^i$ is the highest index i such that $c_i \neq 0$.

4.2.1 Proposition: For polynomials P, Q with coefficients in a field [78] k, the degree of the product is the sum of the degrees:

$$\deg(P \cdot Q) = \deg P + \deg Q$$

4.2.2 Remark: To make this correct even when one of the two polynomials is the 0 polynomial, the 0 polynomial is by convention given degree $-\infty$.

---

[73] Yes, the definition of *prime* rewrites what was a theorem for the ordinary integers as the definition in general, while demoting the lack of proper factors to a slightly more obscure classification, of *irreducibility*.

[74] In the case of the ordinary integers, $\pm p$ are associate, for prime p. We naturally distinguish the positive one of the two. But in more general situations there is not a reliable special choice among associates.

[75] Some sources have attempted to popularize the term *entire* for a ring with no proper zero divisors, but this has not caught on.

[76] If one insisted, one could say that an integral domain is a commutative ring in which 0 is prime, but for practical reasons we want our convention not to include 0 when we speak of prime *elements*. Likewise by convention we do not want *units* to be included when we speak of primes.

[77] The converse is not generally true.

[78] The proof only uses the fact that a product of non-zero elements is necessarily non-zero. Thus, the same conclusion can be reached if the coefficients of the polynomials are merely in an *integral domain*.

Proof: The result is clear if either polynomial is the zero polynomial, so suppose that both are non-zero. Let

$$P(x) = a_m x^m + a_{m-1}x^{m-1} + \ldots + a_2 x^2 + a_1 x + a_0$$

$$Q(x) = b_n x^n + b_{n-1}x^{n-1} + \ldots + b_2 x^2 + b_1 x + b_0$$

where the (apparent) highest-degree coefficients a_m and b_n non-zero. Then in $P \cdot Q$ the highest-degree term is $a_m b_n x^{m+n}$. Since the product of non-zero elements of a field is non-zero, [79] the coefficient of x^{m+n} is non-zero. ///

4.2.3 Corollary: *(Cancellation property)* For polynomials in $k[x]$, with a field k, let $A \cdot P = B \cdot P$ for a non-zero polynomial P. Then $A = B$.

Proof: The equality $AP = BP$ gives $(A - B)P = 0$. Because the degree of the product is the sum of the degrees of the factors,

$$\deg(A - B) + \deg P = \deg 0 = -\infty$$

Since P is non-zero, $\deg P \geq 0$. Then $\deg(A - B) = -\infty$, so $A - B = 0$, and $A = B$. ///

4.2.4 Corollary: The group of units $k[x]^\times$ in the polynomial ring in one variable over a field k is just the group of units $k^\times$ in k. [80]

Proof: Suppose that $P \cdot Q = 1$. Then $\deg P + \deg Q = 0$, so both degrees are 0, that is, P and Q are in k. ///

A polynomial is **monic** if its highest degree coefficient is 1. Since elements of $k^\times$ are units in $k[x]$, any polynomial can be multiplied by a unit to make it monic.

4.2.5 Proposition: (*Euclidean property*) Let k be a field and M a non-zero polynomial in $k[x]$. Let H be any other polynomial in $k[x]$. Then there are unique polynomials Q and R in $k[x]$ such that $\deg R < \deg M$ and

$$H = Q \cdot M + R$$

Proof: [81] Let X be the set of polynomials expressible in the form $H - S \cdot M$ for some polynomial S. Let $R = H - Q \cdot M$ be an element of X of minimal degree. Claim that $\deg R < \deg M$. If not, let a be the highest-degree coefficient of R, let b be the highest-degree coefficient of M, and define

$$G = (ab^{-1}) \cdot x^{\deg R - \deg M}$$

Then

$$R - G \cdot M$$

removes the highest-degree term of R, and

$$\deg(R - G \cdot M) < \deg R$$

But $R - GM$ is still in X, since

$$R - G \cdot M = (H - Q \cdot M) - G \cdot M = H - (Q + G) \cdot M$$

[79] In case this is not clear: let a, b be elements of a field k with $ab = 0$ and $a \neq 0$. Since non-zero elements have inverses, there is a^{-1}, and $a^{-1}ab = a^{-1} \cdot 0 = 0$, but also $a^{-1}ab = b$. Thus, $b = 0$.

[80] We identify the scalars with degree-zero polynomials, as usual. If one is a bit worried about the legitimacy of this, the free-algebra definition of the polynomial ring can be invoked to prove this more formally.

[81] This argument is identical to that for the ordinary integers, as are many of the other proofs here.

By choice of R this is impossible, so, in fact, $\deg R < \deg M$. For uniqueness, suppose

$$H = Q \cdot M + R = Q' \cdot M + R'$$

Subtract to obtain

$$R - R' = (Q' - Q) \cdot M$$

Since the degree of a product is the sum of the degrees, and since the degrees of R, R' are less than the degree of M, this is impossible unless $Q' - Q = 0$, in which case $R - R' = 0$. ///

Compatibly with general terminology, a non-zero polynomial is **irreducible** if it has no proper divisors.

The **greatest common divisor** of two polynomials A, B is the *monic* polynomial g of highest degree dividing both A and B.

4.2.6 Theorem: For polynomials f, g in $k[x]$, the monic polynomial of the form $sf + tg$ (for $s, t \in k[x]$) of smallest degree is the *gcd* of f, g. In particular, greatest common divisors exist.

Proof: Among the non-negative integer values $\deg(sf + tg)$ there is at least one which is minimal. Let $h = sf + tg$ be such, and multiply through by the inverse of the highest-degree coefficient to make h monic. First, show that $h|f$ and $h|g$. We have

$$f = q(sf + tg) + r$$

with $\deg r < \deg(sf + tg)$. Rearranging,

$$r = (1 - qs)f + (-qt)g$$

So r itself is $s'f + t'g$ with $s', t' \in k[x]$. Since $sf + tg$ had the smallest non-negative degree of any such expression, and $\deg r < \deg(sf + tg)$, it must be that $r = 0$. So $sf + tg$ divides f. Similarly, $sf + tg$ divides g, so $sf + tg$ is a divisor of both f and g. On the other hand, if $d|f$ and $d|g$, then certainly $d|sf + tg$.
///

4.2.7 Corollary: Let P be an irreducible polynomial. For two other polynomials A, B, if $P|AB$ then $P|A$ or $P|B$. Generally, if an irreducible P divides a product $A_1 \dots A_n$ of polynomials then P must divide one of the factors A_i.

Proof: It suffices to prove that if $P|AB$ and $P \not| A$ then $P|B$. Since $P \not| A$, and since P is irreducible, the *gcd* of P and A is just 1. Therefore, there are $s, t \in k[x]$ so that

$$1 = sA + tP$$

Then

$$B = B \cdot 1 = B \cdot (sA + tP) = s(AB) + (Bt)P$$

Since $P|AB$, surely P divides the right-hand side. Therefore, $P|B$, as claimed.

4.2.8 Corollary: Irreducible polynomials P in $k[x]$ are prime, in the sense that $P|AB$ implies $P|A$ or $P|B$ for polynomials A and B.

Proof: Let $AB = M \cdot P$, and suppose that P does not divide A. Since P is irreducible, any proper factor is a unit, hence a non-zero constant. Thus, $\gcd(P, A) = 1$, and there are polynomials R, S such that $RP + SA = 1$. Then

$$B = B \cdot 1 = B \cdot (RP + SA) = P \cdot BR + S \cdot AB = P \cdot BR + S \cdot M \cdot P = P \cdot (BR + SM)$$

so B is a multiple of P. ///

4.2.9 Corollary: Any polynomial M in $k[x]$ has a unique factorization (up to ordering of factors) as

$$M = u \cdot P_1^{e_1} \dots P_\ell^{e_\ell}$$

where $u \in k^\times$ is a unit in $k[x]$, the P_i are distinct primes, and the exponents are positive integers.

Proof: [82] First prove existence by induction on degree. [83] Suppose some polynomial F admitted no such factorization. Then F is not irreducible (or the non-factorization is the factorization), so $R = A \cdot B$ with both of A, B of lower degree but not degree 0. By induction, both A and B have factorizations into primes.

Uniqueness is a sharper result, proven via the property that $P|AB$ implies $P|A$ or $P|B$ for prime P. As in the case of integers, given two alleged prime factorizations, any prime in one of them must be equal to a prime in the other, and by cancelling we do an induction to prove that all the primes are the same. ///

4.2.10 Proposition: *(Testing for linear factors)* A polynomial $f(x)$ with coefficients in a field k has a linear factor $x - a$ (with $a \in k$) if and only if $F(a) = 0$.

Proof: If $x - a$ is a factor, clearly $f(a) = 0$. On the other hand, suppose that $f(a) = 0$. Use the division algorithm to write

$$f(x) = Q(x) \cdot (x - a) + R$$

Since $\deg R < \deg(x - a) = 1$, R is a constant. Evaluate both sides at a to obtain

$$0 = f(a) = Q(a) \cdot (a - a) + R = Q(a) \cdot 0 + R = R$$

Therefore, $R = 0$ and $x - a$ divides $f(x)$. ///

4.2.11 Corollary: For polynomial P in $k[x]$, the equation $P(a) = 0$ has no more roots a than the degree of P.

Proof: By the proposition, a root gives a monic linear factor, and by unique factorization there cannot be more of these than the degree. ///

4.2.12 Example: With coefficients not in a field, the intuition that a polynomial equation has no more roots than its degree is inaccurate. For example, with coefficients in $\mathbb{Z}/15$, the equation

$$a^2 - 1 = 0$$

has the obvious roots ± 1, but also the roots 6 and 10. And there are two different factorizations in $(\mathbb{Z}/15)[x]$

$$x^2 - 1 = (x - 1)(x + 1) = (x - 6)(x - 10)$$

4.3 *Ideals*

Let R be a commutative ring with unit 1. An **ideal** in R is an additive subgroup I of R such that $R \cdot I \subset I$. That is, I is an R-submodule of R with (left) multiplication.

[82] It bears emphasizing that the argument here proves unique factorization from the propery of primes that $p|ab$ implies $p|a$ or $p|b$, which comes from the Euclidean property. There are many examples in which a unique factorization result does hold without Euclidean-ness, such as polynomial rings $k[x_1, \dots, x_n]$ in *several* variables over a field, but the argument is more difficult. See *Gauss' Lemma*.

[83] An induction on size completely analogous to the induction on size for the ordinary integers.

4.3.1 Example: One archetype is the following. In the ring $\mathbb{Z}$, for any fixed n, the set $n \cdot \mathbb{Z}$ of multiples of n is an ideal.

4.3.2 Example: Let $R = k[x]$ be the ring of polynomials in one variable x with coefficients in a field k. Fix a polynomial $P(x)$, and let $I \subset R$ be the set of polynomial multiples $M(x) \cdot P(x)$ of $P(x)$.

4.3.3 Example: Abstracting the previous two examples: fix $n \in R$. The set $I = n \cdot R = \{rn : r \in R\}$ of multiples of m is an ideal, the **principal ideal generated by** n. A convenient lighter notation is to write

$$\langle n \rangle = R \cdot n = \text{principal ideal generated by } n$$

4.3.4 Example: In any ring, the **trivial ideal** is $I = \{0\}$. An ideal is **proper** if it is neither the trivial ideal $\{0\}$ nor the whole ring R (which is also an ideal).

4.3.5 Example: If an ideal I contains a unit u in R, then $I = R$. Indeed, for any $r \in R$,

$$r = r \cdot 1 = r \cdot (u^{-1} \cdot u) \in r \cdot u^{-1} \cdot I \subset I$$

For two subsets X, Y of a *ring* R, write

$$X + Y = \{x + y : x \in X,\ y \in Y\}$$

and [84]

$$X \cdot Y = \{\text{finite sums } \sum_i x_i\, y_i : x_i \in X,\ y_i \in Y\}$$

In this notation, for an ideal I in a commutative ring R with 1 we have $R \cdot I = I$.

An integral domain in which every ideal is principal is a **principal ideal domain**. [85]

4.3.6 Corollary: [86] Every ideal I in $\mathbb{Z}$ is principal, that is, of the form $I = n \cdot \mathbb{Z}$. In particular, unless $I = \{0\}$, the integer n is the least positive element of I.

Proof: Suppose I is non-zero. Since I is closed under additive inverses, if I contains $x < 0$ then it also contains $-x > 0$. Let n be the least element of I. Let $x \in I$, take $q, r \in \mathbb{Z}$ with $0 \leq r < n$ such that

$$x = q \cdot n + r$$

Certainly qn is in I, and $-qn \in I$ also. Since $r = x - qn$, $r \in I$. Since n was the smallest positive element of I, $r = 0$. Thus, $x = qn \in n \cdot \mathbb{Z}$, as desired. ///

4.3.7 Corollary: [87] Let k be a field. Let $R = k[x]$ be the ring of polynomials in one variable x with coefficients in k. Then every ideal I in R is principal, that is, is of the form $I = k[x] \cdot P(x)$ for some polynomial P. In particular, $P(x)$ is the monic polynomial of smallest degree in I, unless $I = \{0\}$, in which case $P(x) = 0$.

[84] Note that here the notation $X \cdot Y$ has a different meaning than it does in group theory, since in the present context it is implied that we take all finite sums of products, not just products.

[85] If we do not assume that the ring is a domain, then we certainly may form the notion of *principal ideal ring*. However, the presence of zero divisors is a distraction.

[86] This is a corollary of the Euclidean-ness of $\mathbb{Z}$.

[87] This is a corollary of the Euclidean-ness of $\mathbb{Z}$.

Proof: If $I = \{0\}$, then certainly $I = k[x] \cdot 0$, and we're done. So suppose I is non-zero. Suppose that $Q(x) = a_n x^n + \ldots + a_0$ lies in I with $a_n \neq 0$. Since k is a field, there is an inverse a_n^{-1}. Then, since I is an ideal, the polynomial

$$P(x) = a_n^{-1} \cdot Q(x) = x^n + a_n^{-1} a_{n-1} x^{n-1} + \ldots + a_n^{-1} a_0$$

also lies in I. That is, there is indeed a *monic* polynomial of lowest degree of any element of the ideal. Let $x \in I$, and use the Division Algorithm to get $Q, R \in k[x]$ with $\deg R < \deg P$ and

$$x = Q \cdot P + R$$

Certainly $Q \cdot P$ is still in I, and then $-Q \cdot P \in I$ also. Since $R = x - Q \cdot P$, we conclude that $R \in I$. Since P was the monic polynomial in I of smallest degree, it must be that $R = 0$. Thus, $x = Q \cdot P \in n \cdot k[x]$, as desired. ///

4.3.8 Remark: The proofs of these two propositions can be abstracted to prove that every ideal in a Euclidean ring is principal.

4.3.9 Example: Let R be a commutative ring with unit 1, and fix two elements $x, y \in R$. Then

$$I = R \cdot x + R \cdot y = \{rx + sy : r, s \in R\}$$

is an ideal in R. The two elements x, y are the **generators** of I.

4.3.10 Example: Similarly, for fixed elements $x_1, \ldots, x_n$ of a commutative ring R, we can form an ideal

$$I = R \cdot x_1 + \ldots + R \cdot x_n$$

4.3.11 Example: To construct new, larger ideals from old, smaller ideals proceed as follows. Let I be an ideal in a commutative ring R. Let x be an element of R. Then let

$$J = R \cdot x + I = \{rx + i : r \in R, \ \ i \in I\}$$

Let's check that J is an ideal. First

$$0 = 0 \cdot x + 0$$

so 0 lies in J. Second,

$$-(rx + i) = (-r)x + (-i)$$

so J is closed under inverses. Third, for two elements $rx + i$ and $r'x + i'$ in J (with $r, r' \in R$ and $i, i' \in I$) we have

$$(rx + i) + (r'x + i') = (r + r')x + (i + i')$$

so J is closed under addition. Finally, for $rx + i \in J$ with $r \in R$, $i \in I$, and for $r' \in R$,

$$r' \cdot (rx + i) = (r'r)x + (r'i)$$

so $R \cdot J \subset J$ as required. Thus, this type of set J is indeed an ideal.

4.3.12 Remark: In the case of rings such as $\mathbb{Z}$, where we know that every ideal is principal, the previous construction does not yield any more general type of ideal.

4.3.13 Example: In some rings R, *not* every ideal is principal. We return to an example used earlier to illustrate a failure of unique factorization. Let

$$R = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$$

Let

$$I = \{x \cdot 2 + y \cdot (1 + \sqrt{-5}) : x, y \in R\}$$

These phenomena are not of immediate relevance, but did provide considerable motivation in the historical development of algebraic number theory.

4.4 *Ideals and quotient rings*

Here is a construction of new rings from old in a manner that includes as a special case the construction of $\mathbb{Z}/n$ from $\mathbb{Z}$.

Let R be a commutative ring with unit 1. Let I be an ideal in R. The **quotient ring** R/I is the set of cosets

$$r + I = \{r + i : i \in I\}$$

with operations of addition and multiplication on R/I by

$$(r + I) + (s + I) = (r + s) + I$$

$$(r + I) \cdot (s + I) = (r \cdot s) + I$$

The zero in the quotient is $0_{R/I} = 0 + I$, and the unit is $1_{R/I} = 1 + I$.

4.4.1 Example: The basic example is that $\mathbb{Z}/n$ is the quotient ring $\mathbb{Z}/I$ where $I = n \cdot \mathbb{Z}$.

4.4.2 Remark: It's tedious, but someone should check that the operations of addition and multiplication in $\mathbb{Z}/n$ are *well-defined*: we want the alleged addition and multiplication operations not to depend on the way the coset is *named*, but only on *what it is*. So suppose $r + I = r' + I$ and $s + I = s' + I$. We need to check that

$$(r + s) + I = (r' + s') + I$$

and to prove well-definedness of multiplication check that

$$(r \cdot s) + I = (r' \cdot s') + I$$

Since $r' + I = r + I$, in particular $r' = r' + 0 \in r + I$, so r' can be written as $r' = r + i$ for some $i \in I$. Likewise, $s' = s + j$ for some $j \in I$. Then

$$(r' + s') + I = (r + i + s + j) + I = (r + s) + (i + j + I)$$

The sum $k = i + j$ is an element of I. We claim that for any $k \in I$ we have $k + I = I$. Certainly since I is closed under addition, $k + I \subset I$. On the other hand, for any $x \in I$ we can write

$$x = k + (x - k)$$

with $x - k \in I$, so also $k + I \supset I$. Thus, indeed, $k + I = I$. Thus,

$$(r' + s') + I = (r + s) + I$$

which proves the well-definedness of addition in the quotient ring. Likewise, looking at multiplication:

$$(r' \cdot s') + I = (r + i) \cdot (s + j) + I = (r \cdot s) + (rj + si + I)$$

Since I is an ideal, rj and si are again in I, and then $rj + si \in I$. Therefore, as just observed in the discussion of addition, $rj + si + I = I$. Thus,

$$(r' \cdot s') + I = (r \cdot s) + I$$

and multiplication is well-defined. The proofs that $0 + I$ is the zero and $1 + I$ is the unit are similar.

The **quotient homomorphism**

$$q : R \longrightarrow R/I$$

is the natural map

$$q(r) = r + I$$

The definition and discussion above proves

4.4.3 Proposition: For a commutative ring R and ideal I, the quotient map $R \longrightarrow R/I$ is a (surjective) ring homomorphism. ///

4.5 *Maximal ideals and fields*

Now we see how to make fields by taking quotients of commutative rings by *maximal ideals* (defined just below). This is a fundamental construction.

Let R be a commutative ring with unit 1. [88] An ideal M in R is **maximal** if $M \neq R$ and if for any other ideal I with $I \supset M$ it must be that $I = R$. That is, M is a maximal ideal if there is no ideal strictly larger than M (containing M) except R itself.

4.5.1 Proposition: For a commutative ring R with unit, and for an ideal I, the quotient ring R/I is a *field* if and only if I is a *maximal* ideal.

Proof: Let $x + I$ be a non-zero element of R/I. Then $x + I \neq I$, so $x \notin I$. Note that the ideal $Rx + I$ is therefore strictly larger than I. Since I was already maximal, it must be that $Rx + I = R$. Therefore, there are $r \in R$ and $i \in I$ so that $rx + i = 1$. Looking at this last equation modulo I, we have $rx \equiv 1 \bmod I$. That is, $r + I$ is the multiplicative inverse to $x + I$. Thus, R/I is a field.

On the other hand, suppose that R/I is a field. Let $x \in R$ but $x \notin I$. Then $x + I \neq 0 + I$ in R/I. Therefore, $x + I$ has a multiplicative inverse $r + I$ in R/I. That is,

$$(r + I) \cdot (x + I) = 1 + I$$

From the definition of the multiplication in the quotient, this is $rx + I = 1 + I$, or $1 \in rx + I$, which implies that the ideal $Rx + I$ is R. But $Rx + I$ is the smallest ideal containing I and x. Thus, there cannot be any proper ideal strictly larger than I, so I is maximal. ///

4.6 *Prime ideals and integral domains*

Let R be a commutative ring with unit 1. An ideal P in R is **prime** if $ab \in P$ implies either $a \in P$ or $b \in P$. [89]

4.6.1 Proposition: For a commutative ring R with unit, and for an ideal I, the quotient ring R/I is an *integral domain* [90] if and only if I is a *prime* ideal.

[88] The commutativity allows us to avoid several technical worries which are not the current point, and the presence of 1 likewise skirts some less-than-primary problems. The applications we have in mind of the results of this section do not demand that we worry about those possibilities.

[89] Yes, by this point the property *proven* for prime numbers is taken to be the *definition*.

[90] Again, an integral domain has no zero divisors.

Proof: Let I be prime. Suppose that

$$(x+I)\cdot(y+I) = 0+I$$

Recall that the product in the quotient is not defined exactly as the set of products of elements from the factors, but, rather, in effect,

$$(x+I)\cdot(y+I) = xy+I$$

Then $(x+I)(y+I) = 0+I$ implies that $xy \in I$. By the prime-ness of I, either x or y is in I, so either $x+I = 0+I$ or $y+I = 0+I$.

On the other hand, suppose that R/I is an integral domain. Suppose that $xy \in I$. The definition $(x+I)(y+I) = xy+I$ then says that $(x+I)(y+I) = 0+I$. Since R/I is an integral domain, either $x+I = I$ or $y+I = I$. That is, either $x \in I$ or $y \in I$, and I is prime. ///

4.6.2 Corollary: Maximal ideals are prime. [91]

Proof: If I is a maximal ideal in a ring R, then R/I is a field, from above. Fields are certainly integral domains, so I is prime, from above. ///

4.6.3 Remark: Not all prime ideals are maximal.

4.6.4 Example: Let $R = \mathbb{Z}[x]$ be the polynomial ring in one variable with integer coefficients. Consider the ideal $I = \mathbb{Z}[x]\cdot x$ generated by x. We claim that this ideal is prime, but *not* maximal. Indeed,

$$R/I = \mathbb{Z}[x]/x\mathbb{Z}[x] \approx \mathbb{Z}$$

via the homomorphism

$$P(x)+I \longrightarrow P(0)$$

(One might verify that this map is indeed well-defined, and is a homomorphism.) [92] Since $\mathbb{Z} \approx \mathbb{Z}[x]/I$ is an integral domain, I is prime, but since $\mathbb{Z}$ is not a field, I is not maximal.

4.6.5 Example: Let $R = \mathbb{Z}[x]$ again and let $I = \mathbb{Z}[x]\cdot p$ be the ideal generated by a prime number p. Then

$$R/I = \mathbb{Z}[x]/p\mathbb{Z}[x] \approx (\mathbb{Z}/p)[x]$$

via the map

$$P(x) \longrightarrow (P \text{ with coefficients reduced mod } p)(x)$$

The ring $(\mathbb{Z}/p)[x]$ is an integral domain [93] but not a field, so the ideal is prime but not maximal.

4.6.6 Example: [94] Let k be a field, and consider the polynomial ring $R = k[x,y]$ in two variables. Let $I = k[x,y]\cdot x$ be the ideal generated by x. We claim that this ideal I is prime but not maximal. Indeed,

[91] ... in commutative rings with identity, at least.

[92] This can be verified in different styles. One style is the following. The universality of the polynomial ring assures us that there is a unique $\mathbb{Z}$-algebra homomorphism $e : \mathbb{Z}[x] \longrightarrow \mathbb{Z}$ which sends $x \longrightarrow 0$. Implicit in the $\mathbb{Z}$-algebra homomorphism property is that $n \longrightarrow n$ for $n \in \mathbb{Z}$, so no non-zero integers lie in the kernel of this *evaluation homomorphism* e. Thus, this homomorphism is a surjection to $\mathbb{Z}$.

[93] Since p is prime, $\mathbb{Z}/p$ is a field, so this is a polynomial ring in one variable over a field, which we know is an integral domain.

[94] While the conclusion of this example is correct, the most natural full proof that such things are what they seem requires results we do not yet have in hand, such as Gauss' Lemma.

the quotient R/I is [95] (naturally isomorphic to) $k[y]$ under the evaluation map

$$P(x,y) \longrightarrow P(0,y)$$

Since $k[xy]$ is an integral domain but not a field, we reach the conclusion, in light of the results just above.

4.7 *Fermat-Euler on sums of two squares*

4.7.1 Theorem: [96] A prime integer p is expressible as

$$p = a^2 + b^2$$

if and only if $p = 1 \bmod 4$ (or $p = 2$).

Proof: Parts of this are very easy. Certainly $2 = 1^2 + 1^2$. Also, if an odd [97] prime is expressible as $p = a^2 + b^2$, then, since the squares modulo 4 are just 0 and 1, it must be that one of a, b is odd and one is even, and the sum of the squares is 1 modulo 4.

On the other hand, suppose that $p = 1 \bmod 4$. If p were expressible as $p = a^2 + b^2$ then

$$p = (a + bi)(a - bi)$$

where $i = \sqrt{-1}$ in $\mathbb{C}$. That is, p is expressible as a sum of two squares, if and only if p factors in a particular manner in $\mathbb{Z}[i]$. One might have at some point already observed that the only units in $\mathbb{Z}[i]$ are ± 1 and $\pm i$, so if neither of a, b is 0, then neither of $a \pm bi$ is a unit. We need to analyze the possible factorization of p in $\mathbb{Z}[i]$ a little more closely to understand the close connection to the present issue.

Let $N(a + bi) = a^2 + b^2$ be the usual (square-of) norm. One can check that the only elements of $\mathbb{Z}[i]$ with norm 1 are the 4 units, and norm 0 occurs only for 0. If $p = \alpha \cdot \beta$ is a proper factorization, then by the multiplicative property of N

$$p^2 = N(p) = N(\alpha) \cdot N(\beta)$$

Thus, since neither α nor β is a unit, it must be that

$$N(\alpha) = p = N(\beta)$$

Similarly, α and β must both be irreducibles in $\mathbb{Z}[i]$, since applying N to any proper factorization would give a contradiction. Also, since p is its own complex conjugate,

$$p = \alpha \cdot \beta$$

[95] Certainly if we have a polynomial of the form $xf(x,y)$, replacing x by 0 gives the 0 polynomial in y. On the other hand, it is less clear that $f(0,y) = 0$ implies that f is of the form $f(x,y) = xg(x,y)$ for some polynomial g. The conceptual proof of results of this sort would use the unique factorization property of $k[x,y]$, which follows from the one-variable case via Gauss' lemma. For the present case, with the special factor x (rather than a more general polynomial), a direct approach is still easy. Let $f(x,y) = xg(x,y) + h(x)$ where $h(y)$ is the collection of all monomials in $f(x,y)$ in which x does not appear. Then $f(0,y) = h(y)$. If this is the 0 polynomial in y, then $f(x,y) = xg(x,y)$.

[96] Fermat stated in correspondence that he knew this, roughly around 1650, but there was no recorded argument. About 100 years later Euler reconsidered this and many other unsupported statements of Fermat's, and gave a proof that was publicly available. In this and other cases, it is not clear that Fermat was sufficiently aware of all the things that might go wrong to enable us to be sure that he had a complete proof. It is plausible, but not clear.

[97] The phrase *odd prime* is a standard if slightly peculiar way to refer to prime integers other than 2. Sometimes the import of this is that the prime is *larger* than 2, and sometimes it really is that the prime is *odd*.

implies

$$p = \overline{p} = \overline{\alpha} \cdot \overline{\beta}$$

Since we know that the Gaussian integers $\mathbb{Z}[i]$ are Euclidean and, hence, have unique factorization, it must be that these two prime factors are the same *up to units*. [98]

Thus, either $\alpha = \pm\overline{\alpha}$ and $\beta = \pm\overline{\beta}$ (with matching signs), or $\alpha = \pm i\overline{\alpha}$ and $\beta = \mp i\overline{\beta}$, or $\alpha = u\overline{\beta}$ with u among $\pm 1, \pm i$. If $\alpha = \pm\overline{\alpha}$, then α is either purely imaginary or is real, and in either case its norm is a square, but no square divides p. If $\alpha = \pm i\overline{\alpha}$, then α is of the form $t \pm it$ for $t \in \mathbb{Z}$, and then $N(\alpha) \in 2\mathbb{Z}$, which is impossible.

Thus, $\alpha = u\overline{\beta}$ for some unit u, and $p = uN(\beta)$. Since $p > 0$, it must be that $u = 1$. Letting $\alpha = a + bi$, we have recovered an expression as (proper) sum of two squares

$$p = a^2 + b^2$$

Thus, a prime integer p is a (proper) sum of two squares if and only if it is *not prime* in $\mathbb{Z}[i]$. From above, this is equivalent to

$$\mathbb{Z}[i]/p\mathbb{Z}[x] \text{ is not an integral domain}$$

We grant that for $p = 1 \bmod 4$ there is an integer α such that $\alpha^2 = -1 \bmod p$. [99] That is, (the image of) the polynomial $x^2 + 1$ factors in $\mathbb{Z}/p[x]$.

Note that we can rewrite $\mathbb{Z}[i]$ as

$$\mathbb{Z}[x]/(x^2 + 1)\mathbb{Z}[x]$$

We'll come back to this at the end of this discussion. Then [100]

$$\mathbb{Z}[i]/\langle p\rangle \approx \left(\mathbb{Z}[x]/\langle x^2+1\rangle\right)/\langle p\rangle$$

$$\approx \left(\mathbb{Z}[x]/\langle p\rangle\right)/\langle x^2+1\rangle \approx (\mathbb{Z}/p)[x]/\langle x^2+1\rangle$$

and the latter is *not* an integral domain, since

$$x^2 + 1 = (x - \alpha)(x + \alpha)$$

is not irreducible in $(\mathbb{Z}/p)[x]$. That is, $\mathbb{Z}[i]/\langle p\rangle$ is not an integral domain when p is a prime with $p = 1 \bmod 4$. That is, p is not irreducible in $\mathbb{Z}[i]$, so factors properly in $\mathbb{Z}[i]$, thus, as observed above, p is a sum of two squares. ///

4.7.2 Remark: Let's follow up on the isomorphism

$$\mathbb{Z}[x]/\langle x^2 + 1\rangle \approx \mathbb{Z}[i]$$

[98] This *up to units* issue is nearly trivial in $\mathbb{Z}$, since positivity and negativity give us a convenient handle. But in $\mathbb{Z}[i]$ and other rings with more units, greater alertness is required.

[99] If we grant that there are primitive roots modulo primes, that is, that $(\mathbb{Z}/p)^\times$ is cyclic, then this assertion follows from basic and general properties of cyclic groups. Even without knowledge of primitive roots, we can still give a special argument in this limited case, as follows. Let $G = (\mathbb{Z}/p)^\times$. This group is abelian, and has order divisible by at least 2^2. Thus, for example by Sylow theorems, there is a 2-power-order subgroup A of order at least 4. By unique factorization in polynomial rings, the equation $x^2 - 1 = 0$ has only the solutions ± 1. Thus, there is only a *single* element in A of order 2, and the identity 1 of order 1. Other elements in A must have order a larger power of 2, and then one can arrange elements of order 4. Such things would be 4th roots of 1.

[100] A scrupulous reader should verify that the change in order of quotient-taking is legitimate. It is certainly a good trick, assuming that it works properly.

Since $\mathbb{Z}[x]$ is the free $\mathbb{Z}$-algebra on the generator x, there is a unique $\mathbb{Z}$-algebra homomorphism $\mathbb{Z}[x] \longrightarrow \mathbb{Z}[i]$ taking x to i. We claim that the kernel is identifiable as the principal ideal generated by x^2+1, after which the obvious isomorphism theorem for rings would yield the desired isomorphism.

That is, we claim that if a polynomial $P(x)$ in $\mathbb{Z}[x]$ has the property that $P(i) = 0$, then P is a multiple (in $\mathbb{Z}[x]$) of x^2+1. This is less trivial than in the case of polynomials in one variable over a *field*, but the fact that x^2+1 is *monic* saves us. That is, we can claim that for a *monic* poly $M(x)$, given any other polynomial $P(x) \in \mathbb{Z}[x]$, there are $Q(x)$ and $R(x)$ in $\mathbb{Z}[x]$ with $\deg R < \deg M$, such that

$$P = Q \cdot M + R$$

Indeed, suppose not. Let

$$P(x) = a_n x^n + \ldots + a_0$$

be the polynomial of least degree n which we cannot divide by M and obtain a smaller remainder. Let $m = \deg M$. Necessarily $n \geq m$ or P is itself already of lower degree than M. And, for $n \geq m$,

$$P - a_n \cdot x^{n-m} \cdot M$$

is of strictly lower degree than P, so is expressible as $QM + R$. Then

$$P = (Q + a_n x^{n-m}) \cdot M + R$$

Since the degree of R was of degree at most $n-1$, which is strictly less than n, this contradicts the supposition that P had no such expression.

4.8 *Worked examples*

4.8.1 Example: Let $R = \mathbb{Z}/13$ and $S = \mathbb{Z}/221$. Show that the map

$$f : R \longrightarrow S$$

defined by $f(n) = 170 \cdot n$ is *well-defined* and is a ring homomorphism. (Observe that it does not map $1 \in R$ to $1 \in S$.)

The point is that $170 = 1 \bmod 13$ and $170 = 17 \cdot 10 = 0 \bmod 17$, and $221 = 13 \cdot 17$. Thus, for $n' = n + 13\ell$,

$$170 \cdot n' = 17 \cdot 10 \cdot n + 10 \cdot 17 \cdot 13 = 17 \cdot 10 \cdot n \bmod 13 \cdot 17$$

so the map is well-defined. Certainly the map respects addition, since

$$170(n + n') = 170n + 170n'$$

That it respects multiplication is slightly subtler, but we verify this separately modulo 13 and modulo 17, using unique factorization to know that if $13|N$ and $17|N$ then $(13 \cdot 17)|N$. Thus, since $170 = 1 \bmod 13$,

$$170(nn') = 1 \cdot (nn') = nn' = (170n) \cdot (170n') \bmod 13$$

And, since $17 = 0 \bmod 17$,

$$170(nn') = 0 \cdot (nn') = 0 = (170n) \cdot (170n') \bmod 17$$

Putting these together gives the multiplicativity.

4.8.2 Example: Let p and q be distinct prime numbers. Show directly that there is no field with pq elements.

There are several possible approaches. One is to suppose there exists such a field k, and first invoke Sylow (or even more elementary results) to know that there exist (non-zero!) elements x, y in k with (additive) orders p, q, respectively. That is, $p \cdot x = 0$ (where left multiplication by an ordinary integer means repeated addition). Then claim that $xy = 0$, contradicting the fact that a field (or even integral domain) has no proper zero divisors. Indeed, since p and q are distinct primes, $\gcd(p, q) = 1$, so there are integers r, s such that $rp + sq = 1$. Then

$$xy = 1 \cdot xy = (rp + sq) \cdot xy = ry \cdot px + sx \cdot qy = ry \cdot 0 + sx \cdot 0 = 0$$

4.8.3 Example: Find all the idempotent elements in $\mathbb{Z}/n$.

The idempotent condition $r^2 = r$ becomes $r(r-1) = 0$. For each prime p dividing n, let p^e be the exact power of p dividing n. For the image in $\mathbb{Z}/n$ of an ordinary integer b to be idempotent, it is necessary and sufficient that $p^e | b(b-1)$ for each prime p. Note that p cannot divide both b and $b-1$, since $b - (b-1) = 1$. Thus, the condition is $p^e | b$ *or* $p^e | b-1$, for each prime p dividing n. Sun-Ze's theorem assures that we can choose either of these two conditions for each p as p various over primes dividing n, and be able to find a simultaneous solution for the resulting family of congruences. That is, let $p_1, \ldots, p_t$ be the distinct primes dividing n, and let $p_i^{e_i}$ be the exact power of p_i dividing n. For each p_i choose $\varepsilon_i \in \{0, 1\}$. Given a sequence $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_t)$ of 0s and 1s, consider the collection of congruences $p_i^{e_i} | (b - \varepsilon_i)$, for $i = 1, \ldots, t$. Sun-Ze guarantees that there is a solution, and that it is unique mod n. Thus, each of the 2^t choices of sequences of 0s and 1s gives an idempotent.

4.8.4 Example: Find all the nilpotent elements in $\mathbb{Z}/n$.

For each prime p dividing n, let p^e be the exact power of p dividing n. For the image in $\mathbb{Z}/n$ of an ordinary integer b to be nilpotent, it is necessary and sufficient that for some n sufficiently large $p^e | b^n$ for each prime p. Then surely $p | b^n$, and since p is prime $p | b$. And, indeed, if every prime dividing n divides b, then a sufficiently large power of b will be 0 modulo p^e, hence (by unique factorization, etc.) modulo n. That is, for b to be nilpotent it is necessary and sufficient that every prime dividing n divides b.

4.8.5 Example: Let $R = \mathbb{Q}[x]/(x^2 - 1)$. Find e and f in R, neither one 0, such that

$$e^2 = e \quad f^2 = f \quad ef = 0 \quad e + f = 1$$

(Such e and f are **orthogonal** idempotents.) Show that the maps $p_e(r) = re$ and $p_f(r) = rf$ are ring homomorphisms of R to itself.

Let ξ be the image of x in the quotient. Then $(\xi - 1)(\xi + 1) = 0$. Also note that

$$(\xi - 1)^2 = \xi^2 - 2\xi + 1 = (\xi^2 - 1) - 2\xi + 2 = -2\xi + 2$$

so

$$\left(\frac{\xi - 1}{2}\right)^2 = \frac{\xi^2 - 2\xi + 1}{4} = \frac{(\xi^2 - 1) - 2\xi + 2}{4} = \frac{-\xi + 1}{2}$$

Similarly,

$$\left(\frac{\xi + 1}{2}\right)^2 = \frac{\xi^2 + 2\xi + 1}{4} = \frac{(\xi^2 - 1) + 2\xi + 2}{4} = \frac{\xi + 1}{2}$$

Thus, $e = (-\xi + 1)/2$ and $f = (\xi + 1)/2$ are the desired orthogonal idempotents.

4.8.6 Example: Prove that in $(\mathbb{Z}/p)[x]$ we have the factorization

$$x^p - x = \prod_{a \in \mathbb{Z}/p} (x - a)$$

By Fermat's Little Theorem, the left-hand side is 0 when x is replaced by any of $0, 1, 2, \ldots, p-1$. Thus, by unique factorization in $k[x]$ for k a field (which applies to $\mathbb{Z}/p$ since p is prime), all the factors $x-0$, $x-1$, $x-2$, ..., $x-(p-1)$ divide the left-hand side, and (because these are mutually relatively prime) so does their product. Their product is the right-hand side, which thus at least *divides* the left-hand side. Since degrees add in products, we see that the right-hand side and left-hand side could differ at most by a unit (a polynomial of degree 0), but both are *monic*, so they are identical, as claimed.

4.8.7 Example: Let $\omega = (-1+\sqrt{-3})/2$. Prove that

$$\mathbb{Z}[\omega]/p\mathbb{Z}[\omega] \approx (\mathbb{Z}/p)[x]/(x^2+x+1)(\mathbb{Z}/p)[x]$$

and, as a consequence, that a prime p in $\mathbb{Z}$ is expressible as x^2+xy+y^2 with integers x, y if and only if $p = 1 \bmod 3$ (apart from the single anomalous case $p = 3$).

If a prime is expressible as $p = a^2+ab+b^2$, then, modulo 3, the possibilities for p modulo 3 can be enumerated by considering $a = 0, \pm 1$ and $b = 0, \pm 1 \bmod 3$. Noting the symmetry that $(a,b) \longrightarrow (-a,-b)$ does not change the output (nor does $(a,b) \longrightarrow (b,a)$) we reduce from $3 \cdot 3 = 9$ cases to a smaller number:

$$p = a^2+ab+b^2 = \begin{cases} 0^2+0\cdot 0+0^2 &= 1 \mod 3 \\ 1^2+1\cdot 1+1^2 &= 0 \mod 3 \\ 1^2+1\cdot(-1)+(-1)^2 &= 1 \mod 3 \end{cases}$$

Thus, any prime p expressible as $p = a^2+ab+b^2$ is either 3 or is 1 mod 3.

On the other hand, suppose that $p = 1 \bmod 3$. If p were expressible as $p = a^2+ab+b^2$ then

$$p = (a+b\omega)(a+b\overline{\omega})$$

where $\omega = (-1+\sqrt{-3})/2$. That is, p is expressible as a^2+ab+b^2 if and only if p factors in a particular manner in $\mathbb{Z}[\omega]$.

Let $N(a+b\omega) = a^2+ab+b^2$ be the usual (square-of) norm. To determine the units in $\mathbb{Z}[\omega]$, note that $\alpha \cdot \beta = 1$ implies that

$$1 = N(\alpha) \cdot N(\beta)$$

and these norms from $\mathbb{Z}[\omega]$ are integers, so units have norm 1. By looking at the equation $a^2+ab+b^2 = 1$ with integers a, b, a little fooling around shows that the only units in $\mathbb{Z}[\omega]$ are ± 1, $\pm\omega$ and $\pm\omega^2$. And norm 0 occurs only for 0.

If $p = \alpha \cdot \beta$ is a proper factorization, then by the multiplicative property of N

$$p^2 = N(p) = N(\alpha) \cdot N(\beta)$$

Thus, since neither α nor β is a unit, it must be that

$$N(\alpha) = p = N(\beta)$$

Similarly, α and β must both be irreducibles in $\mathbb{Z}[\omega]$, since applying N to any proper factorization would give a contradiction. Also, since p is its own complex conjugate,

$$p = \alpha \cdot \beta$$

implies

$$p = \overline{p} = \overline{\alpha} \cdot \overline{\beta}$$

Since we know that the (Eisenstein) integers $\mathbb{Z}[\omega]$ are Euclidean and, hence, have unique factorization, it must be that these two prime factors are the same *up to units*.

Thus, either $\alpha = \pm\overline{\alpha}$ and $\beta = \pm\overline{\beta}$ (with matching signs), or $\alpha = \pm\omega\overline{\alpha}$ and $\beta = \pm\omega^2\overline{\beta}$, or $\alpha = \pm\omega^2\overline{\alpha}$ and $\beta = \pm\omega\overline{\beta}$, or $\alpha = u\overline{\beta}$ with u among $\pm 1, \pm\omega, \pm\omega^2$. If $\alpha = \pm\overline{\alpha}$, then α is either in $\mathbb{Z}$ or of the form $t \cdot \sqrt{-3}$ with $t \in \mathbb{Z}$. In the former case its norm is a square, and in the latter its norm is divisible by 3, neither of which can occur. If $\overline{\alpha} = \omega\alpha$, then $\alpha = t \cdot \omega$ for some $t \in \mathbb{Z}$, and its norm is a square, contradiction. Similarly for $\alpha = \pm\omega^2\overline{\alpha}$.

Thus, $\alpha = u\overline{\beta}$ for some unit u, and $p = uN(\beta)$. Since $p > 0$, it must be that $u = 1$. Letting $\alpha = a + b\omega$, we have recovered an expression

$$p = a^2 + ab + b^2$$

with neither a nor b zero.

Thus, a prime integer $p > 3$ is expressible (properly) as $a^2 + ab + b^2$ of two squares if and only if it is *not prime* in $\mathbb{Z}[\omega]$. From above, this is equivalent to

$$\mathbb{Z}[\omega]/\langle p\rangle \text{ is not an integral domain}$$

We grant that for $p = 1 \bmod 3$ there is an integer α such that $\alpha^2 + alf + 1 = 0 \bmod p$. [101] That is, (the image of) the polynomial $x^2 + x + 1$ factors in $(\mathbb{Z}/p)[x]$.

Note that we can rewrite $\mathbb{Z}[\omega]$ as

$$\mathbb{Z}[x]/\langle x^2 + x + 1\rangle$$

Then

$$\mathbb{Z}[\omega]/\langle p\rangle \approx \left(\mathbb{Z}[x]/\langle x^2+1\rangle\right)/\langle p\rangle \approx \left(\mathbb{Z}[x]/\langle p\rangle\right)/\langle x^2+1\rangle \approx (\mathbb{Z}/p)[x]/\langle x^2+1\rangle$$

and the latter is *not* an integral domain, since

$$x^2 + x + 1 = (x - \alpha)(x - \alpha^2)$$

is not irreducible in $(\mathbb{Z}/p)[x]$. That is, $\mathbb{Z}[\omega]/\langle p\rangle$ is not an integral domain when p is a prime with $p = 1 \bmod 3$. That is, p is not irreducible in $\mathbb{Z}[\omega]$, so factors properly in $\mathbb{Z}[\omega]$, thus, as observed above, p is expressible as $a^2 + ab + b^2$. ///

Exercises

4.1 Show that in a commutative ring the set of nilpotent elements is an ideal (the **nilradical** of R). Give an example to show that the set of nilpotent elements may fail to be an ideal in a non-commutative ring.

4.2 Let R be a commutative ring with unit, such that for every $r \in R$ there is an integer $n > 1$ (possibly depending upon r) such that $r^n = r$. Show that every prime ideal in R is maximal.

4.3 Let k be a field. Let P, Q be two polynomials in $k[x]$. Let K be an extension field of k. Show that, if P divides Q in $K[x]$, then P divides Q in $k[x]$.

4.4 Let R be a commutative ring with unit. Show that the set of prime ideals in R has minimal elements under the ordering by inclusion. (*Hint:* You may want to use Zorn's lemma or some other equivalent of the Axiom of Choice.)

[101] If we grant that there are primitive roots modulo primes, that is, that $(\mathbb{Z}/p)^\times$ is cyclic, then this assertion follows from basic and general properties of cyclic groups. Even without knowledge of primitive roots, we can still give a special argument in this limited case, as follows. Let $G = (\mathbb{Z}/p)^\times$. This group is abelian, and has order divisible by 3. Thus, for example by Sylow theorems, there is a 3-power-order subgroup A, and, thus, at least one element of order exactly 3.

4.5 The **radical** of an ideal I in a commutative ring R with unit is

$$\operatorname{rad} I = \{r \in R : r^n \in I \text{ for some } n\}$$

Show that a proper ideal I of a ring is equal to its own radical if and only if it is an intersection of prime ideals.

4.6 Let R be a commutative ring with unit. Check that the **nilradical** N of R, defined to be the set of all nilpotent elements, is

$$\operatorname{nilrad} R = \operatorname{rad}\{0\}$$

Show that R has a *unique* prime ideal if and only if every element of R is either nilpotent or a unit, if and only if R/N is a field.

4.7 Show that a prime p in $\mathbb{Z}$ is expressible as $p = m^2 + 2n^2$ with integers m, n if and only if -2 is a square mod p.

4.8 Let R be a commutative ring with unit. Suppose R contains an *idempotent* element r other than 0 or 1. (That is, $r^2 = r$.) Show that every prime ideal in R contains an idempotent other than 0 or 1.

5. Linear Algebra I: Dimension

5.1 Some simple results
5.2 Bases and dimension
5.3 Homomorphisms and dimension

5.1 *Some simple results*

Several observations should be made. Once stated explicitly, the proofs are easy. [102]

• The intersection of a (non-empty) set of subspaces of a vector space V is a subspace.

Proof: Let $\{W_i : i \in I\}$ be a set of subspaces of V. For w in every W_i, the additive inverse $-w$ is in W_i. Thus, $-w$ lies in the intersection. The same argument proves the other properties of subspaces. ///

The **subspace spanned** by a set X of vectors in a vector space V is the intersection of all subspaces containing X. From above, this intersection is a subspace.

• The subspace spanned by a set X in a vector space V is the collection of all linear combinations of vectors from X.

Proof: Certainly every linear combination of vectors taken from X is in any subspace containing X. On the other hand, we must show that any vector in the intersection of subspaces containing X is a linear combination of vectors in X. Now it is not hard to check that the collection of such linear combinations is *itself* a subspace of V, and contains X. Therefore, the intersection is no larger than this set of linear combinations. ///

[102] At the beginning of the abstract form of this and other topics, there are several results which have little informational content, but, rather, only serve to assure us that the definitions/axioms have not included phenomena too violently in opposition to our expectations. This is not surprising, considering that the definitions have endured several decades of revision exactly to address foundational and other potential problems.

A *linearly independent* set of vectors *spanning* a subspace W of V is a **basis** for W.

5.1.1 Proposition: Given a basis $e_1, \ldots, e_n$ for a vector space V, there is *exactly one* expression for an arbitrary vector $v \in V$ as a linear combination of $e_1, \ldots, e_n$.

Proof: That there is *at least* one expression follows from the spanning property. On the other hand, if

$$\sum_i a_i e_i = v = \sum_i b_i e_i$$

are two expressions for v, then subtract to obtain

$$\sum_i (a_i - b_i) e_i = 0$$

Since the e_i are linearly independent, $a_i = b_i$ for all indices i. ///

5.2 *Bases and dimension*

The argument in the proof of the following fundamental theorem is the *Lagrange replacement principle*. This is the first non-trivial result in linear algebra.

5.2.1 Theorem: Let $v_1, \ldots, v_m$ be a linearly independent set of vectors in a vector space V, and let $w_1, \ldots, w_n$ be a basis for V. Then $m \leq n$, and (renumbering the vectors w_i if necessary) the vectors

$$v_1, \ldots, v_m, w_{m+1}, w_{m+2}, \ldots, w_n$$

are a basis for V.

Proof: Since the w_i's are a basis, we may express v_1 as a linear combination

$$v_1 = c_1 w_1 + \ldots + c_n w_n$$

Not all coefficients can be 0, since v_1 is not 0. Renumbering the w_i's if necessary, we can assume that $c_1 \neq 0$. Since the scalars k are a *field*, we can express w_1 in terms of v_1 and $w_2, \ldots, w_n$

$$w_1 = c_1^{-1} v_1 + (-c_1^{-1} c_2) w_2 + \ldots + (-c_1^{-1} c_2) w_n$$

Replacing w_1 by v_1, the vectors $v_1, w_2, w_3, \ldots, w_n$ span V. They are still linearly independent, since if v_1 were a linear combination of $w_2, \ldots, w_n$ then the expression for w_1 in terms of $v_1, w_2, \ldots, w_n$ would show that w_1 was a linear combination of $w_2, \ldots, w_n$, contradicting the linear independence of $w_1, \ldots, w_n$.

Suppose inductively that $v_1, \ldots, v_i, w_{i+1}, \ldots, w_n$ are a basis for V, with $i < n$. Express v_{i+1} as a linear combination

$$v_{i+1} = a_1 v_1 + \ldots + a_i v_i + b_{i+1} w_{i+1} + \ldots + b_n w_n$$

Some b_j is non-zero, or else v_i is a linear combination of $v_1, \ldots, v_i$, contradicting the linear independence of the v_j's. By renumbering the w_j's if necessary, assume that $b_{i+1} \neq 0$. Rewrite this to express w_{i+1} as a linear combination of $v_1, \ldots, v_i, w_{i+1}, \ldots, w_n$

$$w_{i+1} = (-b_{i+1}^{-1} a_1) v_1 + \ldots + (-b_{i+1}^{-1} a_i) v_i + (b_{i+1}^{-1}) v_{i+1}$$

$$+ (-b_{i+1}^{-1} b_{i+2}) w_{i+2} + \ldots + (-b_{i+1}^{-1} b_n) w_n$$

Thus, $v_1, \ldots, v_{i+1}, w_{i+2}, \ldots, w_n$ span V. Claim that these vectors are linearly independent: if for some coefficients a_j, b_j

$$a_1 v_1 + \ldots + a_{i+1} v_{i+1} + b_{i+2} w_{i+2} + \ldots + b_n w_n = 0$$

then some a_{i+1} is non-zero, because of the linear independence of $v_1, \ldots, v_i, w_{i+1}, \ldots, w_n$. Thus, rearrange to express v_{i+1} as a linear combination of $v_1, \ldots, v_i, w_{i+2}, \ldots, w_n$. The expression for w_{i+1} in terms of $v_1, \ldots, v_i, v_{i+1}, w_{i+2}, \ldots, w_n$ becomes an expression for w_{i+1} as a linear combination of $v_1, \ldots, v_i, w_{i+2}, \ldots, w_n$. But this would contradict the (inductively assumed) linear independence of $v_1, \ldots, v_i, w_{i+1}, w_{i+2}, \ldots, w_n$.

Consider the possibility that $m > n$. Then, by the previous argument, $v_1, \ldots, v_n$ is a basis for V. Thus, v_{n+1} is a linear combination of $v_1, \ldots, v_n$, contradicting their linear independence. Thus, $m \leq n$, and $v_1, \ldots, v_m, w_{m+1}, \ldots, w_n$ is a basis for V, as claimed. ///

Now define the (k-)**dimension** [103] of a vector space (over field k) as the number of elements in a (k-)basis. The theorem says that this number is well-defined. Write

$$\dim V = \text{ dimension of } V$$

A vector space is **finite-dimensional** if it has a finite basis. [104]

5.2.2 Corollary: A linearly independent set of vectors in a finite-dimensional vector space can be augmented to be a basis.

Proof: Let $v_1, \ldots, v_m$ be as linearly independent set of vectors, let $w_1, \ldots, w_n$ be a basis, and apply the theorem. ///

5.2.3 Corollary: The dimension of a *proper* subspace of a finite-dimensional vector space is strictly less than the dimension of the whole space.

Proof: Let $w_1, \ldots, w_m$ be a basis for the subspace. By the theorem, it can be extended to a basis $w_1, \ldots, w_m, v_{m+1}, \ldots, v_n$ of the whole space. It must be that $n > m$, or else the subspace is the whole space. ///

5.2.4 Corollary: The dimension of k^n is n. The vectors

$$\begin{array}{rcl} e_1 & = & (1, 0, 0, \ldots, 0, 0) \\ e_2 & = & (0, 1, 0, \ldots, 0, 0) \\ e_3 & = & (0, 0, 1, \ldots, 0, 0) \\ & \ldots & \\ e_n & = & (0, 0, 0, \ldots, 0, 1) \end{array}$$

are a basis (the **standard basis**).

Proof: Those vectors *span* k^n, since

$$(c_1, \ldots, c_n) = c_1 e_1 + \ldots + c_n e_n$$

[103] This is an instance of terminology that is nearly too suggestive. That is, a naive person might all too easily accidentally assume that there is a connection to the colloquial sense of the word *dimension*, or that there is an appeal to physical or visual intuition. Or one might assume that it is somehow *obvious* that *dimension* is a well-defined invariant.

[104] We proved only the finite-dimensional case of the well-definedness of dimension. The infinite-dimensional case needs *transfinite induction* or an equivalent.

On the other hand, a linear dependence relation

$$0 = c_1 e_1 + \ldots + c_n e_n$$

gives

$$(c_1, \ldots, c_n) = (0, \ldots, 0)$$

from which each c_i is 0. Thus, these vectors are a basis for k^n. ///

5.3 *Homomorphisms and dimension*

Now we see how dimension behaves under homomorphisms.

Again, a vector space **homomorphism** [105] $f : V \longrightarrow W$ from a vector space V over a field k to a vector space W over the same field k is a function f such that

$$\begin{array}{rcll} f(v_1 + v_2) & = & f(v_1) + f(v_2) & (\text{for all } v_1, v_2 \in V) \\ f(\alpha \cdot v) & = & \alpha \cdot f(v) & (\text{for all } \alpha \in k,\ v \in V) \end{array}$$

The **kernel** of f is

$$\ker f = \{v \in V : f(v) = 0\}$$

and the **image** of f is

$$\mathrm{Im}\, f = \{f(v) : v \in V\}$$

A homomorphism is an **isomorphism** if it has a two-sided inverse homomorphism. For vector spaces, a homomorphism that is a bijection is an isomorphism. [106]

• A vector space homomorphism $f : V \longrightarrow W$ sends 0 (in V) to 0 (in W, and, for $v \in V$, $f(-v) = -f(v)$. [107]

5.3.1 Proposition: The kernel and image of a vector space homomorphism $f : V \longrightarrow W$ are vector subspaces of V and W, respectively.

Proof: Regarding the kernel, the previous proposition shows that it contains 0. The last bulleted point was that additive inverses of elements in the kernel are again in the kernel. For $x, y \in \ker f$

$$f(x + y) = f(x) + f(y) = 0 + 0 = 0$$

so $\ker f$ is closed under addition. For $\alpha \in k$ and $v \in V$

$$f(\alpha \cdot v) = \alpha \cdot f(v) = \alpha \cdot 0 = 0$$

so $\ker f$ is closed under scalar multiplication. Thus, the kernel is a vector subspace.

Similarly, $f(0) = 0$ shows that 0 is in the image of f. For $w = f(v)$ in the image of f and $\alpha \in k$

$$\alpha \cdot w = \alpha \cdot f(v) = f(\alpha v) \in \mathrm{Im}\, f$$

[105] Or **linear map** or **linear operator**.

[106] In most of the situations we will encounter, bijectivity of various sorts of homomorphisms is sufficient (and certainly necessary) to assure that there is an inverse map of the same sort, justifying this description of *isomorphism*.

[107] This follows from the analogous result for groups, since V with its additive structure is an abelian group.

For $x = f(u)$ and $y = f(v)$ both in the image of f,

$$x + y = f(u) + f(v) = f(u + v) \in \mathrm{Im}\, f$$

And from above

$$f(-v) = -f(v)$$

so the image is a vector subspace. ///

5.3.2 Corollary: A linear map $f : V \longrightarrow W$ is injective if and only if its kernel is the trivial subspace $\{0\}$.

Proof: This follows from the analogous assertion for groups. ///

5.3.3 Corollary: Let $f : V \longrightarrow W$ be a vector space homomorphism, with V finite-dimensional. Then

$$\dim \ker f + \dim \mathrm{Im}\, f = \dim V$$

Proof: Let $v_1, \ldots, v_m$ be a basis for $\ker f$, and, invoking the theorem, let $w_{m+1}, \ldots, w_n$ be vectors in V such that $v_1, \ldots, v_m, w_{m+1}, \ldots, w_n$ form a basis for V. We claim that the images $f(w_{m+1}), \ldots, f(w_n)$ are a basis for $\mathrm{Im}\, f$. First, show that these vectors *span*. For $f(v) = w$, express v as a linear combination

$$v = a_1 v_1 + \ldots + a_m v_m + b_{m+1} w_{m+1} + \ldots + b_n w_n$$

and apply f

$$\begin{aligned} w &= a_1 f(v_1) + \ldots + a_m f(v_m) + b_{m+1} f(w_{m+1}) + \ldots + b_n f(w_n) \\ &= a_1 \cdot 0 + \ldots + a_m \cdot 0(v_m) + b_{m+1} f(w_{m+1}) + \ldots + b_n f(w_n) \\ &= b_{m+1} f(w_{m+1}) + \ldots + b_n f(w_n) \end{aligned}$$

since the v_is are in the kernel. Thus, the $f(w_j)$'s *span* the image. For linear independence, suppose

$$0 = b_{m+1} f(w_{m+1}) + \ldots + b_n f(w_n)$$

Then

$$0 = f(b_{m+1} w_{m+1} + \ldots + b_n w_n)$$

Then, $b_{m+1} w_{m+1} + \ldots + b_n w_n$ would be in the kernel of f, so would be a linear combination of the v_i's, contradicting the fact that $v_1, \ldots, v_m, w_{m+1}, \ldots, w_n$ is a basis, unless all the b_j's were 0. Thus, the $f(w_j)$ are linearly independent, so are a basis for $\mathrm{Im}\, f$. ///

Exercises

5.1 For subspaces V, W of a vector space over a field k, show that

$$\dim_k V + \dim_k W \;=\; \dim_k (V + W) + \dim_k (V \cap W)$$

5.2 Given two bases $e_1, \ldots, e_n$ and $f_1, \ldots, f_n$ for a vector space V over a field k, show that there is a unique k-linear map $T : V \longrightarrow V$ such that $T(e_i) = f_i$.

5.3 Given a basis $e_1, \ldots, e_n$ of a k-vectorspace V, and given arbitrary vectors $w_1, \ldots, w_n$ in a k-vectorspace W, show that there is a unique k-linear map $T : V \longrightarrow W$ such that $Te_i = w_i$ for all indices i.

5.4 The space $\mathrm{Hom}_k(V, W)$ of k-linear maps from one k-vectorspace V to another, W, is a k-vectorspace under the operation

$$(\alpha \dot{T})(v) = \alpha \cdot (T(v))$$

for $\alpha \in k$ and $T \in \mathrm{Hom}_k(V, W)$. Show that

$$\dim_k \mathrm{Hom}_k(V, W) = \dim_k V \cdot \dim_k W$$

5.5 A **flag** $V_1 \subset \ldots \subset V_\ell$ of subspaces of a k-vectorspace V is simply a collection of subspaces satisfying the indicated inclusions. The **type** of the flag is the list of *dimensions* of the subspaces V_i. Let W be a k-vectorspace, with a flag $W_1 \subset \ldots \subset W_\ell$ of the same *type* as the flag in V. Show that there exists a k-linear map $T : V \longrightarrow W$ such that T restricted to V_i is an isomorphism $V_i \longrightarrow W_i$.

5.6 Let $V_1 \subset V_\ell$ be a flag of subspace inside a finite-dimensional k-vectorspace V, and $W_1 \subset \ldots \subset W_\ell$ a flag inside another finite-dimensional k-vectorspace W. We do not suppose that the two flags are of the same type. Compute the dimension of the space of k-linear homomorphisms $T : V \longrightarrow W$ such that $TV_i \subset W_i$.

6. Fields I

6.1 Adjoining things
6.2 Fields of fractions, fields of rational functions
6.3 Characteristics, finite fields
6.4 Algebraic field extensions
6.5 Algebraic closures

6.1 *Adjoining things*

The general *intention* of the notion of *adjoining* a new element α to a field k is arguably clear. That is, k itself does *not* contain a root of an equation, for example, and we want to enlarge k so that it *does* include such a root. On the face of it, the possibility or legitimacy of doing so depends upon one's philosophical outlook. It is useful to see that the situation is more robust than that. [108]

Let k be a field. Let $k \subset K$ where K is a bigger field. For $\alpha \in K$, define the **field extension (in K) over k generated by α** [109]

$$k(\alpha) = \bigcap_{\text{fields } E \subset K,\ E \supset k,\ \alpha \in E} E$$

It is easy to check that the intersection of subfields of a common field is a field, so this intersection is a field. Rather than a single element, one could as well adjoin any subset of the over-field K. [110]

Before studying $k(\alpha)$ in more detail, consider a different procedure of *adjoining* something: for a commutative

[108] In the 19th century there was widespread confusion or at least concern over issues of existence of *quantities* having various properties. Widespread belief in the legitimacy of the complex numbers was not in place until well into that century, and ironically was abetted by pictorial emphasis on complex numbers as *two-dimensional* things. The advent of the Hamiltonian quaternions in mid-century made the complex numbers seem innocent by comparison.

[109] The notation here is uncomfortably fragile: exchanging the parentheses for any other delimiters alters the meaning.

[110] This definition does *not* give a good computational handle on such field extensions. On the other hand, it is unambiguous and well-defined.

ring R with 1 which is a subring of an R-algebra A, for $\alpha \in A$, one might *attempt* to define [111]

$$R[\alpha] = \{ \text{ polynomials in } \alpha\}$$

One probably understands the intent, that this is

$$R[\alpha] = \{c_0 + c_1\alpha + \ldots + c_n\alpha^n : c_i \in R\}$$

More precisely, a proper definition would be

$$R[\alpha] = \text{the image in } A \text{ of the unique } R\text{-algebra homomorphism sending } x \text{ to } \alpha$$

where we invoke the universal mapping property of $R[x]$.

Specialize R again to be a field k, and let A be a (not necessarily commutative) k-algebra, $\alpha \in A$. Then the natural homomorphism

$$\varphi : k[x] \longrightarrow k[\alpha] \quad (\text{by } x \longrightarrow \alpha)$$

has a kernel which is a principal ideal $\langle f\rangle$. [112] By the usual Isomorphism Theorem the map φ *descends* to the quotient by the kernel, giving an isomorphism

$$\overline{\varphi} : k[x]/\langle f\rangle \approx k[\alpha]$$

If $f = 0$, that is, if the kernel is trivial, then $k[\alpha]$ of course inherits properties of the polynomial ring. [113]

At this point we need to begin using the fact that a k-algebra A is a k-vectorspace. [114] The **degree** of A over k is

$$[A : k] = \text{degree of } A \text{ over } k = \text{ dimension of } A \text{ as } k\text{-vectorspace}$$

If $k[\alpha] \approx k[x]$, then, for example, the various powers of α are linearly independent over k, and $k[\alpha]$ is infinite-dimensional as a k-vectorspace. And there is *no* polynomial $P(x) \in k[x]$ such that $P(\alpha) = 0$. Especially in the simple situation that the k-algebra A is a *field*, such elements α with $k[\alpha] \approx k[x]$ are **transcendental** over k. [115]

On the other hand, a perhaps more interesting situation is that in which the kernel of the natural

$$k[x] \longrightarrow k[\alpha]$$

has non-zero kernel $\langle f\rangle$, with f monic without loss of generality. This f is the **minimal polynomial** of α (in A) over k.

Although our immediate concern is field extensions, there is at least one other useful application of this viewpoint, as follows. Let V be a k-vectorspace, and let A be the k-algebra

$$A = \text{End}_k V$$

[111] Note again the fragility of the notation: $k(\alpha)$ is generally quite different from $k[\alpha]$, although in some useful cases (as below) the two can coincide.

[112] ... since $k[x]$ is a principal ideal domain for k a field. For more general commutative rings R the corresponding discussion is more complicated, though not impossible.

[113] ... to which it is *isomorphic* by the just-demonstrated isomorphism!

[114] By *forgetting* the multiplication in A, if one insists.

[115] This is an essentially *negative* definition: there are no relations.

of k-linear maps (i.e., **endomorphisms**) of V to itself. For $T : V \longrightarrow V$ a k-linear map, we can consider the natural k-algebra map

$$k[x] \longrightarrow \mathrm{End}_k V \quad (\text{by } x \longrightarrow T)$$

We give $\mathrm{End}_k V$ a k-vectorspace structure value-wise by

$$(\alpha \cdot T)(v) = \alpha \cdot (Tv)$$

for $v \in V$ and $\alpha \in k$. If V is finite-dimensional, then $\mathrm{End}_k V$ is also finite-dimensional. [116] In particular, the kernel of the natural map from $k[x]$ cannot be just 0. Let f be the non-zero monic generator for the kernel. Again, [117] this monic is the **minimal polynomial** for T. The general construction shows that for any $P(x) \in k[x]$,

$$P(T) = 0 \in \mathrm{End}_k V \quad \text{if and only if } f \text{ divides } P$$

In particular, if the polynomial equation $f(x) = 0$ has a *root* λ in k, then [118] we can prove that T has **eigenvalue** λ. That is, there is a non-zero vector $v \in V$ (the λ-**eigenvector**) such that

$$Tv = \lambda \cdot v$$

Indeed, let $f(x) = (x - \lambda) \cdot g(x)$ for some $g(x) \in k[x]$. Since g is not the minimal polynomial for T, then there is a vector $w \in V$ such that $g(T) \cdot w \neq 0$. We claim that $v = g(T)w$ is a λ-eigenvector. Indeed,

$$0 = f(T) \cdot w = (T - \lambda) \cdot g(T)w = (T - \lambda) \cdot v$$

and by the previous comment $v = g(T)w$ is not 0. [119]

Returning to field extensions: let K be a field containing a smaller field k, $\alpha \in K$, and let f be the generator for the kernel of the natural map $k[x] \longrightarrow k[\alpha]$. We do assume that f is non-zero, so we can make f monic, without loss of generality. Since f is non-zero, we do call it the **minimal polynomial** of α over k, and, since α *has* a minimal polynomial over k, we say that α is **algebraic** over k. [120] If every element α of a field extension K of k is algebraic over k, then say that the field extension K itself is **algebraic** over k.

Once again, given any polynomial $P(x)$, there are *unique* $Q(x)$ and $R(x)$ with $\deg R < \deg f$ such that

$$P = Q \cdot f + R$$

and

$$P(\alpha) = Q(\alpha) \cdot f(\alpha) + R(\alpha) = Q(\alpha) \cdot 0 + R(\alpha) = R(\alpha)$$

[116] This is not hard to prove: let $e_1, \ldots, e_n$ be a k-basis for V. Then the k-linearity $T(\sum_i c_i e_i) = \sum_i c_i T(e_i)$ shows that T is determined completely by the collection of images Te_i. And $Te_i = \sum_j T_{ij} e_j$ for some collection of n^2 elements T_{ij} of k. Thus, if V is n-dimensional then its endomorphism algebra is n^2-dimensional.

[117] This is terminology completely consistent with linear algebra usage.

[118] From the fact that roots correspond perfectly to linear factors, for polynomials in one variable with coefficients in a field.

[119] Even this brief discussion of minimal polynomials and linear operators should suggest, and correctly so, that use of determinants and invocation of the Cayley-Hamilton theorem, concerning the *characteristic polynomial* of a linear operator, is not exactly to the point.

[120] Again, this situation, where $f(\alpha) = 0$ with a non-zero polynomial f, is in contrast to the case where α satisfies *no* algebraic equation with coefficients in k.

Letting $n = \deg f$, this implies that $1, \alpha, \alpha^2, \ldots, \alpha^{n-1}$ are a k-basis for $k[\alpha]$. [121]

6.1.1 Proposition: For α algebraic over k (all inside K), the ring $k[\alpha]$ is a *field*. [122] That is, for α *algebraic*, $k(\alpha) = k[\alpha]$. The minimal polynomial f of α over k is *irreducible* in $k[x]$. And the degree (dimension) of $k(\alpha)$ over k is

$$[k(\alpha) : k] = \dim_k k(\alpha) = \deg f$$

Proof: First, from above,

$$k[\alpha] \approx k[x]/\langle f \rangle$$

To prove irreducibility, suppose we can write $f = g \cdot h$ with $g, h \in k[x]$ with proper factors. By minimality of f, neither $g(\alpha)$ nor $h(\alpha)$ is 0. But $f(\alpha) = 0$, so $g(\alpha$ and $h(\alpha)$ are zero-divisors, contradiction. [123]

Since $k(\alpha)$ is the smallest field inside the ambient field K containing α and k, certainly $k[\alpha] \subset k(\alpha)$. To prove equality, it would suffice to show that non-zero elements of $k[\alpha]$ have multiplicative inverses in $k[\alpha]$. For polynomial $g(x) \in k[x]$, $g(\alpha) \neq 0$ if and only if the minimal polynomial $f(x)$ of α over k does not divide $g(x)$. Since f is irreducible and does not divide g, there are polynomials r, s in $k[x]$ such that

$$1 = \gcd(f, g) = r(x) \cdot f(x) + s(x) \cdot g(x)$$

so, mapping x to α,

$$1 = r(\alpha) \cdot f(\alpha) + s(\alpha) \cdot g(\alpha) = r(\alpha) \cdot 0 + s(\alpha) \cdot g(\alpha) = s(\alpha) \cdot g(\alpha)$$

That is, $s(\alpha)$ is a multiplicative inverse to $g(\alpha)$, and $k[\alpha]$ is a field. The degree is as asserted, since the polynomials of degree $< \deg f$ are irredundant representatives for the equivalence classes of $k[x]/\langle f \rangle$.
///

6.2 *Fields of fractions, fields of rational functions*

For $k \subset K$ fields and $\alpha \in K$ transcendental over k, it is not true that $k[\alpha] \approx k(\alpha)$, in complete contrast to the case that α is algebraic, discussed just above. [124]

But from elementary mathematics we have the idea that for *indeterminate* [sic] x

$$k(x) = \text{ field of rational functions in } x \ = \{\frac{g(x)}{h(x)} : g, h \in k[x],\ h \neq 0\}$$

We can reconcile this primitive idea with our present viewpoint.

Let R be an *integral domain* (with unit 1) [125] and define the **field of fractions** Q of R to be the collection of ordered pairs (r, s) with $r, s \in R$ and $s \neq 0$, modulo the equivalence relation [126]

$$(r, s) \sim (r', s') \qquad \text{if} \qquad rs' = sr'$$

[121] Indeed, the identity $P = Qf + R$ shows that any polynomial in α is expressible as a polynomial of degree $< n$. This proves spanning. On the other hand, a linear dependence relation $\sum_i c_i \alpha^i = 0$ with coefficient c_i in k is nothing other than a polynomial relation, and our hypothesis is that any such is a (polynomial) multiple of f. Thus, the monomials of degrees less than $\deg f$ are linearly independent.

[122] This should be a little surprising.

[123] Everything is taking place inside the larger field K.

[124] In particular, since $k[\alpha] \approx k[x]$, $k[\alpha]$ is not a field at all.

[125] A definition can be made for more general commutative rings, but the more general definition has more complicated features which are not of interest at the moment.

[126] This would be the usual requirement that two fractions r/s and r'/s' be equal.

The addition is suggested by the usual addition of fractions, namely that

$$(r,s)+(r',s')=(rs'+r's,ss')$$

and the multiplication is the more obvious

$$(r,s)\cdot(r',s')=(rr',ss')$$

One should verify that these operations are well-defined on the quotient Q by that equivalence relation, that Q is a commutative ring with unit (the equivalence class of)$(1,1)$, that $r\longrightarrow(r,1)$ *injects* R to Q. This constructs the field of fractions.

The latter construction is *internal*, in the sense that it constructs a concrete thing in set-theoretic terms, given the original ring R. On the other hand, we can characterize the field of fractions *externally*, by properties of its mappings to other rings or fields. In particular, we have

6.2.1 Proposition: For an integral domain R with unit 1, its field of fractions Q, with the natural inclusion $i:R\longrightarrow Q$, is the unique field (and inclusion of R into it) such that, for any *injective* ring homomorphism $\varphi:R\longrightarrow K$ with a field K, there is a unique $\tilde{\varphi}:Q\longrightarrow K$ such that

$$\varphi\circ i=\tilde{\varphi}$$

Specifically, $\tilde{\varphi}(r,s)=\varphi(r)/\varphi(s)$. [127]

Proof: Indeed, try to define [128]

$$\tilde{\varphi}(r,s)=\varphi(r)/\varphi(s)$$

where the quotient on the right-hand side is in the field K, and the injectivity of φ assure that $s\neq 0$ implies that $\varphi(s)\neq 0$. This is certainly compatible with φ on R, since

$$\tilde{\varphi}(r,1)=\varphi(r)/\varphi(1)=\varphi(r)$$

and the smallest subfield of K containing R certainly must contain all such quotients. The main thing to check is that this definition really is well-defined, namely that if $(r,s)\sim(r',s')$, then

$$\tilde{\varphi}(r,s)=\tilde{\varphi}(r',s')$$

Do this as follows. The equivalence relation is that $rs'=r's$. Applying φ on R gives

$$\varphi(r)\varphi(s')=\varphi(r')\varphi(s)$$

Since φ is injective, for s,s' nonzero in R their images are nonzero in K, so we can divide, to obtain

$$\varphi(r)/\varphi(s)=\varphi(r')/\varphi(s')$$

This proves the well-definedness. That multiplication is preserved is easy, and that addition is preserved is straightforward. ///

To practice categorical arguments, we can also prove, without using formulas or explicit constructions:

[127] Implicitly we must claim that this is well-defined.

[128] What else could it be?

6.2.2 Proposition: Let Q' be a field with inclusion $i' : R \longrightarrow Q'$ such that, for every injective homomorphism $\varphi : R \longrightarrow K$ with a field K, there is a unique $\tilde{\varphi} : Q' \longrightarrow K$ such that

$$\varphi \circ i' = \tilde{\varphi}$$

Then there is a unique isomorphism $j : Q \longrightarrow Q'$ of the field of fractions Q of R (with inclusion $i : R \longrightarrow Q$) to Q' such that

$$i' = j \circ i$$

That is, up to unique isomorphism, there is only one field of fractions of an integral domain.

Proof: First prove that any field map $f : Q \longrightarrow Q$ such that $f \circ i = i$ must be the identity on Q. Indeed, taking $K = Q$ and $f = i : R \longrightarrow K$ in the defining property, we see that the identity map id_Q on Q has the property $\mathrm{id}_K \circ i = i$. The *uniqueness* property assures that any other f with this property must be id_K.

Then let Q' and $i' : R \longrightarrow Q'$ be another pair satisfying the universal mapping condition. Taking $\varphi = i' : R \longrightarrow Q'$ yields $\tilde{\varphi} : Q \longrightarrow Q'$ with $\varphi = \tilde{\varphi} \circ i$. Reversing the roles, taking $\varphi' = i : R \longrightarrow Q$ yields $\tilde{\varphi'} : Q' \longrightarrow Q$ with $\varphi' = \tilde{\varphi'} \circ i'$. Then (by the previous paragraph) $\tilde{\varphi} \circ \tilde{\varphi'} : Q \longrightarrow Q$ must be the identity on Q, and, similarly, $\tilde{\varphi'} \circ \tilde{\varphi} : Q' \longrightarrow Q$; must be the identity on Q'. Thus, $\tilde{\varphi}$ and $\tilde{\varphi'}$ are mutual inverses. This proves the isomorphism of the two objects. [129] ///

Thus, without having a larger field in which the polynomial ring $k[x]$ sits, we simply form the field of fractions of this integral domain, and denote it [130]

$$k(x) = \text{ field of fractions of } k[x] = \text{ rational functions in } x$$

Despite having this construction available, it still may be true that for fields $k \subset K$, there is α in K *transcendental* over k, in the sense (above) that α satisfies no polynomial relation with coefficients in k. [131] In that case, we have the more general definition of $k(\alpha)$ as the intersection of all subfields of K containing k and containing α.

For notational consistency, we should check that $k(\alpha)$ is isomorphic to the field of fractions of $k[\alpha]$. And, indeed, since $k[x]$ injects to $k[\alpha]$ (taking x to α), by the mapping property characterization the field of fractions $k(x)$ of $k[x]$ has a unique injection j to the field $k(\alpha)$ extending the given map. Certainly $k(\alpha) \subset j(k(x))$, since $k(\alpha)$ is the intersection of all subfields of K containin k and α. Thus, the image of the injective map j is exactly $k(\alpha)$, and j is an isomorphism of $k(x)$ to $k(\alpha)$.

6.3 *Characteristics, finite fields*

The linear algebra viewpoint is decisive in understanding many elementary features of fields, for example, the result below on possible cardinalities of finite fields.

[129] The *uniqueness* of the isomorphism also follows from discussion, since if there were two isomorphisms h and h' from Q to Q', then $h' \circ h^{-1} : Q \longrightarrow Q$ would be a non-identity map with the desired property, but only the identity on Q has the universal mapping property.

[130] To say that these are rational *functions* is a bit of a misnomer, but no worse than to refer to polynomial *functions*, which is also misleading but popular.

[131] Again, more precisely, the condition that α be transcendental is that the natural map $k[x] \longrightarrow k[\alpha]$ by $x \longrightarrow \alpha$ has trivial kernel.

First, observe that any ring R is a $\mathbb{Z}$-algebra in a canonical [132] manner, with the action

$$n \cdot r = \begin{cases} \underbrace{r + \ldots + r}_{n} & (n > 0) \\ 0_R & (n = 0) \\ -(\underbrace{r + \ldots + r}_{|n|}) & (n < 0) \end{cases}$$

An easy but tedious induction proves that this $\mathbb{Z}$-algebra structure deserves the name. [133] As evidence for the naturality of this $\mathbb{Z}$-structure, notice that if $f : R \longrightarrow S$ is any ring homomorphism, then f is a $\mathbb{Z}$-algebra homomorphism when the above $\mathbb{Z}$-algebra structures are put on R and S.

When a ring R has an identity 1_R, there is a canonical $\mathbb{Z}$-algebra homomorphism $i : \mathbb{Z} \longrightarrow R$ by

$$i : n \longrightarrow n \cdot 1_R$$

Granting that the $\mathbb{Z}$-algebra structure on R works as claimed, the proof that this is a homomorphism is nearly trivial:

$$i(m+n) = (m+n) \cdot 1_R = m \cdot 1_R + n \cdot 1_R = i(m) + i(n)$$

$$i(m \cdot n) = (m \cdot n) \cdot 1_R = m \cdot (n \cdot 1_R) = m \cdot (1_R \cdot (n \cdot 1_R)) = (m \cdot 1_R) \cdot (n \cdot 1_R) = i(m) \cdot i(n)$$

Now consider the canonical $\mathbb{Z}$-algebra homomorphism $i : \mathbb{Z} \longrightarrow k$ for a field k. [134] If i is injective, then it extends to an injection of the field of fractions $\mathbb{Q}$ of $\mathbb{Z}$ into k. In this case, say k is of **characteristic zero**, and this canonical copy of $\mathbb{Q}$ inside k is the **prime field** inside k. If i is not injective, its kernel is a principal ideal in $\mathbb{Z}$, say $p\mathbb{Z}$ with $p > 0$. Since the image $i(\mathbb{Z})$ is inside a field, it is an integral domain, so $p\mathbb{Z}$ is a (non-zero) prime ideal, which implies that p is prime. This integer p is the **characteristic** of k. We know that $\mathbb{Z}/\langle p \rangle$ is a *field*. Then we see that (by the Isomorphism Theorem for rings) the homomorphism $i : \mathbb{Z} \longrightarrow k$ with kernel $p\mathbb{Z}$ induces an isomorphism

$$\mathbb{Z}/p \approx i(\mathbb{Z}) \subset k$$

This canonical copy of $\mathbb{Z}/p$ inside k is the **prime field** inside k.

A finite field with q elements is often denoted $\mathbb{F}_q$ or $GF(q)$. [135]

6.3.1 Theorem: A finite field K has p^n elements for some prime p and integer n. [136] In particular, let $n = [K : \mathbb{F}_p]$ be the degree of K over its prime field $\mathbb{F}_p \approx \mathbb{Z}/p$ with prime p. Then

$$|K| = p^n$$

[132] This sort of use of *canonical* is meant for the moment to insinuate that there is no whimsical choice involved. A more precise formulation of what *canonical* could mean would require a category-theoretical set-up. We may do this later.

[133] The arguments to prove this are of the same genre as those proving the so-called Laws of Exponents. Here, one must show that $(m+n)r = mr + nr$ and $(mn)r = m(nr)$ for $m, n \in \mathbb{Z}$ and $r \in R$, and $m(rs) = (mr)s$ for $s \in R$.

[134] It is no coincidence that we begin our study of fields by considering homomorphisms of the two simplest interesting rings, $k[x]$ for a field k, and $\mathbb{Z}$, into rings and fields.

[135] This notation begs the question of *uniqueness* (up to isomorphism) of a finite field once its cardinality is specified. We address this shortly.

[136] We will prove *existence* and uniqueness results for finite fields a bit later.

Proof: Let $\mathbb{F}_p$ be the prime field in K. Let $e_1, \ldots, e_n$ be a $\mathbb{F}_p$-basis for the $\mathbb{F}_p$-vectorspace K. Then there are p^n choices of coefficients $c_i \in \mathbb{F}_p$ to form linear combinations

$$\alpha = \sum_{i=1}^{n} c_i\, e_i \in K$$

so K has p^n elements. ///

6.4 *Algebraic field extensions*

The first of the following two examples is amenable to *ad hoc* manipulation, but the second is designed to frustrate naive explicit computation.

6.4.1 Example: Let γ be a root (in some field k of characteristic 0, thus containing the prime field $\mathbb{Q}$) of the equation

$$x^2 - \sqrt{2}x + \sqrt{3} = 0$$

Is γ a root of a polynomial equation with rational coefficients?

In the same spirit as *completing the square*, we can manipulate the equation $x^2 - \sqrt{2}x + \sqrt{3} = 0$ to make the square roots disappear, as follows. Move the x^2 to the opposite side and square both sides, to obtain

$$2x^2 - 2\sqrt{6}x + 3 = x^4$$

Then move everything but the remaining square root to the right-hand side

$$-2\sqrt{6}\, x = x^4 - 2x^2 - 3$$

and square again

$$24x^2 = x^8 - 4x^6 - 2x^4 + 6x^2 + 9$$

and then we find that γ is a root of

$$0 = x^8 - 4x^6 - 2x^4 - 18x^2 + 9$$

It is not so obvious that the original [137]

$$\gamma = \frac{\sqrt{2} \pm \sqrt{2 - 4\sqrt{3}}}{2}$$

are roots. [138]

6.4.2 Example: Let α be a root of the equation

$$x^5 - x + 1 = 0$$

[137] Solving the original quadratic equation directly, by completing the square, for example.

[138] For that matter, it appears that the original equation has exactly two roots, while a degree 8 equation might have 8. Thus, we seem to have introduced 6 *spurious* roots in this process. Of course, an explanation for this is that there are two different square roots of 2 and two different square roots of 3 in k, so really $2 \cdot 2 = 4$ versions of the original quadratic equation, each with perhaps 2 roots in k.

and let β be a root of the equation

$$x^7 - x + 1 = 0$$

Then let γ be a root of the equation

$$x^6 - \alpha x + \beta = 0$$

Is γ a root of a polynomial equation with rational coefficients?

In this second example manipulations at the level of the first example fail. [139] But one might speculate that in answering an existential question it might be possible to avoid explicit computations entirely, as in the proofs of the following results.

6.4.3 Proposition: Let $k \subset K \subset L$ be fields, with $[K : k] < \infty$ and $[L : K] < \infty$. Then

$$[L : k] = [L : K] \cdot [K : k] < \infty$$

In particular, for a K-basis $\{E_i\}$ of L, and for a k-basis e_j of K, the set $\{E_i e_j\}$ is a k-basis for L. [140]

Proof: On one hand, any linear relation

$$\sum_{ij} A_{ij}\, E_i e_j = 0$$

with $A_{ij} \in k$ gives

$$\sum_i (\sum_j A_{ij}\, e_j)\, E_i = 0$$

so for each i we have $\sum_j A_{ij}\, e_j = 0$, by the linear independence of the E_i. And by the linear independence of the e_j we find that $A_{ij} = 0$ for all indices. On the other hand, given

$$\beta = \sum_i b_i E_i \in L$$

with $b_i \in K$, write $b_i = \sum_j a_{ij} e_j$ with $a_{ij} \in k$, and then

$$\beta = \sum_i (\sum_j a_{ij} e_j)\, E_i = \sum_{ij} a_{ij}\, E_i e_j$$

which proves the spanning property. Thus, the elements $E_i e_j$ are a k-basis for L. ///

A field extension K of a field k is **finite** if the degree $[K : k]$ is finite. *Finite* field extensions can be built up by adjoining elements. To economize on parentheses and brackets, [141] write

$$k(\alpha_1, \ldots, \alpha_n) \quad \text{for} \quad k(\alpha_1)(\alpha_2)\ldots(\alpha_n)$$

and

$$k[\alpha_1, \ldots, \alpha_n] \quad \text{for} \quad k[\alpha_1][\alpha_2]\ldots[\alpha_n]$$

[139] The provable limitations of familiar algebraic operations are packaged up in *Galois theory*, a bit later.

[140] The first assertion is merely a qualitative version of the last. Note that this proposition does not mention field *elements* explicitly, but rather emphasizes the vector space structures.

[141] On might worry that this notation glosses over potential issues. But, for example, one can prove that a polynomial ring in two variables really is *naturally* isomorphic to a polynomial ring in one variable over a polynomial ring in one variable.

6.4.4 Proposition: Let K be a field containing k, and suppose that $[K:k] < \infty$. Then any element α in K is algebraic over k, and there are finitely-many $\alpha_1, \ldots, \alpha_n$ such that

$$K = k(\alpha_1, \ldots, \alpha_n) = k[\alpha_1, \ldots, \alpha_n]$$

In particular, *finite* extensions K are necessarily *algebraic*. [142]

Proof: Given $\alpha \in K$, the countably many powers $1, \alpha, \alpha^2, \ldots$ cannot be linearly independent over k, since the whole K is finite-dimensional over k. A linear dependence relation among these powers is a polynomial equation satisfied by α. [143] If K is strictly larger than k, take $\alpha_1 \in K$ but not in k. Then $[k(\alpha_1):k] > 1$, and the multiplicativity

$$[K:k] = [K:k(\alpha_1)] \cdot [k(\alpha_1):k]$$

with $[K:k] < \infty$ implies that

$$[K:k(\alpha_1)] < [K:k]$$

If K is still larger than $k(\alpha_1)$, take α_2 in K not in $k(\alpha_1)$. Again,

$$[K:k(\alpha_1,\alpha_2)] < [K:k(\alpha_1)] < [K:k]$$

These degrees are positive integers, so a decreasing sequence must reach 1 in finitely-many steps (by Well-Ordering). The fact [144] that $k(\alpha) = k[\alpha]$ for α algebraic over k was proven earlier. ///

Let K and L be subfields of a larger field E. The **compositum** $K \cdot L$ of K and L is the smallest subfield of E containing both K and L. [145]

6.4.5 Proposition: Let $k \subset E$ be fields. Let K, L be subfields of K containing k. Suppose that $[K:k] < \infty$ and $[L:k] < \infty$. Then

$$[K \cdot L \,:\, k] \leq [K:k] \cdot [L:k] < \infty$$

In particular, if

$$K = k(\alpha_1, \ldots, \alpha_m) = k[\alpha_1, \ldots, \alpha_m]$$
$$L = k(\beta_1, \ldots, \beta_n) = k[\beta_1, \ldots, \beta_n]$$

then

$$K \cdot L = k(\alpha_1, \ldots, \alpha_m, \beta_1, \ldots, \beta_n) = k[\alpha_1, \ldots, \alpha_m, \beta_1, \ldots, \beta_n]$$

Proof: From the previous proposition, there do exist the α_i and β_j expressing K and L as k with finitely many elements adjoined as in the statement of the proposition. Recall that these mean that

$$K = \text{ intersection of subfields of } E \text{ containing } k \text{ and all } \alpha_i$$

$$L = \text{ intersection of subfields of } E \text{ containing } k \text{ and all } \beta_i$$

[142] The converse is not true. That is, some fields k admit extensions K with the property that every element in K is algebraic over k, but K is infinite-dimensional over k. The rational numbers $\mathbb{Q}$ can be proven to have this property, as do the p-adic numbers $\mathbb{Q}_p$ discussed later. It is not completely trivial to prove this.

[143] A more elegant argument is to map $k[x]$ to K by $x \longrightarrow \alpha$, and note that the kernel must be non-zero, since otherwise the image would be infinite-dimensional over k.

[144] Potentially disorienting and quite substantial.

[145] As with many of these constructions, the notion of *compositum* does not make sense, or at least is not well-defined, unless the two fields lie in a common larger field.

On one hand, $K \cdot L$ contains all the α_i and β_j. On the other hand, since these elements are algebraic over k, we do have

$$k(\alpha_1, \ldots, \alpha_m, \beta_1, \ldots, \beta_n) = k[\alpha_1, \ldots, \alpha_m, \beta_1, \ldots, \beta_n]$$

The left-hand side is a field, by definition, namely the smallest subfield [146] of E containing all the α_i and β_j. Thus, it contains K, and contains L. Thus, we have equality. ///

6.4.6 Proposition: Let $k \subset E$ be fields, and K, L fields between k and E. Let $\alpha \in L$ be algebraic over k. Then

$$[k(\alpha) : k] \geq [K(\alpha) : K]$$

Proof: Since α is algebraic over k, $k(\alpha) = k[\alpha]$, and the degree $[k(\alpha) : k]$ is the degree of the minimal polynomial of α over k. This degree cannot increase when we replace k by the larger field K, and we obtain the indicated inequality. ///

6.4.7 Proposition: Let k be a field, K a field algebraic over k, and L a field containing K. Let $\beta \in L$ be algebraic over K. Then β is algebraic over k.

Proof: Let $M(x)$ be the monic irreducible in $K[x]$ which is the minimal polynomial for β over K. Let $\{\alpha_0, \ldots, \alpha_{n-1}\}$ be the *finite* set (inside K) of coefficients of $M(x)$. Each field $k(\alpha_i)$ is of finite degree over k, so by the previous proposition their compositum $k(\alpha_1, \ldots, \alpha_n)$ is finite over k. The polynomial $M(x)$ is in $k(\alpha_1, \ldots, \alpha_n)[x]$, so β is algebraic over $k(\alpha_1, \ldots, \alpha_n)$. From above, the degree of $k(\alpha_1, \ldots, \alpha_n)(\beta)$ over k is the product

$$[k(\alpha_1, \ldots, \alpha_n)(\beta) : k] = [k(\alpha_1, \ldots, \alpha_n)(\beta) : k(\alpha_1, \ldots, \alpha_n)] \cdot [k(\alpha_1, \ldots, \alpha_n) : k] < \infty$$

Thus, $k(\alpha_1, \ldots, \alpha_n)(\beta)$ is finite over k, and in particular β is algebraic over k.

///

6.4.8 Corollary: Let $k \subset K \subset L$ be fields, with K algebraic over k and L algebraic over K. Then L is algebraic over k.

Proof: This is an element-wise assertion, and for each β in L the previous proposition proves the algebraicity. ///

6.4.9 Remark: An arrangement of fields of the form $k \subset K \subset L$ is sometimes called a **tower** of fields, with a corresponding picture

$$\begin{array}{c} L \\ | \\ K \\ | \\ k \end{array}$$

[146] This discussion would appear to depend perhaps too much upon the larger ambient field E. In one sense, this is true, in that *some* larger ambient field is necessary. On the other hand, if K and L are both contained in a smaller subfield E' of E, we can replace E by E' for this discussion. One may reflect upon the degree to which the outcome genuinely depends upon any difference between E' and E, and how to avoid this concern.

The situation that K and L are intermediate fields between k and E, with compositum KL, is depicted as

$$\begin{array}{c} E \\ | \\ KL \\ / \ \backslash \\ K \quad L \\ \backslash \ / \\ k \end{array}$$

6.5 *Algebraic closures*

A field K is **algebraically closed** if every non-constant polynomial $f(x) \in k[x]$ has at least one root $\alpha \in k$, that is,

$$f(\alpha) = 0$$

Upon division, this algebraic closure property implies that any polynomial in $K[x]$ factors into linear factors in $K[x]$.

Given a field k, a larger field K which is algebraically closed [147] *and* such that every element of K is algebraic over k, is an **algebraic closure of** k. [148]

6.5.1 Theorem: Any field k has an algebraic closure $\overline{k}$, unique up to isomorphism. Any algebraic field extension E of k has at least one injection to $\overline{k}$ (which restricts to the identity on k).

Proof: *(Artin)* Let S be the set of monic irreducibles in $k[x]$, for each $s \in S$ let x_s be an indeterminate, and consider the polynomial ring

$$R = k[\ldots, x_s, \ldots] \quad (s \in S)$$

in S-many variables. [149] We claim that there is at least one maximal proper ideal M in R containing every $f(x_f)$ for $f \in S$. First, one must be sure that the ideal F generated by all $f(x_f)$ is *proper* in R. If F were not proper, there would be elements $r_i \in R$ and irreducibles f_i such that (a finite sum)

$$\sum_{i=1}^{n} r_i \, f_i(x_{f_i}) = 1$$

Make a finite field extension E of k such that all the finitely-many f_i have roots α_i in E, inductively, as follows. First, let $k_1 = k[x]/\langle f_1 \rangle$. Then let F_2 be an *irreducible* factor of f_2 in k_1, and let $k_2 = k_1[x]/\langle F_2 \rangle$.

[147] Note that not only polynomials with coefficients in k must have roots in K, but polynomials with coefficients in K. Thus, one can perhaps imagine a different universe in which one makes a large enough field K such that all polynomials with coefficients in k have roots, but polynomials with coefficients in K need a larger field for their roots. That this does not happen, and that the process of constructing algebraic closures terminates, is the content of the theorem below.

[148] The second requirement is desirable, since we do not want to have algebraic closures be needlessly large. That is, an algebraic closure of k should not contain elements transcendental over k.

[149] This ostentatiously extravagant construction would not have been taken seriously prior to Bourbaki's influence on mathematics. It turns out that once one sacrifices a *little* finiteness, one may as well fill things out symmetrically and accept a *lot* of non-finiteness. Such extravagance will reappear in our modern treatment of tensor products, for example.

And so on, obtaining $E = k_n$. Using the universal mapping property of polynomial rings, we can send x_{f_i} to $\alpha_i \in E$, thus sending $f_i(x_{f_i})$ to 0. [150] Then the relation becomes

$$0 = 1$$

Thus, there is no such relation, and the ideal F is proper.

Next, we claim that F lies in a *maximal* proper ideal M in R. This needs an equivalent of the Axiom of Choice, such as Hausdorff Maximality or Zorn's Lemma. In particular, among all chains of *proper* ideals containing F

$$F \subset \ldots \subset I \subset \ldots$$

there exists a maximal chain. [151] The union of an ascending chain of proper ideals cannot contain 1, or else one of the ideals in the chain would contain 1, and would not be proper. Thus, the union of the ideals in a maximal chain is still proper. If it were not a *maximal* proper ideal then there would be a further (proper) ideal that could be added to the chain, contrary to assumption. Thus, we have a maximal ideal M in R. Thus, $K = R/M$ is a field.

By construction, for monic irreducible (non-constant) f the equation $f(Y) = 0$ has a root in K, namely the image of x_f under the quotient map, since $f(x_f) \in M$ for all irreducibles f. This proves that all non-constant polynomials in $k[x]$ have roots in K.

Now we prove that every element in $\overline{k}$ is algebraic over k. Let α_f be the image of x_f in *kbar*. Since α_f is a zero of f it is algebraic over k. An element β of $\overline{k}$ is a polynomial in finitely-many of the α_fs, say $\alpha_{f_1}, \ldots, \alpha f_n$. That is, $\beta \in k[\alpha_1, \ldots, \alpha_n]$, which is a field since each α_i is algebraic over k. Since (for example) the compositum (inside $\overline{k}$) of the algebraic extensions $k(\alpha_{f_i}) = k[\alpha_{f_i}]$ is algebraic, β is algebraic over k.

Next, we prove that non-constant $F(x) \in \overline{k}[x]$ has a zero in $\overline{k}$ (hence, it has *all* zeros in $\overline{k}$). The coefficients of F involve some finite list $\alpha_{f_1}, \ldots, \alpha_{f_n}$ out of all α_f, and $F(x)$ has a zero in $\overline{k}(\alpha_{f_1}, \ldots, \alpha_{f_n})[x]/\langle F \rangle$. Thus, since β is algebraic over an algebraic extension of k, it is algebraic over k, and, thus, is a root of a polynomial in $k[x]$.

Now consider an algebraic extension E of k, and show that it admits an imbedding into $\overline{k}$. First, if $\alpha \in E$, let f be the minimal polynomial of α over k, and let β be a zero of f in $\overline{k}$. Map $k[x] \longrightarrow \overline{k}$ by sending $x \longrightarrow \beta$. The kernel is exactly the ideal generated by f, so (by an isomorphism theorem) the homomorphism $k[x] \longrightarrow \overline{k}$ descends to an injection $k[\alpha] \longrightarrow \overline{k}$. This argument can be repeated to extend the inclusion $k \subset \overline{k}$ to any extension $E = k(\alpha_1, \ldots, \alpha_n)$ with α_i algebraic over k. We use an equivalent of the Axiom of Choice to complete the argument: consider the collection of ascending chains of fields E_i (containing k) inside E admitting families of injections $\psi_i : E_i \longrightarrow \overline{k}$ with the compatibility condition that

$$\psi_j \,|_{E_i}| = \psi_i \quad \text{for} \quad E_i \subset E_j$$

We can conclude that there is a *maximal* chain. Let E' be the union of the fields in this maximal chain. The field E' imbeds in $\overline{k}$ by ψ_i on E_i, and the compatibility condition assures us that this is well-defined. We claim that $E' = E$. Indeed, if not, there is $\alpha \in E$ that is not in E'. But then the first argument shows that $E'(\alpha)$ does admit an imbedding to $\overline{k}$ extending the given one of E', contradiction. Thus, $E' = E$ and we have, in fact, imbedded the whole algebraic extension E to $\overline{k}$.

Last, we prove that any other algebraic closure K of k is isomorphic to $\overline{k}$. [152] Indeed, since K and $\overline{k}$ are algebraic over k, we have at least one injection $K \longrightarrow \overline{k}$, and at least one injection $\overline{k} \longrightarrow K$, but there

[150] No, we have no idea what happens to the r_i, but we don't care.

[151] Maximal in the sense that there is no other proper ideal J containing F that either contains or is contained in every element of the (maximal) chain.

[152] Note that we do not claim uniqueness of the isomorphism. Indeed, typically there are many different maps of a given algebraic closure $\overline{k}$ to itself that fix the underlying field k.

is no reason to think that our capricious construction assures that these are mutual inverses. A different mechanism comes into play. Consider K imbedded into $\overline{k}$. Our claim is that K is necessarily all of $\overline{k}$. Indeed, any element of $\overline{k}$ is algebraic over k, so is the zero of a polynomial f in $k[x]$, say of degree n, which has all n roots in the subfield K of $\overline{k}$ because K is algebraically closed. That is, every element of the overfield $\overline{k}$ is actually in the subfield K, so the two are equal. ///

Exercises

6.1 Let γ be a root of the equation $x^2 + \sqrt{5}x + \sqrt{2} = 0$ in an algebraic closure of $\mathbb{Q}$. Find an equation with *rational* coefficients having root γ.

6.2 Let γ be a root of the equation $x^2 + \sqrt{5}x + \sqrt[3]{2} = 0$ in an algebraic closure of $\mathbb{Q}$. Find an equation with *rational* coefficients having root γ.

6.3 Find a polynomial with rational coefficients having a root $\sqrt{2} + \sqrt{3}$.

6.4 Find a polynomial with rational coefficients having a root $\sqrt{2} + \sqrt[3]{5}$.

6.5 Let γ be a root of $x^5 - x + 1 = 0$ in an algebraic closure of $\mathbb{Q}$. Find a polynomial with rational coefficients of which $\gamma + \sqrt{2}$ is a root.

6.6 Show that the field obtained by adjoining $\sqrt{2}$, $\sqrt[4]{2}$, $\sqrt[8]{2}$, $\sqrt[16]{2}$, ..., $\sqrt[2^n]{2}$, ..., to $\mathbb{Q}$ is *not* of finite degree over $\mathbb{Q}$.

7. Some Irreducible Polynomials

7.1 Irreducibles over a finite field
7.2 Worked examples

Linear factors $x - \alpha$ of a polynomial $P(x)$ with coefficients in a field k correspond precisely to roots $\alpha \in k$ of the equation $P(x) = 0$. This follows from unique factorization in $k[x]$. [153] Here we also look at some special higher-degree polynomials, over *finite* fields, where we useful structural interpretation of the polynomials. [154]

Here we take for granted the existence of an algebraic closure $\overline{k}$ of a given field, as a fixed universe in which to consider roots of polynomial equations.

7.1 *Irreducibles over a finite field*

7.1.1 Proposition: Let (non-constant) $M(x)$ be an irreducible in $k[x]$, with field k. Let I be the ideal generated in $k[x]$ by $M(x)$. Let α be the image of x in the field $K = k[x]/I$. Then α is a root of the equation $M(x) = 0$. [155]

Proof: The salient aspects of the ring structure in the quotient can be summarized by the point that the quotient map $k[x] \longrightarrow k[x]/I$ is a ring homomorphism, in fact, a k-algebra homomorphism. Thus, for any polynomial f,

$$f(x) + I = f(x + I)$$

In particular,

$$M(x + I) = M(x) + I = I = 0 + I$$

[153] And this unique factorization follows from the *Euclidean*-ness of the polynomial ring.

[154] All these are *cyclotomic* polynomials, that is, divisors of $x^n - 1$ for some n. A systematic investigation of these polynomials is best done with a little more preparation. But they do provide accessible examples immediately.

[155] This is immediate, when one looks at the proof, but deserves complete explicitness.

which shows that $x + I$ is a root of the equations. ///

7.1.2 Proposition: [156] Let $P(x)$ be a polynomial in $k[x]$ for a field k. The equation $P(x) = 0$ has a root α generating [157] a degree d extension K of k if and only if $P(x)$ has a degree d irreducible factor $f(x)$ in $k[x]$.

Proof: Let α be a root of $P(x) = 0$ generating a degree d extension [158] $k(\alpha) = k[\alpha]$ over k. Let $M(x)$ be the minimal polynomial for α over k. Let

$$P = Q \cdot M + R$$

in $k[x]$ with $\deg R < \deg M$. Then, evaluating these polynomials at α, $R(\alpha) = 0$, but the minimality of the degree of M with this property assures that $R = 0$. That is, M divides P.

On the other hand, for an irreducible (monic, without loss of generality) $M(x)$ dividing $P(x)$, the quotient $K = k[x]/\langle M(x)\rangle$ is a field containing (a canonical copy of) k, and the image α of x in that extension is a root of $M(x) = 0$. Letting $P = Q \cdot M$,

$$P(\alpha) = Q(\alpha) \cdot M(\alpha) = Q(\alpha) \cdot 0 = 0$$

showing that $P(x) = 0$ has root α. ///

The first two examples use only the correspondence between linear factors and roots in the ground field.

7.1.3 Example: $x^2 + 1$ is irreducible over $k = \mathbb{Z}/p$ for any prime $p = 3 \bmod 4$.

Indeed, if $x^2 + 1$ had a linear factor then the equation $x^2 + 1 = 0$ would have a root α in k. This alleged root would have the property that $\alpha^2 = -1$. Thus, $\alpha \neq 1$, $\alpha \neq -1$, but $\alpha^4 = 1$. That is, the order of α in $k^\times$ is 4. But the order of $(\mathbb{Z}/p)^\times$ is $p - 1$. The hypothesis $p = 3 \bmod 4$ was exactly designed to deny the existence of an element of order 4 in $(\mathbb{Z}/p)^\times$. Thus, $x^2 + 1$ is irreducible in such $k[x]$.

7.1.4 Example: $x^2 + x + 1$ is irreducible over $k = \mathbb{Z}/p$ for any prime $p = 2 \bmod 3$.

If $x^2 + x + 1$ had a linear factor then $x^2 + x + 1 = 0$ would have a root α in k, and, since

$$x^3 - 1 = (x - 1)(x^2 + x + 1)$$

$\alpha^3 = 1$ but $\alpha \neq 1$ since $1 + 1 + 1 \neq 0$ in k. That is, the order of α in $k^\times$ is 3. But the order of $(\mathbb{Z}/p)^\times$ is $p - 1$, and the hypothesis $p = 2 \bmod 3$ exactly precludes any element of order 3 in $(\mathbb{Z}/p)^\times$. Thus, $x^2 + x + 1$ is irreducible in such $k[x]$.

7.1.5 Example: $P(x) = x^4 + x^3 + x^2 + x + 1$ is irreducible over $k = \mathbb{Z}/p$ for prime $p \neq \pm 1 \bmod 5$ and $p \neq 5$. Note that

$$x^5 - 1 = (x - 1)(x^4 + x^3 + x^2 + x + 1)$$

Thus, any root of $P(x) = 0$ has order [159] 5 or 1 (in whatever field it lies). The only element of order 1 is the identity element 1. If $P(x)$ had a linear factor in $k[x]$, then $P(x) = 0$ would have a root in k. Since

[156] This assertion should not be surprising, when one looks at the technique of the proof, which is nearly identical to the proof that linear factors correspond to roots in the base field.

[157] As earlier, the field extension $k(\alpha)$ *generated by* α makes sense only inside a fixed larger field. Throughout the present discussion we fix an algebraic closure of any ground field k and consider extensions inside that algebraic closure.

[158] Since the degree of the extension is finite, it is equal to polynomials in α over k, as we saw earlier.

[159] By Lagrange.

$1+1+1+1+1 \neq 0$ in k, 1 is not a root, so any possible root must have order 5. [160] But the order of $k^\times = (\mathbb{Z}/p)^\times$ is $p-1$, which is not divisible by 5, so there is no root in the base field k.

If $P(x)$ had an irreducible *quadratic* factor $q(x)$ in $k[x]$, then $P(x) = 0$ would have a root in a quadratic extension K of k. Since $[K:k] = 2$, the field K has p^2 elements, and

$$K^\times = p^2 - 1 = (p-1)(p+1)$$

By Lagrange, the order of any element of $K^\times$ is a divisor of p^2-1, but 5 does not divide p^2-1, so there is no element in K of order 5. That is, there is no quadratic irreducible factor.

By additivity of degrees in products, lack of factors up to half the degree of a polynomial assures that the polynomial is irreducible. Thus, since the quartic $x^4+x^3+x^2+x+1$ has no linear or quadratic factors, it is irreducible.

7.1.6 Example: $P(x) = x^6+x^5+x^4+x^3+x^2+x+1$ is irreducible over $k = \mathbb{Z}/p$ for prime $p = 3 \bmod 7$ or $p = 5 \bmod 7$.

Note that

$$x^7 - 1 = (x-1)(x^6+x^5+x^4+x^3+x^2+x+1)$$

Thus, any root of $P(x) = 0$ has order 7 or 1 (in whatever field it lies). The only element of order 1 is the identity element 1. If $P(x)$ had a linear factor in $k[x]$, then $P(x) = 0$ would have a root in k. Since $1+1+1+1+1+1+1 \neq 0$ in k, 1 is not a root, so any possible root must have order 7. But the order of $k^\times = (\mathbb{Z}/p)^\times$ is $p-1$, which is not divisible by 7, so there is no root in the base field k.

If $P(x)$ had an irreducible *quadratic* factor $q(x)$ in $k[x]$, then $P(x) = 0$ would have a root in a quadratic extension K of k. Since $[K:k] = 2$, the field K has p^2 elements, and

$$|K^\times| = p^2 - 1 = (p-1)(p+1)$$

By Lagrange, the order of any element of $K^\times$ is a divisor of p^2-1, but 7 divides neither $3^2-1 = 8 = 1 \bmod 7$ nor $5^2-1 = 24 = 3 \bmod 7$, so there is no element in K of order 7. That is, there is no quadratic irreducible factor.

If $P(x)$ had an irreducible *cubic* factor $q(x)$ in $k[x]$, then $P(x) = 0$ would have a root in a cubic extension K of k. Since $[K:k] = 3$, the field K has p^3 elements, and

$$|K^\times| = p^3 - 1$$

By Lagrange, the order of any element of $K^\times$ is a divisor of p^3-1, but 7 divides neither $3^3-1 = 26 = 5 \bmod 7$ nor $5^3-1 = -8 = -1 \bmod 7$, so there is no element in K of order 7. That is, there is no cubic irreducible factor.

By additivity of degrees in products, lack of factors up to half the degree of a polynomial assures that the polynomial is irreducible. Thus, since the sextic $x^6+x^5+x^4+x^3+x^2+x+1$ has no linear, quadratic, or cubic factors, it is irreducible.

7.1.7 Example: $P(x) = (x^{11}-1)/(x-1)$ is irreducible over $k = \mathbb{Z}/p$ for prime p of order 10 (multiplicatively) mod 11. That is, modulo $p = 2, 6, 7, 8 \bmod 11$ this polynomial is irreducible. [161]

[160] The only other positive divisor of 5, thinking of Lagrange.

[161] By this point, one might have guessed that the irreducibility will be assured by taking primes p such that $p^d \neq 1$ for $d < 10$. The fact that there are such primes can be verified in an *ad hoc* fashion by simply looking for them, and Dirichlet's theorem on primes in arithmetic progressions assures that there are infinitely many such. The presence of primitive roots $2, 6, 7, 8$ (that is, generators for the cyclic group $(\mathbb{Z}/11)^\times$) modulo 11 is yet another issue, when we replace 11 by a different prime.

Again, any root of $P(x) = 0$ has order 11 or 1 (in whatever field it lies). The only element of order 1 is the identity element 1. If $P(x)$ had a linear factor in $k[x]$, then $P(x) = 0$ would have a root in k. Since $11 \neq 0$ in k, 1 is not a root, so any possible root must have order 11. But the order of $k^\times = (\mathbb{Z}/p)^\times$ is $p - 1$, which is not divisible by 11, so there is no root in the base field k.

If $P(x)$ had an irreducible degree d factor $q(x)$ in $k[x]$, then $P(x) = 0$ would have a root in a degree d extension K of k. The field K has p^d elements, so

$$|K^\times| = p^d - 1$$

By Lagrange, the order of any element of $K^\times$ is a divisor of $p^d - 1$, but 11 divides none of $p - 1$, $p^2 - 1$, $p^3 - 1$, $p^4 - 1$, ..., $p^9 - 1$, by design.

7.2 *Worked examples*

7.2.1 Example: *(Lagrange interpolation)* Let $\alpha_1, \ldots, \alpha_n$ be *distinct* elements in a field k, and let $\beta_1, \ldots, \beta_n$ be any elements of k. Prove that there is a unique polynomial $P(x)$ of degree $< n$ in $k[x]$ such that, for all indices i,

$$P(\alpha_i) = \beta_i$$

Indeed, letting

$$Q(x) = \prod_{i=1}^{n} (x - \alpha_i)$$

show that

$$P(x) = \sum_{i=1}^{n} \frac{Q(x)}{(x - \alpha_i) \cdot Q'(\alpha_i)} \cdot \beta_i$$

Since the α_i are distinct,

$$Q'(\alpha_i) = \prod_{j \neq i} (\alpha_i - \alpha_j) \neq 0$$

(One could say more about purely algebraic notions of derivative, but maybe not just now.) Evaluating $P(x)$ at $x \longrightarrow \alpha_i$,

$$\frac{Q(x)}{(x - \alpha_j)} \text{ evaluated at } x \longrightarrow \alpha_i = \begin{cases} 1 & (\text{for } j = i) \\ 0 & (\text{for } j = i) \end{cases}$$

Thus, all terms but the ith vanish in the sum, and the ith one, by design, gives β_i. For uniqueness, suppose $R(x)$ were another polynomial of degree $< n$ taking the same values at n distinct points α_i as does $Q(x)$. Then $Q - R$ is of degree $< n$ and vanishes at n points. A non-zero degree ℓ polynomial has at most ℓ zeros, so it must be that $Q - R$ is the 0 polynomial.

7.2.2 Example: *(Simple case of partial fractions)* Let $\alpha_1, \ldots, \alpha_n$ be *distinct* elements in a field k. Let $R(x)$ be any polynomial in $k[x]$ of degree $< n$. Show that there exist unique constants $c_i \in k$ such that in the field of rational functions $k(x)$

$$\frac{R(x)}{(x - \alpha_1) \ldots (x - \alpha_n)} = \frac{c_1}{x - \alpha_1} + \ldots + \frac{c_n}{x - \alpha_n}$$

In particular, let

$$Q(x) = \prod_{i=1}^{n} (x - \alpha_i)$$

and show that

$$c_i = \frac{R(\alpha_i)}{Q'(\alpha_i)}$$

We might emphasize that the field of rational functions $k(x)$ is most precisely the *field of fractions* of the polynomial ring $k[x]$. Thus, in particular, equality $r/s = r'/s'$ is exactly equivalent to the equality $rs' = r's$ (as in elementary school). Thus, to test whether or not the indicated expression performs as claimed, we test whether or not

$$R(x) = \sum_i \left(\frac{R(\alpha_i)}{Q'(\alpha_i)} \cdot \frac{Q(x)}{x - \alpha_i} \right)$$

One might notice that this is the previous problem, in case $\beta_i = R(\alpha_i)$, so its correctness is just a special case of that, as is the uniqueness (since $\deg R < n$).

7.2.3 Example: Show that the ideal I generated in $\mathbb{Z}[x]$ by $x^2 + 1$ and 5 is *not* maximal.

We will show that the quotient is not a field, which implies (by the standard result proven above) that the ideal is not maximal (proper).

First, let us make absolutely clear that the quotient of a ring R by an ideal $I = Rx + Ry$ generated by two elements can be expressed as a two-step quotient, namely

$$(R/\langle x \rangle)/\langle \bar{y} \rangle \approx R/(Rx + Ry)$$

where the $\langle \bar{y} \rangle$ is the principal ideal generated by the *image* $\bar{y}$ of y in the quotient $R/\langle x \rangle$. The principal ideal generated by y in the quotient $R/\langle x \rangle$ is the set of cosets

$$\langle \bar{y} \rangle = \{(r + Rx) \cdot (y + Rx) : r \in R\} = \{ry + Rx : r \in R\}$$

noting that the multiplication of cosets in the quotient ring is *not* just the element-wise multiplication of the cosets. With this explication, the natural map is

$$r + \langle x \rangle = r + \langle x \rangle \longrightarrow r + \langle x \rangle + \langle y \rangle' = r + (Rx + Rx)$$

which is visibly the same as taking the quotient in a single step.

Thus, first

$$\mathbb{Z}[x]/\langle 5 \rangle \approx (\mathbb{Z}/5)[x]$$

by the map which reduces the coefficients of a polynomial modulo 5. In $(\mathbb{Z}/5)[x]$, the polynomial $x^2 + 1$ *does* factor, as

$$x^2 + 1 = (x - 2)(x + 2)$$

(where these 2s are in $\mathbb{Z}/5$, not in $\mathbb{Z}$). Thus, the quotient $(\mathbb{Z}/5)[x]/\langle x^2 + 1 \rangle$ has proper zero divisors $\bar{x} - 2$ and $\bar{x} + 2$, where $\bar{x}$ is the image of x in the quotient. Thus, it's not even an integral domain, much less a field.

7.2.4 Example: Show that the ideal I generated in $\mathbb{Z}[x]$ by $x^2 + x + 1$ and 7 is *not* maximal.

As in the previous problem, we compute the quotient in two steps. First,

$$\mathbb{Z}[x]/\langle 7 \rangle \approx (\mathbb{Z}/7)[x]$$

by the map which reduces the coefficients of a polynomial modulo 7. In $(\mathbb{Z}/7)[x]$, the polynomial $x^2 + x + 1$ *does* factor, as

$$x^2 + x + 1 = (x - 2)(x - 4)$$

(where 2 and 4 are in $\mathbb{Z}/7$). Thus, the quotient $(\mathbb{Z}/7)[x]/\langle x^2+x+1\rangle$ has proper zero divisors $\bar{x}-2$ and $\bar{x}-4$, where $\bar{x}$ is the image of x in the quotient. Thus, it's not even an integral domain, so certainly not a field.

Exercises

7.1 Show that x^2+x+1 is irreducible in $\mathbb{F}_5[x]$, and in $\mathbb{F}_{29}[x]$.

7.2 Show that x^3-a is irreducible in $\mathbb{F}_7[x]$ unless $a=0$ or ± 1.

7.3 Determine how x^5+1 factors into irreducibles in $\mathbb{F}_2[x]$.

7.4 Exhibit an irreducible quintic in $\mathbb{F}_{11}[x]$.

7.5 Show that the ideal generated by x^2-x+1 and 13 in $\mathbb{Z}[x]$ is *not* maximal.

7.6 Show that the ideal generated by x^2-x+1 and 17 in $\mathbb{Z}[x]$ *is* maximal.

7.7 Let $\alpha_1,\ldots,\alpha_n$ be distinct elements of a field k not of characteristic 2. Show that there are elements $a_1,b_1,\ldots,a_n,b_n$ in k such that

$$\frac{1}{(x-\alpha_1)^2\ldots(x-\alpha_n)^2} = \frac{a_1}{x-\alpha_1}+\frac{b_1}{(x-\alpha_1)^2}+\ldots+\frac{a_n}{x-\alpha_n}+\frac{b_n}{(x-\alpha_n)^2}$$

8. Cyclotomic polynomials

8.1 Multiple factors in polynomials
8.2 Cyclotomic polynomials
8.3 Examples
8.4 Finite subgroups of fields
8.5 Infinitude of primes $p = 1 \bmod n$
8.6 Worked examples

8.1 *Multiple factors in polynomials*

There is a simple device to detect repeated occurrence of a factor in a polynomial with coefficients in a field.

Let k be a *field*. For a polynomial

$$f(x) = c_n x^n + \ldots + c_1 x + c_0$$

with coefficients c_i in k, *define* the **(algebraic) derivative** [162] $Df(x)$ of $f(x)$ by

$$Df(x) = nc_n x^{n-1} + (n-1)c_{n-1}x^{n-2} + \ldots + 3c_3 x^2 + 2c_2 x + c_1$$

Better said, D is by definition a k-linear map

$$D : k[x] \longrightarrow k[x]$$

defined on the k-basis $\{x^n\}$ by

$$D(x^n) = nx^{n-1}$$

8.1.1 Lemma: For f, g in $k[x]$,

$$D(fg) = Df \cdot g + f \cdot Dg$$

[162] Just as in the calculus of polynomials and rational functions one is able to evaluate all limits algebraically, one can readily prove (without reference to any limit-taking processes) that the notion of *derivative* given by this formula has the usual properties.

8.1.2 Remark: Any k-linear map T of a k-algebra R to itself, with the property that

$$T(rs) = T(r) \cdot s + r \cdot T(s)$$

is a k-linear **derivation** on R.

Proof: Granting the k-linearity of T, to prove the derivation property of D is suffices to consider basis elements x^m, x^n of $k[x]$. On one hand,

$$D(x^m \cdot x^n) = Dx^{m+n} = (m+n)x^{m+n-1}$$

On the other hand,

$$Df \cdot g + f \cdot Dg = mx^{m-1} \cdot x^n + x^m \cdot nx^{n-1} = (m+n)x^{m+n-1}$$

yielding the product rule for monomials. ///

A field k is **perfect** if either the characteristic of k is 0 [163] or if, in characteristic $p > 0$, there is a pth root $a^{1/p}$ in k for every $a \in k$. [164]

8.1.3 Proposition: Let $f(x) \in k[x]$ with a field k, and P an irreducible polynomial in $k[x]$. If P^e divides f then P divides $\gcd(f, Df)$. If k is *perfect* and $e - 1 \neq 0$ in k, there is a converse: [165] if P^{e-1} divides both f and Df then P^e divides f.

Proof: On one hand, suppose $f = P^e \cdot g$ with ≥ 2. By the product rule,

$$Df = eP^{e-1}DP \cdot g + P^e \cdot Dg$$

is a multiple of P^{e-1}. [166] This was the easy half.

On the other hand, for the harder half of the assertion, suppose P^{e-1} divides both f and Df. Write

$$f/P^{e-1} = Q \cdot P + R$$

with $\deg R < \deg P$. Then $f = QP^e + RP^{e-1}$. Differentiating,

$$Df = DQ\, P^e + eQP^{e-1}DP + DR\, P^{e-1} + R(e-1)P^{e-2}\, DP$$

By hypothesis P^{e-1} divides Df. All terms on the right-hand side except possibly $R(e-1)P^{e-2}\, DP$ are divisible by P^{e-1}, so P divides $R(e-1)P^{e-2}\, DP$. Since P is irreducible, either $e - 1 = 0$ in k, or P divides R, or P divides DP. If P divides R, P^e divides f, and we're done.

If P does not divide R then P divides DP. Since $\deg DP < \deg P$, if P divides DP then $DP = 0$. This would require that all the exponents of x occurring with non-zero coefficient are divisible by the characteristic p, which must be positive. So P is of the form

$$P(x) = a_{pm}x^{pm} + a_{p(m-1)}x^{p(m-1)} + a_{p(m-2)}x^{p(m-2)} + \ldots + a_{2p}x^{2p} + a_p x^p + a_0$$

[163] as for $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$

[164] As is the case for finite fields such as $\mathbb{Z}/p$, by Fermat's Little Theorem.

[165] In particular, this converse holds if the characteristic of k is 0.

[166] This half does not need the irreducibility of P.

Using the perfect-ness of the field k, each a_i has a pth root b_i in k. Because the characteristic is $p > 0$,

$$(A + B)^p = A^p + B^p$$

Thus, $P(x)$ is the pth power of

$$b_{pm}x^n + b_{p(m-1)}x^{(m-1)} + b_{p(m-2)}x^{(m-2)} + \ldots + b_{2p}x^2 + b_px + b_0$$

If P is a pth power it is not irreducible. Therefore, for P irreducible DP is not the zero polynomial. Therefore, $R = 0$, which is to say that P^e divides f, as claimed. ///

8.2 *Cyclotomic polynomials*

For $b \neq 0$ in a field k, the **exponent** of b is the smallest positive integer n (if it exists) such that $b^n = 1$. That is, b is a root of $x^n - 1$ but not of $x^d - 1$ for any smaller d. We construct polynomials $\Phi_n(x) \in \mathbb{Z}[x]$ such that

$$\Phi_n(b) = 0 \text{ if and only if } b \text{ is of exponent } n$$

These polynomials Φ_n are **cyclotomic polynomials**.

8.2.1 Corollary: The polynomial $x^n - 1$ has no repeated factors in $k[x]$ if the field k has characteristic *not* dividing n.

Proof: It suffices to check that $x^n - 1$ and its derivative nx^{n-1} have no common factor. Since the characteristic of the field does not to divide n, $n \cdot 1_k \neq 0$ in k, so has a multiplicative inverse t in k, and

$$(x^n - 1) - (tx) \cdot (nx^{n-1}) = -1$$

and $\gcd(x^n - 1, nx^{n-1}) = 1$. ///

Define the nth **cyclotomic polynomial** $\Phi_n(x)$ by

$$\Phi_1(x) = x - 1$$

and for $n > 1$, inductively,

$$\Phi_n(x) = \frac{x^n - 1}{lcm \text{ of all } x^d - 1 \text{ with } 0 < d < n,\ d \text{ dividing } n}$$

with the least common multiple *monic*.

8.2.2 Theorem:

- Φ_n is a monic polynomial with integer coefficients. [167]
- For α in the field k, $\Phi_n(\alpha) = 0$ if and only if $\alpha^n = 1$ and $\alpha^t \neq 1$ for all $0 < t < n$.
- $\gcd(\Phi_m(x), \Phi_n(x)) = 1$ for $m < n$ with neither m nor n divisible by the characteristic of the field k.
- The degree of $\Phi_n(x)$ is $\varphi(n)$ (Euler's phi-function)
- Another description of $\Phi_n(x)$:

$$\Phi_n(x) = \frac{x^n - 1}{\prod_{1 \leq d < n, d|n} \Phi_d(x)}$$

[167] More properly, if the ambient field k is of characteristic 0, then the coefficients lie in the copy of $\mathbb{Z}$ inside the prime field $\mathbb{Q}$ inside k. If the ambient field is of positive characteristic, then the coefficients lie inside the prime field (which is the natural image of $\mathbb{Z}$ in k). It would have been more elegant to consider the cyclotomic polynomials as polynomials in $\mathbb{Z}[x]$, but this would have required that we wait longer.

• $x^n - 1$ factors as

$$x^n - 1 = \prod_{1 \le d \le n, d|n} \Phi_d(x)$$

Proof: We know that $d|n$ (and $d > 0$) implies that $x^d - 1$ divides $x^n - 1$. Therefore, by unique factorization, the least common multiple of a collection of things each dividing $x^n - 1$ also divides $x^n - 1$. Thus, the indicated *lcm* does divide $x^n - 1$.

For α in k, $x - \alpha$ divides $\Phi_n(x)$ if and only if $\Phi_n(\alpha) = 0$. And $\alpha^t = 1$ if and only if $x - \alpha$ divides $x^t - 1$. The definition

$$\Phi_n(x) = \frac{x^n - 1}{\textit{lcm of all } x^d - 1 \text{ with } 0 < d < n,\ d \text{ dividing } n}$$

shows first that $\Phi_n(\alpha) = 0$ implies $\alpha^n = 1$. Second, if $\alpha^t = 1$ for any proper divisor t of n then $x - \alpha$ divides $x^t - 1$, and thus $x - \alpha$ divides the denominator. But $x^n - 1$ has no repeated factors, so $x - \alpha$ dividing the denominator would prevent $x - \alpha$ dividing $\Phi_n(x)$, contradiction. That is, $\Phi_n(\alpha) = 0$ if and only if α is of order n.

To determine the *gcd* of Φ_m and Φ_n for neither m nor n divisible by the characteristic of k, note that Φ_m divides $x^m - 1$ and Φ_n divides $x^n - 1$, so

$$\gcd(\Phi_m, \Phi_n) \quad \text{divides} \quad \gcd(x^m - 1, x^n - 1)$$

We claim that for m, n two integers (divisible by the characteristic or not)

$$\gcd(x^m - 1, x^n - 1) = x^{gcd(m,n)} - 1$$

Prove this claim by induction on the maximum of m and n. Reduce to the case $m > n$, wherein

$$x^m - 1 - x^{m-n} \cdot (x^n - 1) = x^{m-n} - 1$$

For g a polynomial dividing both $x^m - 1$ and $x^n - 1$, g divides $x^{m-n} - 1$. By induction,

$$\gcd(x^{m-n} - 1, x^n - 1) = x^{gcd(m-n,n)} - 1$$

But

$$\gcd(m, n) = \gcd(m - n, n)$$

and

$$x^m - 1 = x^{m-n} \cdot (x^n - 1) + x^{m-n} - 1$$

so

$$\gcd(x^m - 1, x^n - 1) = \gcd(x^{m-n} - 1, x^n - 1)$$

and induction works. Thus,

$$\gcd(x^m - 1, x^n - 1) = x^{gcd(m,n)} - 1$$

Since

$$d \le m < n$$

d is a *proper* divisor of n. Thus, from

$$\Phi_n(x) = \frac{x^n - 1}{\textit{lcm of all } x^d - 1 \text{ with } 0 < d < n,\ d \text{ dividing } n}$$

we see that $\Phi_n(x)$ divides $(x^n - 1)/(x^d - 1)$. Since $x^n - 1$ has no repeated factors, $\Phi_n(x)$ has no factors in common with $x^d - 1$. Thus, $\gcd(\Phi_m, \Phi_n) = 1$.

Next, use induction to prove that

$$x^n - 1 = \prod_{1 \le d \le n,\, d|n} \Phi_d(x)$$

For $n = 1$ the assertion is true. From the definition of Φ_n,

$$x^n - 1 = \Phi_n(x) \cdot \mathrm{lcm}\{x^d - 1 : d|n, 0 < d < n\}$$

By induction, for $d < n$

$$x^d - 1 = \prod_{0 < e \le d, e|d} \Phi_e(x)$$

Since for $m < n$ the *gcd* of Φ_m and Φ_n is 1,

$$\mathrm{lcm}\{x^d - 1 : d|n, 0 < d < n\} = \prod_{d|n, d<n} \Phi_d(x)$$

Thus,

$$x^n - 1 = \Phi_n(x) \cdot \prod_{d|n, d<n} \Phi_d(x)$$

as claimed.

Inductively, since all lower-index cyclotomic polynomials have integer coefficients [168] and are monic, and $x^n - 1$ is monic with integer coefficients, the quotient of $x^n - 1$ by the product of the lower ones is monic with integer coefficients.

The assertion about the degree of Φ_n follows from the identity (see below) for Euler's phi-function

$$\sum_{d|n, d>0} \varphi(d) = n$$

This completes the proof of the theorem. ///

8.2.3 Proposition: Let $\varphi(x)$ be Euler's phi-function

$$\varphi(x) = \sum_{1 \le \ell \le x; \gcd(\ell, x) = 1} 1$$

Then for m and n relatively prime

$$\varphi(mn) = \varphi(m) \cdot \varphi(n) \qquad \text{(weak multiplicativity)}$$

For p prime and ℓ a positive integer

$$\varphi(p^\ell) = (p-1) \cdot p^{\ell - 1}$$

And

$$\sum_{d|n, d>0} \varphi(d) = n$$

Proof: By unique factorization, for $\gcd(m, n) = 1$,

$$\gcd(t, mn) = \gcd(t, m) \cdot \gcd(t, n)$$

[168] Or, more properly, coefficients in the canonical image of $\mathbb{Z}$ in the field k.

so, t is prime to mn if and only if t is prime to both m and n. The gcd of m and n is the smallest positive integer of the form $rm + sn$. By Sun-Ze,

$$f : \mathbb{Z}/m \oplus \mathbb{Z}/n \longrightarrow \mathbb{Z}/mn$$

by

$$f : (x, y) \longrightarrow rmy + snx$$

is a *bijection*, since m and n are coprime. From $rm + yn = 1$, $rm = 1 \bmod n$ so rm is prime to n, and $sn = 1 \bmod m$ so sn is prime to m. Thus, $rmy + snx$ has a common factor with m if and only if x does, and $rmy + snx$ has a common factor with n if and only if y does. Thus, f gives a bijection

$$\{x : 1 \leq x < m, \gcd(x, m) = 1)\} \times \{y : 1 \leq y < n, \gcd(y, n) = 1)\}$$

$$\longrightarrow \{z : 1 \leq z < mn, \gcd(z, mn) = 1)\}$$

and $\varphi(mn) = \varphi(m) \cdot \varphi(n)$. This reduces calculation of $\varphi()$ to calculation for prime powers p^e. An integer x in $1 \leq x < p^e$ is prime to p^e if and only if it is not divisible by p, so there are

$$\varphi(p^e) = p^e - p^{e-1} = (p - 1)p^{e-1}$$

such x, as claimed.

To prove

$$\sum_{d|n, d>0} \varphi(d) = n$$

start with n a prime power p^e, in which case

$$\sum_{d|p^e} \varphi(d) = \sum_{0 \leq k \leq e} \varphi(p^k) = 1 + \sum_{1 \leq k \leq e} (p - 1)p^{k-1} = 1 + (p - 1)(p^e - 1)/(p - 1) = p^e$$

Let $n = p_1^{e_1} \dots p_t^{e_t}$ with distinct primes p_i. Then

$$\sum_{d|n} \varphi(d) = \prod_{i=1,\dots,t} \left(\sum_{d|p_i^{e_i}} \varphi(d) \right) = \prod_{i=1,\dots,t} \varphi(p_i^{e_i}) = \varphi(\prod_i p_i^{e_i}) = \varphi(n)$$

This proves the desired identity for φ. ///

8.3 *Examples*

For prime p, the factorization of $x^p - 1$ into cyclotomic polynomials is boring

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + x^2 + x + 1$$

For $n = 2p$ with odd prime p

$$\Phi_{2p}(x) = \frac{x^{2p} - 1}{\Phi_1(x)\,\Phi_2(x)\,\Phi_p(x)} = \frac{x^{2p} - 1}{\Phi_2(x)\,(x^p - 1)} = \frac{x^p + 1}{x + 1}$$

$$= x^{p-1} - x^{p-2} + x^{p-3} - \dots + x^2 - x + 1$$